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Master Thesis

Impact of Automated Thermal Zoning on Building Energy Simulations: Ground-Truth Validation and Sensitivity Analysis

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List of Abbreviations

Abbreviation	**Meaning**
AixLib	Modelica library for building and energy system simulation
BES	Building Energy Simulation
BSP	Binary Space Partitioning
CityGML	City Geography Markup Language
DPI	Dots per inch
GT	Ground truth
IoU	Intersection over Union
KPI	Key Performance Indicator
LoD	Level of Detail
LoD2	Level of Detail 2
LPG	LoadProfileGenerator
MSD	Modified Swiss Dwellings
MZA	Multizone Assignment Algorithm
OSM	OpenStreetMap
PDF	Portable Document Format
RC	Resistance–Capacitance
ROM	Reduced-Order Model
TABULA	Typology Approach for Building Stock Energy Assessment
TEASER	Tool for Energy Analysis and Simulation for Efficient Retrofit
TRY	Test Reference Year
UBEM	Urban Building Energy Modelling
VDI	Verein Deutscher Ingenieure
WWR	Window-to-Wall Ratio

1. Introduction

Buildings account for a substantial share of final energy consumption and greenhouse gas emissions worldwide. Therefore, the building sector plays an important role in the transition towards lower operational energy use and improved urban thermal comfort. As urban areas move towards climate-neutral building stocks, building energy simulation becomes increasingly important for evaluating renovation strategies and their effect on building performance. However, such analyses must often be carried out not only for individual buildings, but also for neighbourhoods, districts, and entire urban building stocks.

Urban building energy modelling (UBEM) addresses this need by extending building energy simulation from individual buildings to larger building stocks. It can support urban energy planning and policy-related scenario analysis (Reinhart and Cerezo Davila 2016). However, the reliability of UBEM results depends strongly on how individual buildings are represented within the aggregated model. Building-level simplifications can affect the accuracy of simulated performance indicators and, in turn, influence the interpretation of urban-scale energy assessments.

A major limitation of current UBEM workflows is the restricted availability of detailed internal building information. Common urban data sources, such as CityGML and LoD2 building models, usually describe the external building form rather than the internal spatial organisation. Still, they do not typically include internal layouts or dwelling-level structures (Biljecki et al. 2016). As a result, buildings are often modelled as single thermal zones or with simplified zoning assumptions. While these simplifications reduce modelling effort, they can neglect relevant internal thermal differences within multi-family residential buildings.

Thermal zoning is therefore a central modelling decision in building energy simulation. It defines the spatial units used for thermal calculation and operational assumptions. Previous studies show that zoning granularity can influence simulation reliability and computational effort (Shin and Haberl 2019). This is particularly relevant for multi-family residential buildings, where separate dwellings and stairwells may have different thermal conditions and may interact through interzonal heat exchange.

At urban scale, manually defining suitable thermal zones for many buildings is not practical because detailed floor plans are usually unavailable and the modelling effort would be too high. Automated thermal zone generation methods therefore provide a way to improve model detail while maintaining scalability. Existing methods such as Autozoner generate thermal zones for buildings with unknown or limited interior space definitions (Dogan et al. 2016). However, generic zoning approaches do not fully represent the dwelling–stairwell organisation that is important for multi-family residential buildings.

The Multizone Assignment Algorithm (MZA) addresses this limitation by generating thermal zoning for individual dwellings and the stairwell from limited geometric and auxiliary input data (Waiz et al. 2025). In the wider MZA workflow, the generated zones can be enriched with thermal and operational assumptions and transferred to TEASER–AixLib for dynamic building energy simulation (Remmen et al. 2018). The method therefore offers a promising pathway for scalable multi-zone modelling of residential buildings when detailed internal layouts are not available.

Despite this potential, the applicability of the MZA depends on whether the generated dwelling–stairwell zoning is both geometrically reliable and simulation-relevant. Since the MZA infers internal structures from limited input, it is necessary to test whether the generated zones reproduce the reference zoning structure at the intended level of abstraction. Geometric deviations alone do not show whether zoning differences are relevant for simulation outputs. Therefore, their influence on selected energy and comfort indicators must be tested through simulation.

In addition to zoning uncertainty, MZA-based simulation also depends on uncertain input enrichment. Limited urban geometry must be converted into simulation-ready models based on assumptions about construction and envelope properties, operation, occupancy, and weather conditions. Previous studies show that input uncertainty can influence urban building energy simulation results and that sensitivity analysis is useful for identifying influential assumptions (Prataviera et al. 2022). Therefore, evaluating the MZA requires both geometric validation of the generated zones and sensitivity analysis of the resulting modelling uncertainties.

Against this background, this thesis aims to evaluate the suitability of the MZA for generating simulation-relevant dwelling–stairwell thermal zoning structures for multi-family residential buildings from limited geometric input. The thesis combines geometric validation with simulation-based sensitivity analysis. The validation compares MZA-generated zones with reference zoning structures derived from the Modified Swiss Dwellings dataset and German real estate floor plans. The simulation analysis then evaluates how controlled zoning variants and uncertain input parameters influence selected energy and comfort indicators.

The thesis first reviews the theoretical background and state of research on UBEM, thermal zoning, automated thermal zone generation, the MZA workflow, reference floor plan data, and sensitivity analysis. Based on this literature review, the research gap and research framework are identified. A methodology is then developed to address this gap by combining geometric validation of MZA-generated dwelling–stairwell zones with simulation-based sensitivity analysis. The resulting validation and simulation outputs are presented and subsequently discussed with respect to MZA reliability, the relevance of zoning deviations in the simulation, and the influence of uncertain input parameters.

2. Theoretical background and state of research

This chapter establishes the theoretical and methodological background for evaluating automated thermal zoning in building energy simulation. It first discusses thermal zoning as a modelling decision and explains how the granularity of zoning affects the balance among simulation detail, data requirements, and modelling effort. The chapter then examines the limitations of geometric data in urban building energy modelling. It reviews automated approaches for generating thermal zones from limited input data, with particular focus on the Multizone Assignment Algorithm. Since the thesis combines geometric validation with simulation-based sensitivity analysis, the chapter also reviews reference floor plan data for constructing ground-truth zoning representations and discusses sensitivity analysis as a method for identifying influential input parameters. The chapter concludes by synthesising the reviewed literature to identify a research gap and derive the research questions, hypotheses, and thesis aim.

2.1. Urban building energy modelling

Urban building energy modelling (UBEM) extends building energy modelling from individual buildings to groups of buildings, districts, or entire cities, enabling estimation of energy demand at larger spatial scales. Reinhart and Cerezo Davila (2016) describe UBEM as a bottom-up modelling approach in which physical building energy models are applied to groups of buildings to estimate operational energy use at neighbourhood or city scale.

A central limitation of UBEM is the restricted availability of detailed building information. Urban-scale datasets usually provide only high-level geometric and semantic information, such as building footprints, heights, numbers of storeys, use types, or years of construction, rather than the detailed input required for individual building simulation. Chen and Hong (2018) note that detailed 3D geometry and thermal zoning are difficult to obtain for each building at urban scale. Therefore, typical UBEM geometry inputs are often limited to GIS-based building geometry.

CityGML and related semantic 3D city models provide an important basis for organising urban-scale building geometry. As illustrated in Figure 2.1, CityGML buildings can be represented at different Levels of Detail (LoD), ranging from footprint representations to detailed exterior and interior building models. LoD2 is particularly relevant for urban building energy modelling because it is commonly available for large urban datasets and provides the external building volume and roof geometry required for envelope-based simulation.

Higher levels of detail, especially LoD3 and LoD4, can include façade openings and interior structures, but such information is rarely consistently available at the district or city scale. As a result, LoD2-based models still typically require additional methods to infer the internal spatial divisions required for thermal zone boundaries (Biljecki et al. 2016). Although Agugiaro et al. (2018) show that Energy ADE can represent simulation-relevant building information, the absence of internal layouts still requires automated zoning methods.

LoD2-based urban datasets often lack key thermal simulation attributes, such as construction

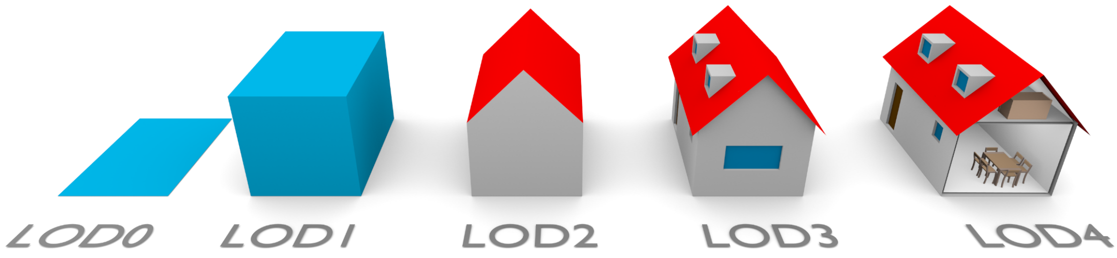


Figure 2.1: CityGML Levels of Detail (LoD) for building models, ranging from a two-dimensional footprint representation in LoD0 to an interior building model in LoD4. Adapted from Biljecki et al. (2016).

age, retrofit state, envelope properties, window areas, ventilation, schedules, internal gains, and setpoints. Therefore, simulation-ready models require both spatial enrichment through thermal zoning and typological enrichment of thermal and operational parameters.

TEASER illustrates how limited urban input data can be enriched for simulation by generating reduced-order thermal models from available geometry and typological assumptions (Remmen et al. 2018). However, such workflows still require predefined thermal zones and therefore depend on a prior zoning step. Since UBEM applies building energy modelling to large numbers of buildings, manually defining missing thermal zone structures is not scalable. Automated zoning methods are therefore needed to infer approximate apartment, corridor and stairwell structures when they are missing from urban-scale datasets.

2.2. Thermal zones as modelling units

Thermal zoning is a primary modelling decision in building energy simulation because it divides a building's internal space into heat-balance units. Shin and Haberl (2019) describe a thermal zone as a space or group of spaces with sufficiently similar heating and cooling requirements, thermal behaviour, usage patterns, and control requirements to be represented by one control device, such as a thermostat or temperature sensor. Therefore, a thermal zone is not only a geometric subdivision, but also a heat-transfer and control concept. If spaces with different thermal behaviour are merged into a single zone, these differences are simplified to a single common zone condition. The zoning decision therefore affects how zone-level thermal and operational conditions are represented in the simulation model. Johari et al. (2022) use a similar definition in urban building energy modelling, where a thermal zone is a portion of a building with HVAC demand sufficiently similar to be represented by a single controlling unit or sensor.

2.2.1. Zoning granularity and simulation purpose

Zoning granularity describes the spatial level at which a building is divided into thermal zones, depending on the simulation purpose, available data, and required accuracy. Reducing the number of zones is a common simplification strategy, but it must be evaluated because it affects modelling reliability, computational effort, and simulation accuracy (Johari et al. 2022).

Elhadad et al. (2020) similarly show that lower zoning detail can reduce simulation effort, but strong aggregation may cause deviations in heating, cooling, and comfort-related results.

This trade-off is also demonstrated by Jansen et al. (2021), who compare different zoning strategies using a TEASER-based reduced-order model and an EnergyPlus reference model. They show that suitable zoning strategies can keep annual heating and cooling demand within approximately ±10% while reducing computational effort by up to 97%. In contrast, unsuitable discretisation may produce unrealistic local thermal behaviour. Kühn et al. (2024) further relate zoning granularity to the distinction between high-order models and reduced-order models: high-order models provide greater spatial detail, while reduced-order models simplify the physical representation and aggregate zones, for example at flat or whole-building level.

For automated zoning workflows, zoning granularity is not only a deliberate modelling choice. Still, it can also appear as a deviation from the intended internal structure, for example when dwellings are merged, over-subdivided, displaced, or when shared circulation areas are represented differently. Such deviations may affect simulation outputs by altering façade allocation, interzonal heat-transfer areas, zone temperatures, or the treatment of unheated shared spaces (Brès et al. 2017).

2.2.2. Zoning levels

Building energy simulation can use different levels of thermal zoning, ranging from detailed room-level models to highly aggregated single-zone models. The appropriate zoning level depends on the simulation purpose, available input data, and acceptable modelling effort. Table 2.1 summarises the main zoning levels discussed in the literature, together with their typical purpose, limitations, and supporting references.

These findings show that each zoning level serves a distinct simulation purpose and entails different limitations. In the context of UBEM, this poses a specific challenge: more detailed dwelling- and stairwell-level zoning may improve the representation of multi-family residential buildings, but the internal spatial information required for such zoning is usually unavailable in urban-scale datasets. This gap motivates the need for automated thermal zone generation methods that can infer suitable zoning structures from limited geometric input.

2.3. Automated thermal zone generation

Several automated approaches have been developed to generate internal spatial or thermal zone representations when detailed building information is unavailable. Some methods focus on the procedural generation of interior layouts, such as Lopes et al. (2010), who generate room layouts for different building types and multi-storey buildings. In building energy modelling, however, the aim is often not to reconstruct a complete architectural floor plan, but to define suitable simulation zones. For this purpose, Dogan et al. (2016) propose Autozoner, which subdivides building massing models into perimeter and core thermal zones when interior space definitions are unknown. More recently, Xiang et al. (2024) propose the Convex Partition Zoner (CPZ), which automatically generates perimeter and core zones from building floor

Table 2.1: Levels of thermal zoning discussed in literature

Zoning level	Description	Simulation purpose	Limitation	References
Room level	Each room is modelled as an individual zone.	Individual building simulation when detailed input is available.	Requires internal layouts and high modelling effort.	Kühn et al. (2024); Elhadad et al. (2020)
Dwelling /flat level	Each flat is represented as a single zone.	Individual- or district-scale multi-family building simulation.	Does not represent room-level variation within each dwelling.	Kühn et al. (2024)
Storey level	Each floor is represented as one zone.	Simplified building, city or district simulations with vertical differences.	Cannot distinguish between different dwellings on the same floor.	Johari et al. (2022); Elhadad et al. (2020)
Single-zone	The entire building is represented as a single zone.	Large-scale neighbourhood, city, or building-stock simulations.	Averages differences between dwellings, orientations, floors, and shared spaces.	Johari et al. (2022); Elhadad et al. (2020)
Perimeter–core	The building is divided into perimeter and core zones.	Early stage design or automated urban modelling.	Does not represent the internal structure of multi-family buildings.	Shin and Haberl (2019); Dogan et al. (2016)

geometries when actual HVAC zones are unknown or not yet designed.

At the urban scale, automated model-generation workflows also use limited geospatial data to produce simulation-ready energy models. For example, Cerezo Davila et al. (2016) generate a citywide UBEM for Boston from existing geospatial datasets and archetype libraries, while Chen et al. (2017) introduce CityBES for automatically generating and simulating urban building energy models from city datasets.

For multi-family residential buildings, generic perimeter–core or floor-level zoning does not fully capture the dwelling and stairwell organisation that shapes dwelling-level thermal behaviour, interzonal heat transfer, and user profiles. Recent work by Waiz et al. (2025) addresses this limitation through the Multizone Assignment Algorithm (MZA), which generates dwelling- and stairwell-based thermal zone structures from limited geometry within the TEASER–AixLib simulation framework. The following section therefore discusses the MZA in more detail.

2.4. Multizone Assignment Algorithm

The objective of the MZA is to infer an approximate dwelling- and stairwell-based thermal zoning from limited geometric and semantic input. This is particularly relevant for multi-family residential buildings and dwelling blocks, where a single-zone representation can mask differences between individual dwellings and shared access spaces. These differences are relevant because heat transfer to unheated neighbouring zones, interzonal heat exchange between dwellings, varying occupancy profiles, and user-controlled setpoint temperatures can influence heat-load prediction.

At the level relevant for this thesis, the important output is not a complete architectural floor plan, but an abstracted thermal zoning of the floor area into dwelling zones and stairwell zones. This distinction is important for the validation approach of the present thesis, because the generated zones are evaluated at dwelling–stairwell level instead of room or architectural detail level.

For the present thesis, it is also important to distinguish between the MZA as a complete simulation workflow and the MZA as a spatial zone generation method. The full workflow includes data aggregation, preprocessing, automatic zoning, thermal enrichment, and dynamic simulation in TEASER–AixLib. The geometric and topological validation in this thesis focuses mainly on the spatial output of the zoning step: the number, geometry, location, and adjacency of the generated dwelling and stairwell zones.

2.4.1. Workflow structure

The MZA workflow can be divided into three conceptual layers: data aggregation and preprocessing, automatic zoning, and enrichment and simulation. The data aggregation and preprocessing layer prepares the geometric, semantic, and typological information required for the zoning process. In contrast, the enrichment and simulation layer translates the generated zones into thermal simulation models. Both layers depend strongly on available regional data, typological assumptions, and modelling choices. The automatic zoning layer, in contrast, represents the MZA's stairwell spatial logic and can be discussed separately once the required zoning inputs are provided. Figure 2.2 provides a conceptual overview of the MZA workflow.

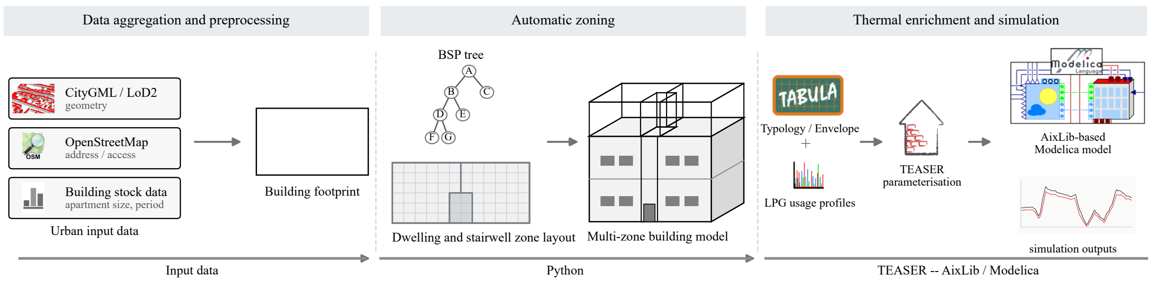


Figure 2.2: Schematic overview of the MZA workflow from limited urban input data to automatic zoning and thermal simulation.

Data aggregation and preprocessing

Data aggregation and preprocessing prepare the building information required before automatic zoning can be applied. In the original MZA workflow, geodata from OpenStreetMap (OSM) (OpenStreetMap contributors 2024) are combined with CityGML data to provide a more complete description of the buildings under investigation. CityGML provides the LoD2-type external building geometry used by the workflow, while OSM can supplement this information with attributes such as building type, roof type, and the address point of a building.

In addition to CityGML and OSM, the MZA workflow uses external building-stock information to populate attributes that are typically missing from urban geometry data. This additional dataset is based on building-property classifications derived from German census data, as reported by Blanco et al. (2023). It supports enriching building data with construction year typologies and related building-stock attributes. These enriched attributes are then used in the wider MZA workflow to support assumptions about construction type, envelope properties, average dwelling size, and the expected number of dwelling units.

A relevant part of this preprocessing is the rule-based preparation of the stairwell location. The MZA workflow estimates the stairwell location using available external information, such as the address point or street layout. It orients it orthogonally to the nearest possible external wall. This is important because the stairwell can represent an unheated shared zone that influences heat demand, and because its position defines an access boundary that later constrains the automatic generation of the dwelling zone. The stairwell dimensions are estimated from the floor area and with reference to stair standards. However, the authors note that future refinements should more explicitly account for the construction year and building type.

This part of the workflow is context-dependent because it relies on the availability and quality of external geodata, address information, regional building-stock assumptions, and assumptions about stairwell placement.

Automatic zoning

Based on the input information prepared in the preceding workflow layer, the MZA then applies the automatic zoning step to generate dwelling and stairwell zones. The MZA framework contains several zoning options, including manual zoning, perimeter–core zoning, graph-based clustering, and the heuristic automatic zoning algorithm. However, the MZA paper primarily focuses on an automated heuristic approach to generating dwelling zones in multi-family buildings.

In the heuristic approach, automatic zoning begins with the geometric discretisation of the building footprint area using Binary Space Partitioning (BSP), which recursively subdivides the base surface into smaller spatial elements up to a predefined granularity (Fuchs et al. 1980). As illustrated in Figure 2.3, the subdivision process can be represented as a tree structure, where each split creates new child nodes until the final partitions are reached. These final partitions, referred to as leaves, form the basic spatial units for subsequent zone assignment.

After the floor area has been discretised, additional spatial information is assigned to the BSP leaves. This includes leaf adjacency, contact with the external building boundary, contact with the stairwell or access boundary, and blocked edges that should not be crossed. These relations are required because the automatic zoning procedure does not generate independent free-form polygons. Instead, it generates candidate dwelling regions within the residential floor area and later assigns the discretised leaves to these regions. The stairwell itself is not treated as a dwelling seed. Its prepared position defines the access boundary from which the

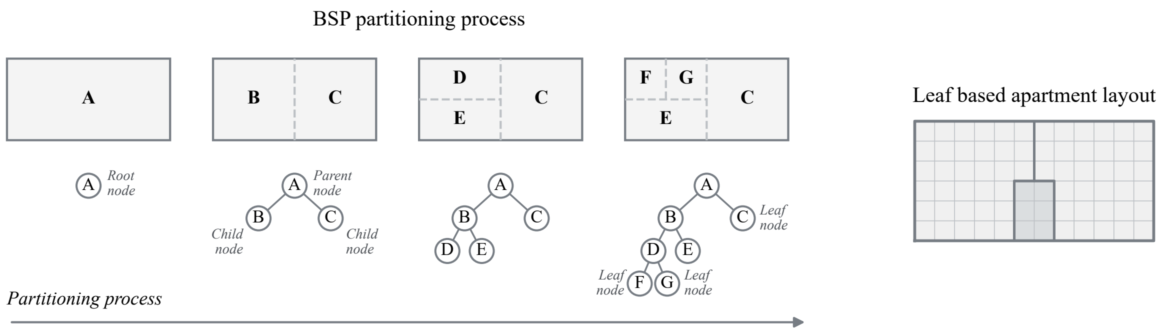


Figure 2.3: Conceptual illustration of the BSP partitioning process in the MZA workflow. The floor area is recursively subdivided into a tree structure, and the resulting leaf nodes form the spatial units for subsequent zone assignment.

residential sub-areas are generated.

The heuristic zoning logic is based on Voronoi seed placement (Okabe et al. 2000). For a required number of dwelling zones, seed points are placed along the boundary between the stairwell area and the remaining residential floor area. Each seed represents one required dwelling zone per floor. The number of required dwelling zones is taken from the preceding data aggregation and preprocessing step, where it is estimated from auxiliary information such as the average dwelling size or the expected number of households per building volume.

Several seed configurations are tested iteratively. Depending on the available access geometry, seeds are placed either along one access boundary or distributed across the two most relevant access boundaries. Initial configurations use regular spacing, while later iterations introduce random variation to explore alternative dwelling-zone layouts.

For each seed configuration, a Voronoi diagram is generated and clipped to the residential free region, excluding the stairwell and blocked edges. Each clipped cell represents a candidate dwelling region. Candidate solutions are accepted only if all cells satisfy the main spatial constraints, including sufficient contact with the access boundary and external facade, and if no empty or invalid cells are produced. Among the valid candidates, the algorithm selects the configuration that provides the best balance between compact zone geometry and similar dwelling areas.

After the best polygon-level candidate has been selected, the BSP leaves are assigned to the generated Voronoi cells based on their largest intersection area, with nearest-seed or representative-point assignment used as a fallback. The resulting leaf-based zoning is then post-processed to improve zone connectivity, remove local geometric irregularities, and preserve the main spatial constraints. These operations are only accepted if the main constraints remain satisfied. Overall, the method combines BSP discretisation, Voronoi-based candidate generation, objective-based selection, and local post-processing to generate connected dwelling-level thermal zones around the prepared stairwell geometry (Waiz et al. 2025).

Enrichment and simulation

After the automatic zoning step, the MZA workflow continues with the enrichment and simulation stage. In this stage, the generated multi-zone geometry is translated into a thermal simulation model. This step is carried out through TEASER (Remmen et al. 2018) and the Modelica library AixLib (Müller et al. 2016). TEASER is used to parameterise reduced-order building models in accordance with VDI 6007-1 (VDI 2015), while AixLib provides the Modelica-based simulation environment for the resulting thermal model.

This enrichment step assigns thermal and operational properties to the generated zones, including construction type, construction year, material assumptions, envelope parameters, set temperatures, and usage conditions. In the original MZA workflow, TABULA-based German building typology data support the assignment of construction-period classes, material parameters, and window areas when detailed window information is unavailable (Loga et al. 2016).

In these models, thermally conductive building components are represented using resistance–capacitance (RC) elements, which describe heat storage and heat transfer through building elements with reduced computational effort. This allows the model to describe heat storage and heat transfer through building elements with reduced computational effort. Since the MZA explicitly generates several neighbouring zones, interzonal heat transfer becomes relevant. The workflow therefore extends the reduced-order model with additional thermal connections between adjacent zones, enabling representation of heat exchange among dwellings, floors, and unheated stairwell zones (Groesdonk et al. 2023; Waiz et al. 2025).

Internal gains and usage profiles are also assigned in this simulation layer. In addition to standardised usage assumptions, the MZA workflow allows the use of LoadProfileGenerator (LPG) (Pflugradt et al. 2022) profiles to describe occupants, appliances, and lighting with more detailed, household-specific behaviour. These profiles can then be distributed across the generated dwelling zones, so that different dwellings may receive different occupancy and internal gain schedules.

2.4.2. Evaluation of the MZA workflow

Waiz et al. (2025) show that the MZA can generate consistent zoning outputs for several investigated building floor plans and can support more detailed heat load prediction than a single-zone model. However, this additional spatial detail increases computational effort, with the multi-zone model including heat transfer requiring up to 750% more simulation time than the single-zone model.

The authors identify further refinement of the zoning methodology, extensive validation using real buildings, and improved treatment of building-stock and construction-period assumptions as important directions for future work (Waiz et al. 2025). These findings indicate that the MZA should be evaluated at multiple levels. Geometric and topological validation is required to determine whether the generated dwelling and stairwell zones correspond to realistic internal building structures. However, geometric comparison alone cannot show whether observed zoning deviations are relevant for simulation outputs.

A generated zoning layout may differ from the reference floor plan structure. Still, the importance of this difference depends on whether it changes performance indicators such as heating demand, peak load, or overheating. Therefore, the following section first discusses floor plan data as a basis for constructing reference dwelling–stairwell geometries for geometric and topological validation. At the same time, the subsequent sensitivity analysis assesses the relevance of the identified zoning deviations and uncertain enrichment inputs to the simulation.

2.5. Reference floor plan data for geometric validation

To validate the zoning output of the Multizone Assignment Algorithm (MZA), a reference representation of the internal dwelling and stairwell structure is required. Floor plans provide an important intermediate source of spatial information. Although they are not thermal zone models themselves, they describe the internal organisation of buildings and can be abstracted into dwelling and stairwell reference zones.

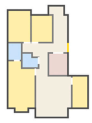
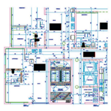
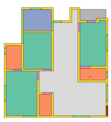
Pizarro et al. (2022) describe floor plans as two-dimensional documents that contain geometric, topological, and semantic information. These information types are relevant to MZA validation because the reference zoning must capture the zone's geometry, connectivity, and spatial meaning. However, floor plan data is often difficult to use directly. Kalervo et al. (2019) explain that floor plans are commonly rasterised for publication or real estate use, which removes structured geometric and semantic information.

Similarly, Pizarro et al. (2022) emphasise that floor plan analysis is challenging because drawings differ in notation, symbols, scale, and style. Public floor plan datasets also vary considerably in annotation type, access, size, and target task. In particular, Pizarro et al. (2023) note that many existing datasets focus on single-unit plans. In contrast, multi-unit plans introduce a different abstraction layer, as they include multiple dwellings, corridors, stairwells, and common areas.

Table 2.2 summarises selected floor plan datasets and their main application scopes. These datasets differ in their target tasks and levels of annotation, ranging from floor plan generation and image parsing to multi-unit plan recognition and simulation-oriented spatial reasoning (Wu et al. 2019; LIFULL Co., Ltd. and National Institute of Informatics 2017; Kalervo et al. 2019; Pizarro et al. 2023).

More recent datasets, such as the Modified Swiss Dwellings (MSD) dataset and ResPlan, provide richer structured representations than image-only floor plan collections. They include geometric, semantic, structural, or graph-based information, which makes them more suitable for tasks such as floor plan generation, spatial reasoning, simulation-oriented analysis, and 3D conversion (Engelenburg et al. 2024; Abouagour and Garyfallidis 2025).

Table 2.2: Selected floor plan datasets and their main representations and scopes, with example layouts adapted from the respective dataset publications.

Dataset	**Example**	**Representation**	**Scope**
RPLAN		Raster-based annotated residential layouts	Large-scale data-driven residential floor plan generation.
LIFULL		Real estate floor plan images and vectorised subsets	Floor plan generation, raster to vector conversion, and layout learning from real estate plan data.
CubiCasa5K		Raster images with SVG polygon annotations	Floor plan image parsing, object recognition, and semantic segmentation.
MLSTRUCT–FP		High-resolution raster floor plans with wall and slab annotations	Multi-unit floor plan recognition and wall vectorisation.
MSD		Image, vector, semantic, structural, and graph-based representations	Floor plan generation of building complexes, including multi-apartment residential layouts.
ResPlan		Vector graph residential floor plans	Structured residential floor plan learning, spatial reasoning, simulation-oriented applications, and 3D conversion.

Overall, the literature shows that floor plans contain the geometric, topological, and semantic information needed to reconstruct the internal organisation of residential buildings (Pizarro et al. 2022). However, this information is often not directly available in a structured form, because floor plans are commonly distributed as heterogeneous documents (Kalervo et al. 2019; Pizarro et al. 2023).

This creates a methodological gap for MZA validation. The MZA output must be compared with a reference representation at the same abstraction level, namely, dwelling and stairwell zones, including their geometry, location, and spatial relationships. Since existing floor plans and datasets do not directly provide this reference zoning representation, a dedicated validation dataset has to be constructed by interpreting, filtering, and abstracting available floor plan sources.

2.6. Simulation uncertainty and sensitivity analysis

Building energy simulation models rely on many assumptions about geometry, construction, operation, occupancy, and system behaviour. Since these inputs are often incomplete, estimated, or uncertain, simulation outputs should not be interpreted solely as deterministic results from a single fixed model configuration. Sensitivity analysis provides a way to examine how variations in input assumptions influence model outputs and to identify which parameters are most important for reliable performance assessment. This section discusses zoning variants as a modelling uncertainty, then introduces input uncertainty in building energy simulation, and finally reviews sensitivity analysis methods for identifying influential parameters.

2.6.1. Zoning variants as a modelling uncertainty

As discussed in Section 2.2, thermal zoning affects how heat transfer, control assumptions, and spatial temperature differences are represented in building energy simulation. Previous studies show that zoning assumptions and interzonal heat exchange can influence predicted energy use, thermal comfort, and model accuracy, especially in multi-family residential buildings where flats and shared spaces interact thermally (O'Brien et al. 2011).

However, the MZA-based workflow introduces a more specific type of zoning uncertainty. In this case, the zoning structure is not manually defined or selected only as a general level of detail; it is generated automatically from limited geometric and auxiliary input, and may therefore deviate from reference dwelling and stairwell structures. Such deviations can include under-zoning, over-zoning, alternative dwelling–stairwell configurations, displaced stairwell locations, and different representations of stairwell spaces. Geometric validation alone cannot determine whether such deviations meaningfully affect simulation outputs. Therefore, simulation-based analysis is needed to examine whether MZA-related zoning variants lead to meaningful differences in outputs such as annual heating demand, peak heating demand, and overheating hours under uncertain input conditions.

2.6.2. Input uncertainty in Urban building energy simulation

Building energy simulation depends strongly on the quality and completeness of input data. This is especially relevant in urban building energy modelling, where building-specific information is often incomplete and must be estimated through typological or statistical enrichment. Malhotra et al. (2022) show that attributes such as year of construction, building usage, and refurbishment state are important enrichment inputs for CityGML-based heating demand modelling.

Such enrichment introduces input uncertainty because the assigned values may not correspond exactly to the real building. Prataviera et al. (2022) show that input uncertainty can affect urban building energy simulation results and argue that uncertainty and sensitivity analysis are needed to identify the assumptions that most strongly influence simulated energy demand. In the MZA-based workflow, this is relevant because the generated zoning structure is combined with uncertain enrichment inputs such as construction age, retrofit state, envelope properties, ventilation assumptions, setpoints, internal gains, and occupancy profiles.

Sensitivity analysis provides a systematic way to evaluate this problem. Rather than interpreting one fixed simulation configuration as a deterministic result, it examines how variations in uncertain inputs affect the selected outputs. In this thesis, sensitivity analysis is therefore used to evaluate both MZA-related zoning deviations and uncertain enrichment inputs with respect to annual heating demand, peak heating demand, and overheating hours.

2.6.3. Sensitivity analysis methods

Several sensitivity analysis methods are available for building energy applications. Tian (2013) distinguishes between local and global sensitivity analysis methods. Local methods are simple and computationally efficient, but they only explore variations around a base case and do not capture interactions between inputs. Global methods explore a wider input space and are therefore more suitable when several uncertain parameters may interact.

Tian (2013) describes sensitivity analysis in building performance analysis as a sequential workflow. The process starts by defining input variations or probability distributions, followed by the creation and execution of building energy models. The resulting simulation outputs are then collected and analysed using suitable sensitivity analysis methods, before the sensitivity indices are interpreted and presented. Figure 2.4 summarises this general workflow.

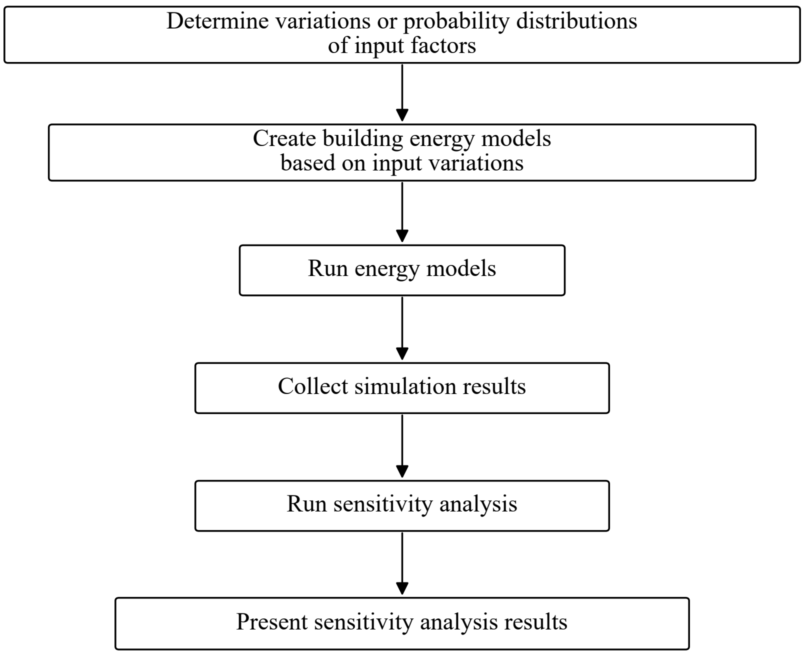


Figure 2.4: Typical workflow for sensitivity analysis in building performance analysis. Adapted from Tian (2013).

2.7. Research gap and framework

Despite these developments, some gaps remain. First, the generated zoning geometry of the MZA has not yet been systematically validated against independent reference dwelling–stairwell structures. Since the algorithm infers internal dwelling and stairwell zones from limited external geometry and auxiliary input data, its spatial output must be compared with reference ground-truth zoning at the same abstraction level.

Second, it remains unclear whether MZA-related zoning deviations are relevant for simulation outputs. Such deviations can arise when limited input data lead to an incorrect number of zones, alternative dwelling–stairwell arrangements, shifted stairwell locations, or changed stairwell representation. Existing studies show that zoning granularity and interzonal heat transfer can influence building performance simulation. Still, they do not directly connect these automated zoning deviations to performance indicators such as heating demand, peak load, or overheating.

Third, the influence of uncertain thermal enrichment inputs in an MZA-based workflow remains unclear. Although LoD2 data provide external building geometry, several simulation inputs still have to be estimated, including construction age, retrofit state, envelope and window properties, ventilation, setpoints, internal gains, and occupancy profiles. These assumptions may strongly affect the simulation results and therefore need to be evaluated through sensitivity analysis.

This thesis addresses these gaps by combining geometric validation with simulation-based sensitivity analysis. The validation evaluates whether the MZA can reproduce reference dwelling–stairwell structures, while the simulation analysis examines whether MZA-related zoning deviations and uncertain enrichment inputs influence selected simulation outputs. The thesis does not validate the complete thermal simulation workflow against measured energy performance, but focuses on MZA-generated zoning structures and their relevance for selected simulation outputs.

The central research question of this thesis is:

> To what extent can the Multizone Assignment Algorithm generate simulation-relevant thermal zoning structures for multi-family residential buildings from limited geometric input, and which missing or uncertain building information most strongly influences the resulting building energy simulation outputs?

This central question is addressed through three connected research questions:

1. **RQ1:** To what extent can the MZA reproduce dwelling and stairwell-based reference zoning structures in multi-family residential buildings using limited geometric input?
2. **RQ2:** How do MZA-related zoning deviations identified through geometric validation, such as reduced or increased zoning granularity, alternative dwelling–stairwell configurations, and stairwell representation, influence selected building performance indicators under uncertain input conditions?

3. **RQ3:** Which missing or uncertain input information required by the MZA-based simulation workflow has the strongest influence on simulation outputs?

Based on the literature review, the thesis hypothesises that the MZA can reproduce the dominant dwelling- and stairwell-level zoning structure of multi-family residential buildings with sufficient geometric and structural topological agreement. It is further expected that MZA-related zoning deviations are more relevant for peak and comfort-related outputs than for annual heating demand. In addition, missing or uncertain simulation-relevant enrichment inputs required by the MZA-based workflow are expected to strongly influence the selected simulation outputs.

3. Research Methodology

3.1. Validation

In the standard MZA workflow, the external building geometry is available, while the internal dwelling and stairwell structure is unknown. Therefore, an independent reference zoning is required to assess the spatial quality of the generated zones.

The broader target context of this validation is multi-family residential building modelling in the DACH region, where automated thermal zoning methods are relevant because detailed internal layouts are often unavailable at urban scale. This regional focus is appropriate because the MZA workflow is developed for Central European urban building data and uses enrichment assumptions such as German building-stock information and TABULA-based typologies (Loga et al. 2016).

The validation focuses on the geometric and topological reconstruction capability of the automatic zoning process in the MZA workflow. It evaluates whether the algorithm can reproduce the number, geometry, location, and adjacency of dwelling and stairwell zones from reduced geometric input. It does not validate the auxiliary input assumptions, the thermal enrichment procedure, or the subsequent dynamic simulation stage. Auxiliary input assumptions, such as the estimated average dwelling size, are not validated as independent statistical models; however, their influence is considered when interpreting zoning deviations. For example, under-zoning or over-zoning may indicate a limitation of the zone count estimation step rather than a direct failure of the spatial allocation algorithm.

For this thesis, the reference ground-truth is defined at the dwelling–stairwell level. Each dwelling is treated as an individual thermal zone, and each stairwell as a separate one. This abstraction corresponds to the intended output level of the MZA and captures the dominant internal organisation of multi-family residential buildings without requiring room-level detail (Biljecki et al. 2016).

Based on the methodological gap identified in Section 2.5, the validation uses floor plans as source material for constructing reference zoning data. The floor plans are not treated as ground-truth directly. Instead, they are converted into a common reference representation that contains the geometry, location, and spatial relationships of dwelling and stairwell zones. The following sections describe the selected data sources and the methodology to derive the reference zoning geometries.

3.1.1. Data sources for reference zoning

To construct the reference zoning used for validation, this thesis uses two data sources: the Modified Swiss Dwellings (MSD) dataset (Engelenburg et al. 2024) and real estate floor plan documents (ImmoScout24 2026). As discussed in Section 2.5, several possible floor plan sources were considered, including sources containing single-family residential layouts. For the validation, however, only cases matching the defined target population were retained, namely multi-family residential buildings in Europe with identifiable dwelling units and stairwell spaces.

Due to the limited availability of openly accessible European floor plan datasets with sufficient information for dwelling and stairwell reference zoning, this study uses selected Swiss and German floor plan data. The MSD dataset is used as the main reference source because it provides a larger and more structured set of Swiss residential floor plans with geometric and topological information. German real estate floor plans are used as a smaller, complementary source because they provide additional examples of real buildings from the German context. Together, the two sources provide validation from different perspectives: MSD enables reproducible, graph-based extraction of reference zoning from structured Swiss residential floor plan data, whereas the German real estate floor plans provide additional validation cases derived from real building documents.

The resulting sample is interpreted as a purposive Swiss–German validation sample within the broader DACH residential building context, rather than as a statistically representative sample of all DACH residential building typologies.

3.1.2. Modified Swiss Dwellings dataset

The first reference source used in this study is the Modified Swiss Dwellings (MSD) dataset. Engelenburg et al. (2024) introduced MSD as a benchmark dataset for floor plan generation of multi-family residential building complexes. MSD focuses on realistic medium to large-scale residential floor plans and explicitly includes both single and multi-family residential buildings, thereby providing a substantially richer basis for understanding and studying the spatial relations in multi-family residential buildings.

MSD is derived from the earlier Swiss Dwelling (SD) dataset (Standfest et al. 2022) and was produced through a structured cleaning and curation process, which includes the exclusion of non-floor plan geometries, floor plans with non-residential-like details, near duplicate cases originating from the same plan ID, and the filtering of very small layouts by retaining only medium to large-scale plans. After additional cleaning steps, the final curated dataset comprised 5,372 floor plans. These filtering steps are important for the present work because they indicate that the dataset has already been restricted to residentially meaningful and spatially complex cases, which makes it more suitable as a source for ground-truth abstraction than a raw unfiltered plan archive (Engelenburg et al. 2024).

A major strength of the dataset is that it does not store floor plans as images alone. As illustrated in Figure 3.1, MSD provides several complementary representations of the same building layout, including polygonal room geometries, semantic room type labels, zone type labels, structural information, and access graph representations.

The graph-based representation is particularly relevant for this thesis because it enables dwelling boundaries and shared circulation areas to be inferred from connectivity information rather than from image interpretation alone.

A limitation of the MSD dataset is that it does not provide metadata such as address information or year of construction. In the standard MZA workflow, this metadata can be used as auxiliary

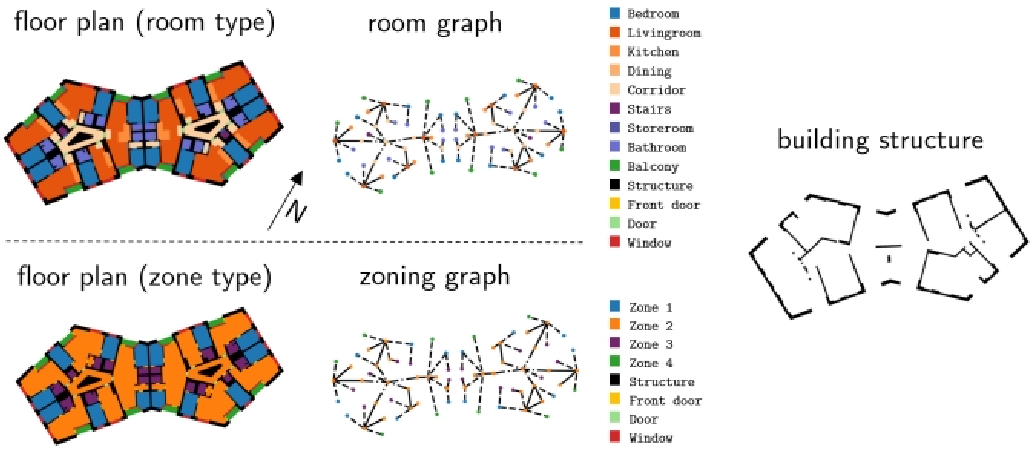


Figure 3.1: Data representations provided in the MSD dataset. Adapted from Engelenburg et al. (2024).

input for building-specific enrichment, such as estimating the average dwelling size or the expected number of dwellings from external datasets. However, since the validation in this thesis focuses on the geometric and topological reconstruction of dwelling and stairwell zones, the absence of this metadata does not affect the extraction of the reference ground-truth zoning from MSD. Instead, it limits only the reproduction of the building-specific enrichment step based on external datasets, while the zoning validation remains primarily driven by the geometric properties of the building footprint and the derived reference dwelling–stairwell structure.

Since MSD lacks the required enrichment input data, the estimated average dwelling size is derived from floor plan characteristics during MZA input preparation. This adaptation is used only to provide the auxiliary information required by the MZA setup and is not part of the reference ground-truth extraction. Thus, the missing metadata is treated as a dataset limitation to be addressed through preprocessing, while validation remains focused on comparing MZA-generated zones with the extracted reference thermal zones. The detailed preprocessing procedure is described in Section 3.1.5.

Ground-truth extraction

In the provided MSD dataset, the graph representations are stored as serialised objects that encode the room graph, node attributes, and edge connectivity.

In this graph, nodes correspond to spatial areas (rooms), and edges represent spatial connectivity between them. Each node contains a set of attributes, including a polygonal geometry representing the room shape, a semantic room-type label (e.g., bedroom, kitchen, corridor), and a centroid location. Edges encode adjacency relationships and are further characterised by connectivity types that describe how two spaces are linked. In particular, the dataset distinguishes between different connection types, such as direct passage, door connections, and dwelling entrance connections (front doors). The latter represent transitions between shared circulation spaces and private dwelling interiors and therefore play a key role in identifying dwelling boundaries.

The generation of the reference ground-truth from MSD is illustrated in Figure 3.2. The workflow begins from the original room level graph representation of a selected floor plan, shown in Figure 3.2(a). Here, the building was described as a collection of individual rooms, each with its own semantic category and adjacency relation.

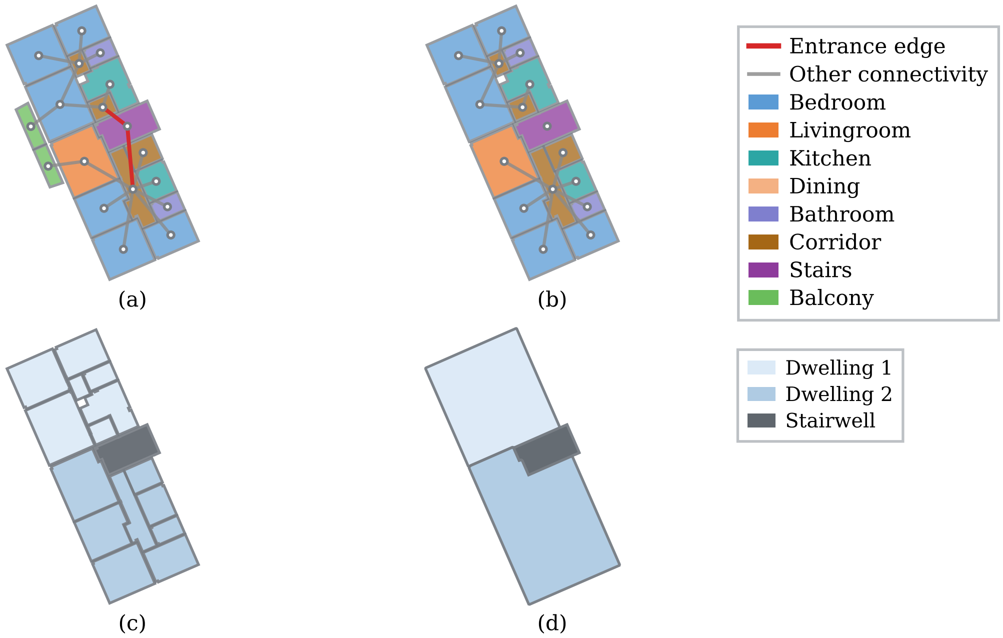


Figure 3.2: Graph-based extraction of reference ground-truth of thermal zones from the MSD dataset. (a) Room-level representation. (b) Preprocessed graph after auxiliary space and entrance edge removal. (c) Classification of dwellings and stairwells. (d) Merged dwelling and stairwell geometries (reference ground-truth of thermal zones).

Since the objective of the study was to derive thermally relevant residential units, two preprocessing steps were applied, as shown in Figure 3.2(b). First, auxiliary spaces such as balconies were removed from the graph because they do not contribute to the definition of thermally relevant interior zones. Second, to separate dwelling units from shared spaces, all edges corresponding to dwelling entrance connections are removed from the graph. In the dataset, these edges are explicitly labelled as entrance edges. Dwelling units are then identified based on a set of private room types, including bedrooms, living rooms, kitchens, dining rooms, and bathrooms.

After the entrance connections are removed, the graph is split into connected components, and each component represents a group of rooms that remain mutually reachable through the remaining graph edges without passing through an dwelling entrance. A component is classified as a dwelling unit if it contains at least one node belonging to a private residential room type. This classification is illustrated in Figure 3.2(c). This criterion ensures that only components corresponding to actual residential units are identified as dwellings. All remaining nodes that are not assigned to any dwelling component are classified as belonging to the stairwell. In this way, the stairwell is defined topologically as the set of spaces located before the dwelling entrances in the connectivity graph.

Following this topological classification, the room geometries are aggregated to form thermal zone geometries. To ensure spatial consistency, buffering operations are applied during the merging process to close small gaps between adjacent polygons and obtain spatially consistent zone outlines. This geometric regularisation is consistent with the chosen level of ground-truth abstraction, in which the objective is not to reconstruct detailed internal wall thicknesses, but to derive thermally relevant zones for validation. Similar simplifications are used in automatic thermal zoning approaches for buildings with unknown or limited definitions of interior space (Dogan et al. 2016). The resulting merged reference geometries are shown in Figure 3.2(d).

For the present study, the full MSD dataset was filtered to retain only buildings that followed the defined validation scope. Specifically, only buildings with between two and six dwellings and between one and two stairwells were retained, resulting in an initial subset of 406 cases, which were then processed for ground-truth generation. Manual visual inspection was then used as a quality control step to refine this subset. During this step, duplicate or near-duplicate layouts were removed, along with cases showing ambiguous dwelling–stairwell separation, unreliable graph-derived zoning, unsuitable footprint geometry, or centrally placed stairwells incompatible with the current MZA stairwell placement assumptions. After this refinement, 200 floor plans were selected from the remaining eligible cases to form the final validation dataset. The sample was capped at this size to keep the validation workflow manageable within the scope of the thesis. These MSD cases were then used as the reference dataset for validating the MZA workflow. A complete overview of the selected layouts is provided in Appendix A.1.

3.1.3. Real estate floor plans

The second reference source used in this study comprises architectural floor plans collected from publicly available German real estate listings on ImmoScout24 (2026) and from other private sources. These sources mainly provide floor plan documents in PDF or image-based formats, typically as part of property advertisements or building documentation. Unlike the MSD dataset, these floor plans are not provided as structured or machine-readable graph data. Instead, they are available as drawing-based documents that visually describe the internal arrangement of dwellings, stairwells, and shared spaces within residential buildings and therefore must be interpreted and extracted before they can be used for validation. The variability of these source documents is illustrated in Figure 3.3, which shows examples of the different real estate floor plan formats used in this study.

As shown in Figure 3.3, the real estate floor plan data are heterogeneous in terms of graphical representation, drawing quality, level of detail, and available metadata. The examples include dimensioned architectural drawings, scanned plans, marketing style layouts, and annotated PDF floor plans. Despite these differences, the floor plans contain architectural information, and in many cases the documents include scale information, room names, area values, total floor area, and the year of construction. The scale and area information can support converting the extracted geometries into actual physical dimensions. The year of construction is not used directly to construct the polygonal ground-truth geometry. Still, it is relevant to the MZA

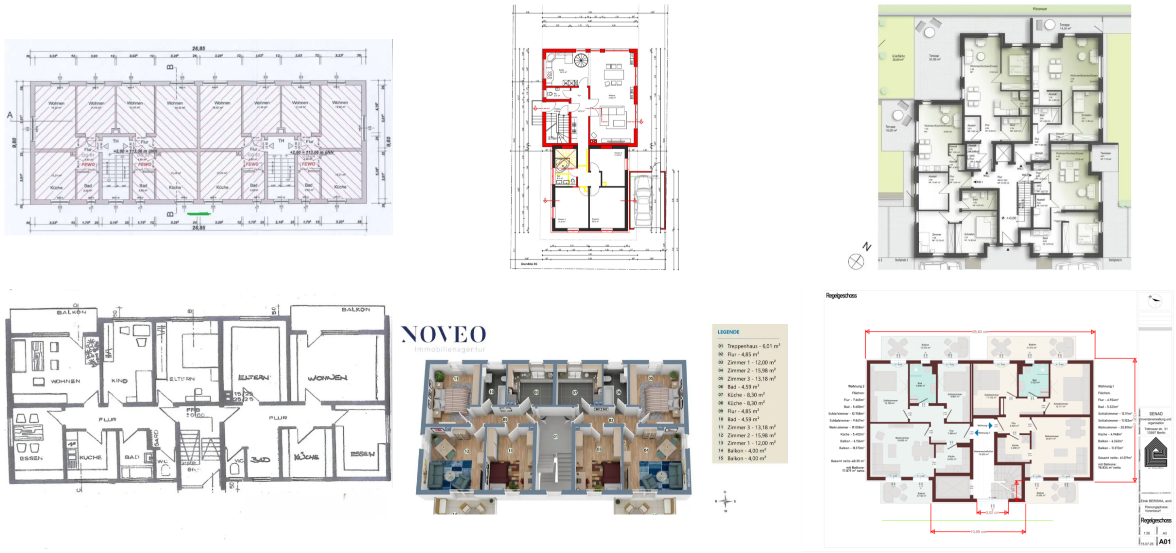


Figure 3.3: Examples of real estate floor plan documents used as reference sources.

setup because it helps assign age-related assumptions to buildings for the zoning workflow. The collected floor plan documents were retained together with basic case metadata to ensure traceability during the subsequent digitisation and validation workflow.

Because these collected documents vary in drawing quality, scale information, and available metadata, only clearly interpretable multi-family residential floor plans are selected for the validation dataset. Floor plans with unclear dwelling boundaries, insufficient drawing quality, missing or ambiguous scale information, or layouts that could not be reliably interpreted were excluded. In this way, the real estate floor plans provide German reference cases within the broader European multi-family residential context considered in this thesis.

Although the real estate plans do not contain explicit connectivity graphs or semantic labels, they provide sufficient architectural information to identify the dwelling and stairwell structure of multi-family residential buildings. Therefore, the real estate floor plans provide a basis for constructing a reference ground-truth of thermal zones to validate the MZA output.

Ground-truth extraction

Unlike the MSD branch, where the reference ground-truth is derived from an existing graph representation, the real estate branch begins with PDF floor plans. The generation of the reference ground-truth from real estate floor plans is illustrated in Figure 3.4. Since the source documents are available as drawing-based floor plans, the workflow begins by converting the selected PDF page into an image representation, as shown in Figure 3.4(a). The selected PDF page is rasterised at 300 dpi. This resolution was chosen as a practical minimum for document digitisation, following archival digitisation guidance that recommends 300 ppi for ordinary paper/textual documents, and it provides sufficient visual detail for manual tracing of dwelling and stairwell boundaries (National Archives and Records Administration 2023). The rasterised image is then displayed in an interactive Matplotlib (Hunter 2007) interface that serves as a manual digitisation environment.

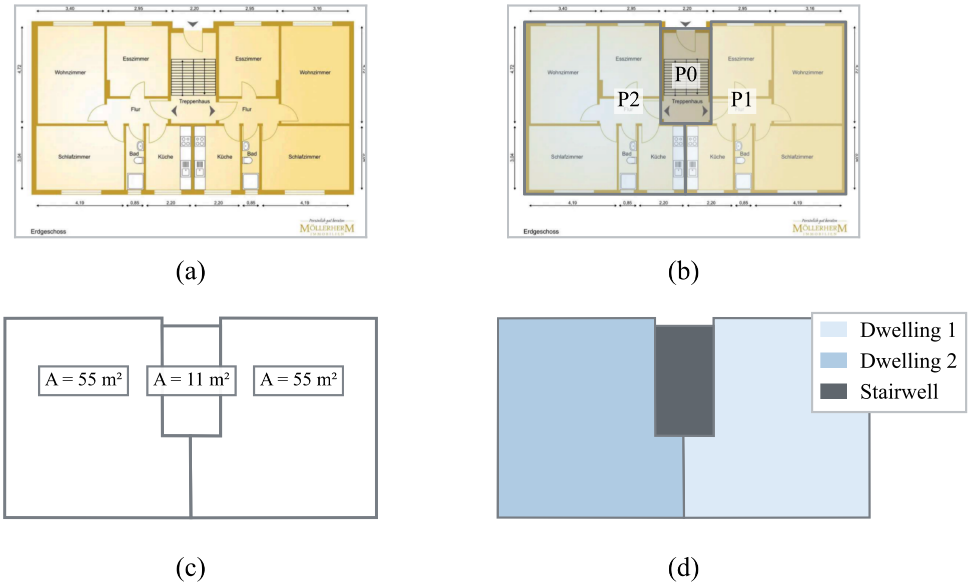


Figure 3.4: Extraction of reference ground-truth from real estate floor plans. (a) Real estate PDF floor plan in the interactive interface. (b) Manual digitisation of dwelling and stairwell polygons. (c) Scaling of polygons to physical metric coordinates. (d) Classification into dwellings and stairwells (reference ground-truth of thermal zones).

Within this interface, the user traces each dwelling or stairwell region by selecting successive vertices along the visible boundary of the floor plan, as shown in Figure 3.4(b). After digitisation, the extracted polygons are still represented in image coordinates and therefore require conversion into physical metric coordinates before they can be used as reference ground-truth geometries. For this, a pixel-to-metre conversion factor is required. Depending on the information available in the source document, three scaling strategies are supported, as summarised in Table 3.1.

Table 3.1: Scaling methods used to convert digitised real estate floor plan polygons from image coordinates to metric coordinates

Scaling method	**Available information**	**Pixel-to-metre conversion factor**
Known scale method	A drawing scale is explicitly provided, such as 1:50 or 1:100. The rasterisation resolution D is known in dpi, and n denotes the scale denominator.	$s_{\text{scale}} = \frac{25.4}{D} \cdot \frac{n}{1000}$
Reference line method	A dimension line or reference element of known real-world length is visible in the floor plan. The user selects the two endpoints of this element in the image.	$s_{\text{ref}} = \frac{L_{\text{real}}}{d_{\text{px}}}$
Area-based method	The total floor area is available from the real estate document. The union area of the digitised polygons is calculated in pixel units.	$s_{\text{area}} = \sqrt{\frac{A_{\text{real}}}{A_{\text{px}}}}$

Here, L_{real} is the known real-world reference length, d_{px} is the measured image length in pixels, A_{real} is the known real-world area, and A_{px} is the corresponding digitised polygon area in pixel units. The three methods differ only in how the conversion factor s is obtained.

After selecting the appropriate scaling method, the resulting pixel-to-metre conversion factor is denoted as s. Each polygon vertex is converted from image coordinates to metric coordinates as:

$$(x_{\mathrm{m},i}, y_{\mathrm{m},i}) = s(x_{\mathrm{px},i}, y_{\mathrm{px},i}) \tag{3.1}$$

where $(x_{\mathrm{px},i}, y_{\mathrm{px},i})$ are the image coordinates and $(x_{\mathrm{m},i}, y_{\mathrm{m},i})$ are the corresponding metric coordinates. The same conversion factor is applied to all digitised polygons, preserving their relative geometry and adjacency. The resulting scaled geometries are shown in Figure 3.4(c).

After scaling, the area of each polygon is calculated from its metric vertex coordinates and, where possible, compared with the area information in the original real estate document. This provides a consistency check for the digitised and scaled geometries. The drawing-based extraction uses the same level of abstraction as the validation, and the resulting reference geometries are shown in Figure 3.4(d).

For the present study, 40 real estate floor plans were selected in accordance with the validation scope defined at the beginning of this chapter. The selection focused on floor plans that provided sufficient information for reliable extraction of dwelling and stairwell data, including readable dwelling boundaries, identifiable stair or circulation stairwell spaces, and usable scale, dimension, or area information. Floor plans with unclear dwelling boundaries, ambiguous stairwell spaces, insufficient drawing quality, missing or unreliable scale information, or layouts that could not be interpreted consistently were not included.

The collected real estate floor plans were then digitised, scaled, and classified into dwelling and stairwell geometries. These cases were used as reference floor plans for validating the MZA workflow. A complete overview of the selected real estate layouts is provided in Appendix A.2.

3.1.4. Extracted ground-truth

Although the MSD and real estate floor plan used different extraction procedures, both were converted into a common reference ground-truth zoning representation. The output of both branches consisted of:

- the number of dwelling units,
- the geometry of each dwelling unit,
- the number of stairwells,
- the geometry of each stairwell, and
- the relative spatial arrangement of the dwelling and stairwell geometries within the same floor plan coordinate system.

This common representation preserved not only the individual zone geometries but also their positions, adjacencies, and overall arrangement within the building footprint. It therefore made it possible to apply the subsequent MZA input preparation workflow consistently to both datasets and later compare the MZA-generated zones with the corresponding reference geometries.

3.1.5. Preparation of input data for MZA

Following the extraction of the reference ground-truth thermal zones, a separate input dataset was prepared for the Multizone Assignment Algorithm (MZA). This step corresponds to the data aggregation and preprocessing stage of the original MZA workflow discussed in Section 2.4.1, in which limited external building information is transformed into the geometric and auxiliary inputs required for automatic zoning. In the original workflow, this preprocessing is based on sources such as CityGML, OSM, address information, and regional building stock assumptions. In the present validation setting, however, these external data sources were unavailable for the floor plan cases. Therefore, an equivalent prepared input was generated from the extracted geometry of the reference floor plan.

The input preparation converted the extracted dwelling and stairwell thermal zone polygons into a simplified external footprint, cleaned and oriented the footprint to a consistent stairwell/access-side reference, and inserted the footprint into an existing CityGML building model by replacing the original footprint. In addition, supporting input assumptions required by the MZA, such as average dwelling size, were prepared where necessary.

Generation of simplified footprint

For each validation case, the stored ground-truth thermal zones first consisted of separate dwelling and stairwell polygons, as shown in Figure 3.5(a). These polygons were merged into a single footprint that dissolved shared boundaries between adjacent zones and produced one external floor plan outline, as shown in Figure 3.5(b). This simplified footprint therefore represented the limited geometric input provided to the MZA.

These polygons were merged into a single footprint using a geometric union operation that dissolved shared boundaries between adjacent zones and produced one external floor plan outline, as shown in Figure 3.5(b).

After the simplified footprint was generated, its orientation was standardised, as the MZA uses not only the footprint shape but also the access- or street-facing side to infer the internal stairwell location. Since the footprints derived from floor plan data were stored in local coordinate systems without address metadata, the extracted stairwell geometry was used as a preprocessing reference to identify the likely access-facing facade, rotate it parallel to the x-axis, and position the stairwell side consistently at the lower side of the oriented footprint, as shown in Figure 3.5(c)–(d).

To create an MZA-compatible input model, the oriented footprint was inserted into an existing CityGML building object serving as the template. This was necessary because the MZA workflow expects a CityGML-based input model with georeferenced building context, including position, address-related information, and vertical structure. The template therefore provided the spatial and vertical building context, while its original footprint was replaced with the footprint derived from the floor plan data, as shown in Figure 3.6.

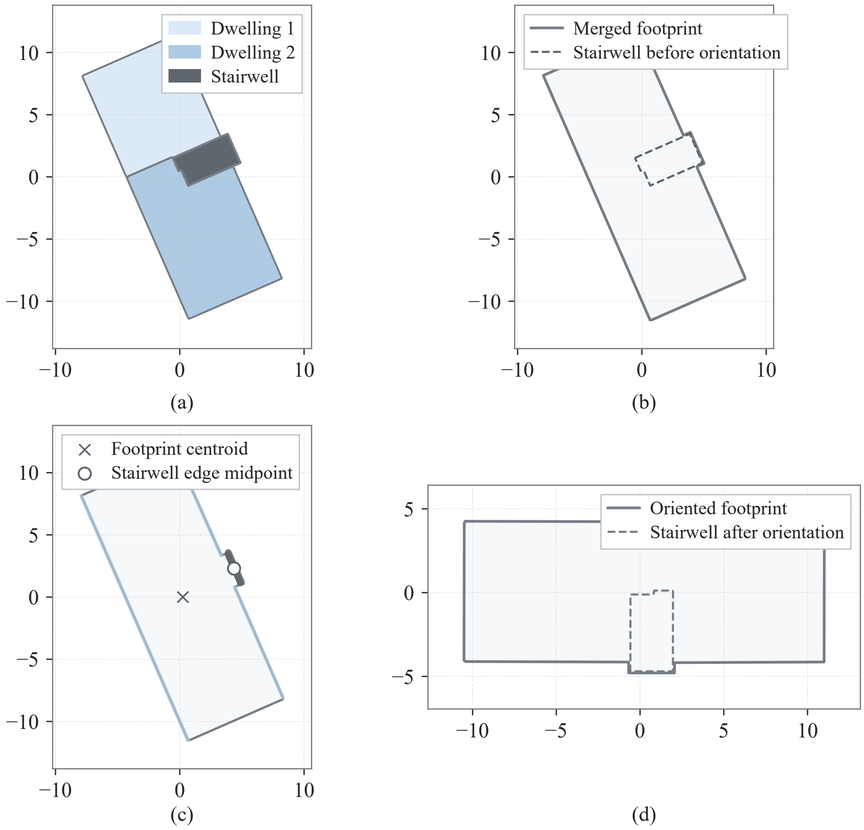


Figure 3.5: Orientation standardisation workflow for an example building, showing the identification of the stairwell side facade and the rotation of the merged footprint for MZA input preparation.

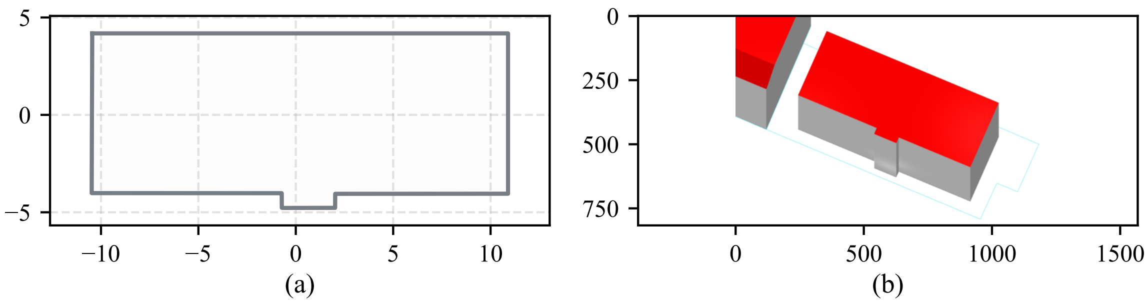


Figure 3.6: CityGML footprint replacement workflow. (a) Oriented external footprint exported as a GeoJSON intermediate file. (b) CityGML/LoD2 building after replacing the original footprint with the footprint derived from floor plan data while retaining the template building's vertical structure.

Estimation of missing auxiliary input

In addition to the external footprint geometry, the MZA workflow requires auxiliary information to constrain the expected dwelling subdivision of the building. Depending on the available input configuration, this information can be provided either as an estimated average dwelling area or as an expected number of dwellings. In the standard MZA workflow, such information can be derived from address-related and building stock attributes, for example through construction period-based assumptions. Therefore, after the geometric replacement step, a separate preparation of the average dwelling size input was required for the MZA runs.

For the real estate floor plan cases, the year of construction was available from the source documents or listing metadata and was therefore retained as part of the MZA input. Because the MSD dataset lacks construction year, address, and regional building stock metadata, the average dwelling area input had to be estimated separately. To avoid introducing reference ground-truth information into the automated zoning process, the known dwelling counts and measured dwelling areas were not used directly as inputs to the MZA.

Based on the MSD processing workflow described in Section 3.1.2, the filtered and subsetted MSD cases were converted into a building-level estimation dataset containing dwelling areas, stairwell areas, and external footprint areas. This dataset contained 400 buildings which were not used for the validation. The external footprint area A_{fp} was used as the predictor, while the mean dwelling area was used as the target variable. The mean was selected because the resulting dwelling size estimate is later used by the MZA workflow to derive the expected number of dwellings from the available floor area. It is therefore consistent with the area balance logic of the MZA input preparation workflow.

Several candidate models were compared to estimate the mean dwelling area from the external footprint area. These included a constant baseline model using one average value for all buildings, a simple linear regression model, and quantile-based piecewise linear regression models with different numbers of footprint-area regimes. The models were evaluated using a 70/30 split of the estimation dataset. The mean absolute error (MAE) was used as the primary selection metric.

The comparison showed that the quantile-based piecewise linear models performed better than the constant baseline and the simple linear regression. Among the tested models, the four-regime piecewise linear model produced the lowest MAE and was therefore selected for the final implementation. The detailed model comparison results are provided in Appendix A.3.

The selected model divides the footprint-area range into four regimes using quantile-based breakpoints. Within each regime, a separate linear relationship is fitted between the external footprint area and the mean dwelling area. This allows the estimation model to account for different relationships between footprint size and representative dwelling size in small, medium, and large floor plan cases.

The selected model was then refitted on the full 400 buildings dataset in order to obtain the final regression parameters used for the MZA input preparation workflow. The piecewise linear model is expressed as:

$$\hat{A}_{\mathrm{apt},r} = \alpha_r + \beta_r A_{\mathrm{fp}} \quad \text{for} \quad A_{\mathrm{fp}} \in I_r \tag{3.2}$$

where $\hat{A}_{\mathrm{apt},r}$ is the estimated average dwelling area in regime r, A_{fp} is the external footprint area, I_r is the footprint-area interval assigned to regime r, and α_r and β_r are the fitted intercept and slope for that regime. The footprint-area intervals were defined using quantile-based breakpoints, so that the regimes contained approximately balanced numbers of buildings. Piecewise regression was used because it allows different footprint-area ranges to have different linear relationships between external footprint area and average dwelling area.

Figure 3.7 shows the final refitted model together with the MSD derived data points and the regime breakpoints. The scatter in the data indicates that external footprint area alone explains only part of the variation in mean dwelling area. This is expected, because the same external footprint area can correspond to different dwelling counts, dwelling size distributions, and stairwell area proportions. The model is therefore interpreted as a pragmatic auxiliary input estimation step rather than as a complete explanatory model of dwelling size.

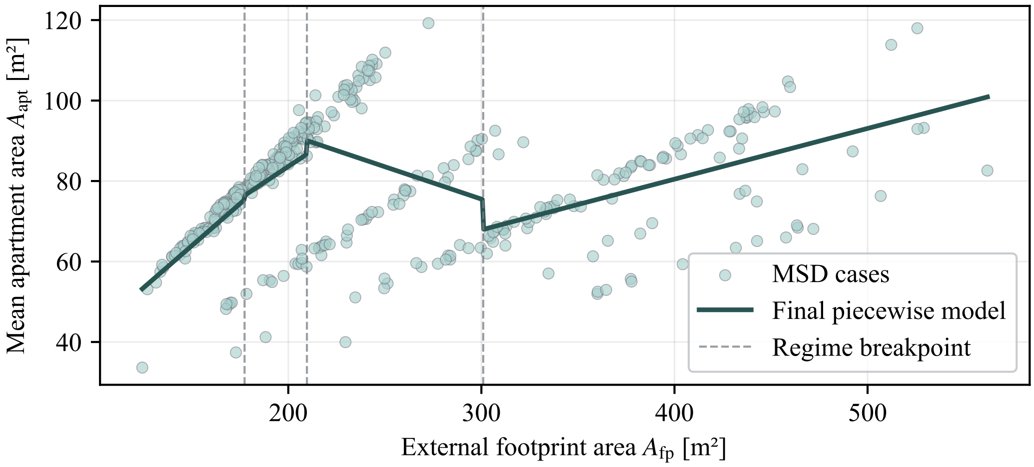


Figure 3.7: Final quantile-based piecewise linear model used to estimate the missing average dwelling area input for the MSD cases. The vertical dashed lines indicate the footprint-area breakpoints defining the four regimes.

The final refitted four-regime model is defined in Equation 3.3, including the footprint-area interval, fitted intercept, and slope for each regime.

$$\hat{A}_{\mathrm{dw}} = \begin{cases} 1.2430 + 0.417820A_{\mathrm{fp}}, & A_{\mathrm{fp}} \leq 177.3575, \\ 20.0725 + 0.317681A_{\mathrm{fp}}, & 177.3575 < A_{\mathrm{fp}} \leq 209.6500, \\ 123.7882 - 0.160915A_{\mathrm{fp}}, & 209.6500 < A_{\mathrm{fp}} \leq 300.9475, \\ 30.0455 + 0.126055A_{\mathrm{fp}}, & A_{\mathrm{fp}} > 300.9475. \end{cases} \tag{3.3}$$

The dataset-specific MZA inputs were then prepared and applied. For the real estate cases, the oriented CityGML file and the available year of construction were provided to the MZA workflow, whereas for the MSD cases, the oriented CityGML file was used together with the integrated piecewise model to estimate the missing mean dwelling-area input from the external footprint area.

3.1.6. Comparison and validation metrics

For each validation case, the ground-truth layout was defined as the set of ground-truth dwelling and stairwell polygons, denoted as G_i, where $i = 1, \ldots, N_{\mathrm{GT}}$. The MZA output consisted of a set of predicted thermal zone polygons, denoted as P_j, where $j = 1, \ldots, N_{\mathrm{pred}}$. Here, N_{GT} is the number of ground-truth zones and N_{pred} is the number of predicted zones. Since the MZA output uses generic zone identifiers rather than the same semantic labels as the ground-truth data, a geometric matching procedure was required before zone-level errors could be calculated.

Before calculating the comparison metrics, the predicted geometry was brought into the same local coordinate frame as the ground-truth geometry. This step was necessary because the MZA output was generated through the CityGML-based workflow, whereas the reference ground-truth geometries were stored in floor plan based local coordinates.

After alignment, the geometric overlap between each ground-truth zone G_i and each predicted zone P_j was quantified using the intersection over union (IoU). These pairwise IoU values formed an overlap matrix between the ground-truth layout and the MZA-predicted layout. To assign predicted zones to ground-truth zones, a one-to-one geometric matching was performed using the Hungarian assignment algorithm (Kuhn 1955). The pairwise IoU values were converted into a cost matrix,

$$C_{ij} = 1 - \mathrm{IoU}_{ij}, \tag{3.4}$$

so that minimising the total assignment cost is equivalent to maximising the total geometric overlap between matched zones. After the Hungarian assignment had produced the optimal one-to-one pairing, each ground-truth zone G_i was linked to the predicted zone P_{j^*} assigned to it by the optimisation. Assigned pairs with an IoU below the predefined minimum threshold were not accepted as valid matches, so that very small incidental overlaps were treated as unmatched rather than as correct zone assignments.

The comparison metrics used in the validation are summarised in Table 3.2. The resulting matching information was used to calculate the zone-level and building-level comparison metrics.

3.1.7. Implementation setup and result evaluation

Figure 3.8 summarises the implemented validation workflow. The workflow is organised as a data-transfer pipeline that starts from the two floor plan source formats, converts both into a common ground-truth representation, prepares the corresponding MZA-compatible input files, and finally compares the MZA-generated zoning output with the stored ground-truth geometries. The figure also indicates the main file formats used to transfer data between processing stages.

The validation workflow converts MSD and real estate floor plan sources into a common ground-truth structure of dwelling and stairwell polygons, which is stored as `.pkl` files and

Table 3.2: Summary of comparison metrics used for evaluating the MZA-generated zoning against the ground-truth.

Metric	Definition	Interpretation
IoU	$\mathrm{IoU}_{ij} = \dfrac{\mathrm{area}(G_i \cap P_j)}{\mathrm{area}(G_i \cup P_j)}$	Overlap between ground-truth zone G_i and predicted zone P_j.
Mean overall IoU	$\overline{\mathrm{IoU}}_{\mathrm{overall}} = \dfrac{1}{N_{\mathrm{GT}}} \sum_{i=1}^{N_{\mathrm{GT}}} \mathrm{IoU}_i$	Building-level geometric agreement, including missed ground-truth zones as zero-IoU contributions.
Area error [%]	$\mathrm{AreaError}_i = \dfrac{\lvert \mathrm{area}(P_{j^*}) - \mathrm{area}(G_i) \rvert}{\mathrm{area}(G_i)} \cdot 100$	Difference between the predicted and ground-truth zone area.
Centroid distance [m]	$d_i = \lVert c(G_i) - c(P_{j^*}) \rVert$	Spatial displacement of the predicted zone relative to the ground-truth zone.
Zone-count deviation	$\Delta N = N_{\mathrm{pred}} - N_{\mathrm{GT}}$	Difference between predicted and ground-truth zone counts.
Topology error	$E_{\mathrm{missing}} = E_{\mathrm{GT}} \setminus E_{\mathrm{pred}}$, $E_{\mathrm{extra}} = E_{\mathrm{pred}} \setminus E_{\mathrm{GT}}$	Missing or additional dwelling–stairwell adjacency relations after zone matching.

Note. N_{matched} denotes the number of valid matched zones, N_{FN} denotes the number of unmatched ground-truth zones, and N_{FP} denotes the number of unmatched predicted zones. The function area$(\cdot)$ denotes polygon area, $c(\cdot)$ denotes the polygon centroid, and E_{GT} and E_{pred} denote the dwelling–stairwell adjacency relations in the ground-truth and predicted layouts, respectively.

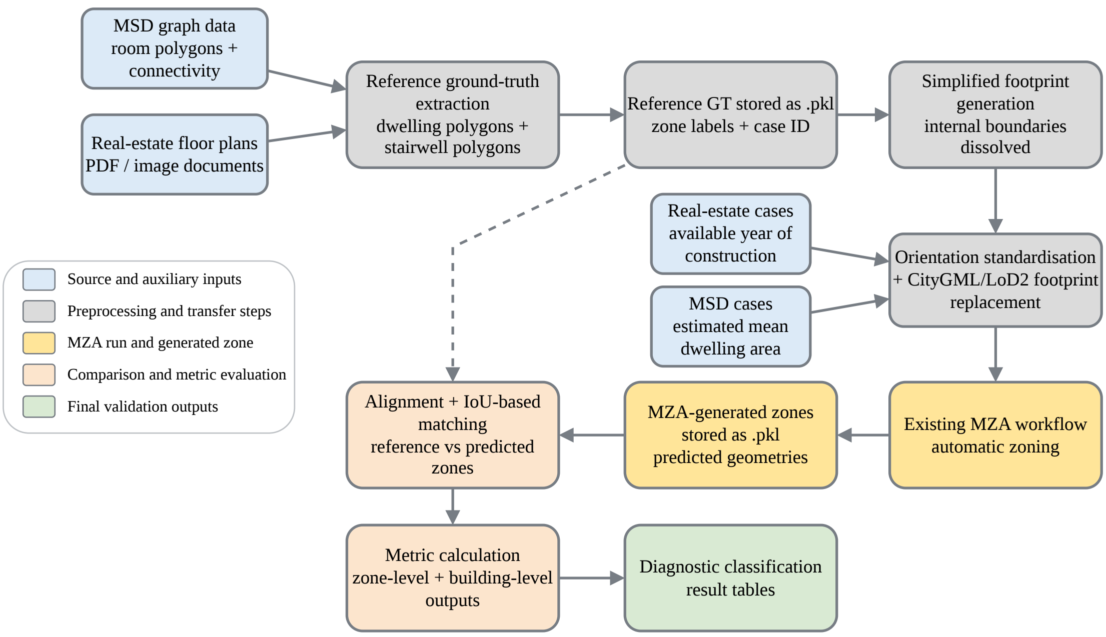


Figure 3.8: Implemented validation workflow from ground-truth extraction to MZA input preparation, generated-zone comparison, and diagnostic result evaluation.

used both for MZA input generation and later comparison.

The ground-truth geometries are simplified into external footprints, inserted into CityGML template buildings, processed by the MZA workflow, and compared with the predicted zones using alignment, matching, and metric calculation procedures to produce zone-level, building-level, and overlay outputs.

The implementation and evaluation scripts used for the validation processing pipeline are provided in an accompanying GitHub repository (Thomas 2026).

Result evaluation and diagnostic classification

The calculated zone-level and building-level metrics were used to assign a final diagnostic outcome to each validation case. The diagnostic classification served as a case-level interpretation of the geometric and topological validation results. It did not introduce an additional metric, but combined the structural, topological, and geometric indicators into one interpretable result for each building.

The rules in Table 3.3 were applied sequentially from top to bottom. Therefore, later geometric categories were evaluated only for cases that had already passed the structural and topology checks. This avoids repeating the same zero-error conditions in every row and reflects the hierarchy used in the diagnostic classification.

Table 3.3: Hierarchical diagnostic rules used to interpret the comparison metrics.

Diagnostic category	Evaluation stage	Additional rule applied at this stage
Structural mismatch	Structural check	$N_{FN} > 0$ and $N_{FP} > 0$
Under-zoned	Structural check	$N_{FN} > 0$ or $\Delta N < 0$
Over-zoned	Structural check	$N_{FP} > 0$ or $\Delta N > 0$
Correct count but topology mismatch	Topology check	$E_{missing} > 0$ or $E_{extra} > 0$
Weak geometric match	Geometric check	$\overline{\mathrm{IoU}}_{overall} < 0.50$
Correct topology but high area error	Geometric check	$\overline{\mathrm{AreaError}} > 40\%$
Correct topology but shifted zones	Geometric check	$\bar{d}_{centroid} > 3.0\,\mathrm{m}$
Good geometric and topological match	Geometric check	$\overline{\mathrm{IoU}}_{overall} \geq 0.70$, $\overline{\mathrm{AreaError}} \leq 40\%$, and $\bar{d}_{centroid} \leq 3.0\,\mathrm{m}$
Moderate geometric match	Remaining valid cases	No previous diagnostic rule is triggered

Selected ground-truth prediction overlay figures were used to support the interpretation of the diagnostic results. These overlays were mainly reviewed for failure cases. They made it possible to relate the numerical diagnosis to the spatial layout and to identify deviation mechanisms such as merged dwellings, split dwelling regions, displaced stairwells, incorrectly allocated areas, or missing stairwell–dwelling adjacency.

The final validation interpretation combined the building-level diagnostic table, the diagnostic category distribution, and representative spatial overlays.

3.2. Sensitivity analysis

The sensitivity analysis part of the thesis builds on the geometric validation described in Section 3.1. The validation evaluates whether the MZA can reproduce reference dwelling–stairwell zoning structures from limited geometric input. However, geometric validation alone does not show whether observed zoning deviations are relevant for building simulation outputs. A case may show geometric or topological deviations, but their thermal consequences depend on how the resulting zoning structure affects the heat-balance representation, interzonal exchange, operational assumptions, and stairwell treatment.

The sensitivity analysis therefore uses controlled model variants to study the effect of zoning deviations and zoning-related simplifications on simulation outputs. These variants are not interpreted as direct reproductions of individual failed validation cases. Instead, they represent the main mechanisms behind possible MZA-related deviations, including aggregation, reduced or increased zoning granularity, alternative layout arrangements, and changes in stairwell representation.

A deterministic comparison would show the effect of zoning variants only for one selected input configuration. However, the MZA-based workflow is intended for limited-information urban modelling, where both the internal zoning and several thermal or operational inputs are uncertain. Therefore, the sensitivity analysis evaluates whether zoning-related differences remain relevant when physical input uncertainty and stochastic user-profile variability are also considered.

In addition to comparing zoning model variants, the sensitivity analysis evaluates the influence of uncertain input parameters used during thermal enrichment and simulation. These inputs include missing, simplified, or uncertain assumptions such as construction age, envelope properties, window parameters, ventilation, setpoints, internal gains, weather data, and occupancy profiles. Morris screening is used to identify the most influential inputs, while Sobol analysis is used to quantify the contribution of the selected inputs to output variability.

The sensitivity analysis is therefore treated as the simulation-based continuation of the validation framework. The geometric validation first assesses whether the MZA reproduces the ground-truth dwelling and stairwell structure, while the sensitivity analysis examines whether the main zoning-deviation mechanisms and uncertain enrichment inputs lead to relevant changes in the selected simulation outputs. The following subsections describe the simulation setup, zoning-variant comparison, Morris screening, and Sobol analysis.

3.2.1. Simulation setup

For the sensitivity analysis, the zoning variants were implemented using the same TEASER–AixLib workflow introduced in Section 2.4.1. The workflow converts MZA-compatible building geometry and zoning information into reduced-order Modelica models and applies the required typological and operational enrichment for dynamic annual simulation.

Construction-related thermal properties were assigned through the TABULA building typology (Loga et al. 2016) in the TEASER workflow. Other simulation inputs were treated as separate operational or modelling assumptions: heating setpoints were defined as operational control assumptions, the WWR factor was applied as a facade/window-area modifier, and occupancy-related internal gains were represented using stochastic household profiles generated with the LoadProfileGenerator (Pflugradt et al. 2022).

The LoadProfileGenerator is an agent-based tool that creates time-dependent residential profiles for occupancy, appliance use, and lighting based on household and occupant characteristics. Such profiles are relevant to multi-family house simulations because different households can exhibit distinct occupancy patterns, internal gains, and temperature requirements (Kühn et al. 2024). For the present study, these generated household profiles were assigned to residential zones and scaled using an internal-gain factor.

Interzonal heat transfer was retained in the multi-zone variants because the comparison depends on temperature differences between dwellings, neighbouring zones, and the stairwell. This is relevant for the present analysis because the zoning variants differ in their zone count, adjacency relations, and stairwell treatment. These differences can affect internal heat exchange and therefore annual heating demand, peak heating demand, and overheating hours.

Figure 3.9 adapts the general sensitivity analysis workflow shown in Figure 2.4 to the MZA-based simulation context. In this thesis, the input variations consist of controlled zoning model variants, sampled uncertain input configurations, and stochastic occupancy assumptions. These are combined within the same annual TEASER–AixLib simulation framework and post-processed into run-level performance indicators for zoning-variant comparison and input-sensitivity analysis.

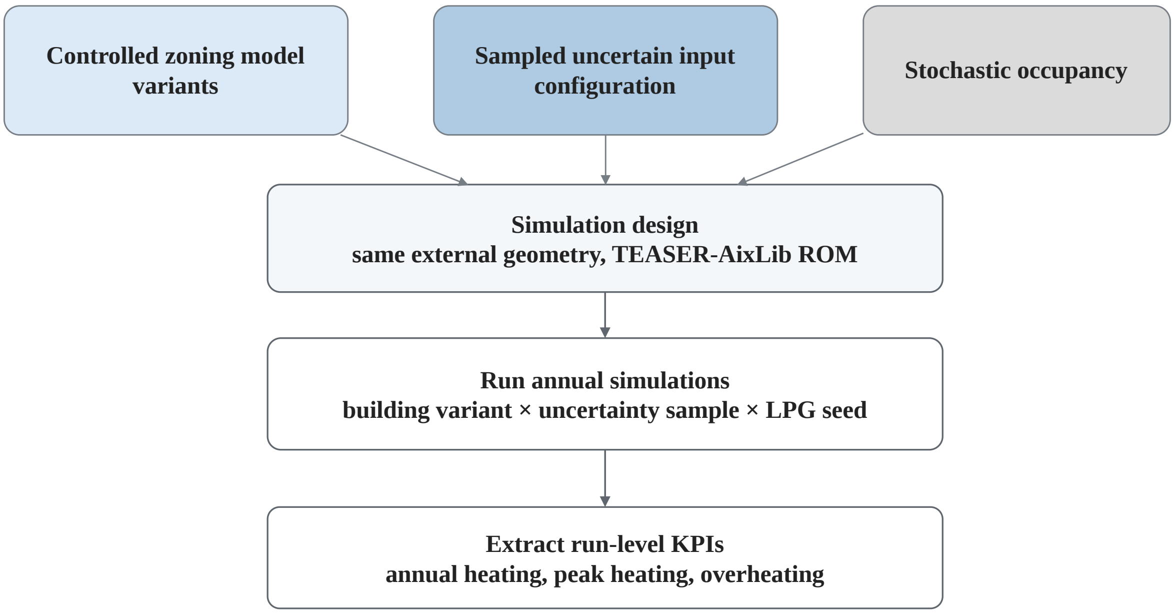


Figure 3.9: Workflow of the zoning-variant sensitivity analysis setup. Zoning variants, uncertain inputs, and stochastic occupancy profiles are combined in annual TEASER–AixLib simulations. The outputs are then processed into performance indicators for zoning comparison.

For the sensitivity comparison, the same external building geometry was used across the zoning variants. For each annual simulation run, the selected zoning variant was enriched with the corresponding uncertainty sample and stochastic occupancy seed. The variants therefore differed mainly in how the internal thermal zoning and stairwell representation were defined. This made it possible to interpret matched differences between variants as effects of zoning structure and stairwell treatment, while the uncertainty sampling captured the effect of missing or uncertain enrichment inputs.

3.2.2. Zoning model variants

The zoning variants used in the sensitivity analysis are summarised in Table 3.4, while their schematic layout arrangements are shown in Figure 3.10. The simulation case represents a five-storey multi-family residential building.

Table 3.4: Definition of zoning variants.

Variant	Residential zoning	stairwell	stairwell state	Total zones
V1	Aggregated building	No	–	1
V2	One zone per storey	Yes	Unheated	6
V3	Two zones per storey, Layout A	Yes	Unheated	11
V4	Two zones per storey, Layout B	Yes	Unheated	11
V5	Four zones per storey, Layout A	Yes	Unheated	21
V6	Four zones per storey, Layout B	Yes	Unheated	21
V7	Two zones per storey, Layout A	No	–	10
V8	Two zones per storey, Layout A	Yes	Heated	11

For variants with an explicit stairwell representation, the stairwell or circulation area was modelled as a separate thermal zone. The stairwell geometry was generated using the same MZA-compatible assumptions as in the simulation workflow, including stairwell dimensions. In the standard multi-zone variants, the stairwell was treated as unheated, while one variant tested the effect of modelling it as heated. This distinction is important because the stairwell can act as an intermediate thermal zone between dwellings and may influence interzonal heat exchange and heating demand.

The residential parts of the building were subdivided into different levels of zoning granularity. The variants range from a fully aggregated single-zone model to storey-level, dwelling-level, and finer residential subdivisions. Additional variants with the same number of zones but different layout arrangements were included to test whether the spatial placement of internal zone boundaries affects the simulation outputs. These layout variants are not intended to reproduce exact floor plan geometry. Instead, they provide controlled alternatives to test whether changes in adjacency relations and separating surfaces between zones influence heating demand, peak heating demand, and overheating hours.

Some residential subdivisions, especially the finer-granularity variants, may appear geometrically simplified or irregular when compared with realistic dwelling layouts. For the simulation analysis, however, the main purpose of these variants is not architectural realism, but the controlled comparison of thermal zoning mechanisms. The relevant simulation effect is therefore primarily linked to the number of zones, their adjacency relationships, the size of internal

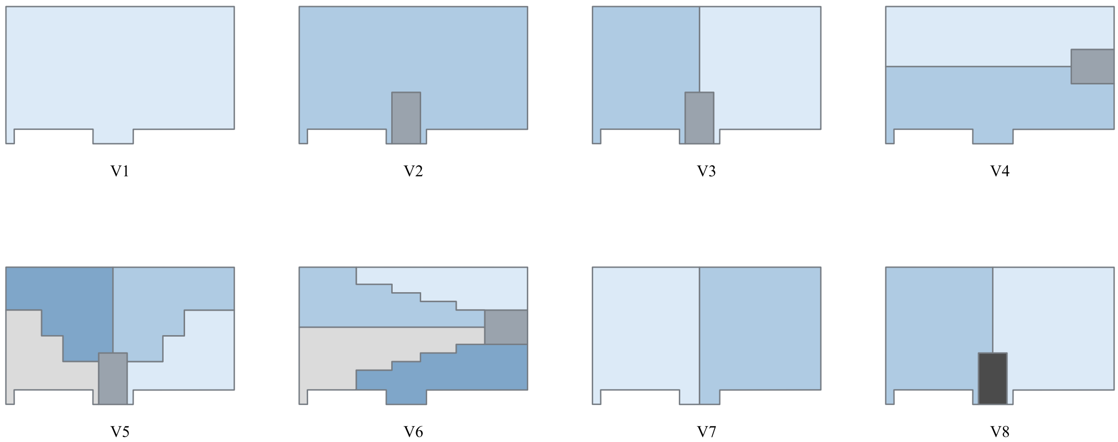


Figure 3.10: Schematic layout of the zoning model variants used in the sensitivity analysis.

separating surfaces, and the thermal treatment of the stairwell.

The zoning variants are not intended to directly reproduce individual validation failure cases. Instead, they provide controlled analogues of the main zoning mechanisms observed in the validation results. V1 and V2 represent aggregated or under-zoned configurations, while V5 and V6 represent finer subdivision relative to the two-zone-per-floor variants. The pairs V3–V4 and V5–V6 test layout-related deviations at fixed zone count, including shifted boundaries, area redistribution, and alternative stairwell placement. Variants V7 and V8 examine changes to the stairwell's representation and thermal treatment. In this way, the variants isolate the simulation effects of selected zoning-deviation mechanisms without reproducing individual failed cases exactly.

3.2.3. Boundary conditions and input uncertainties

The sensitivity analysis varied input assumptions that are required during the thermal enrichment and simulation stages of the MZA-based workflow. These inputs represent information that is either missing, simplified, or uncertain in limited-information urban modelling workflows. The uncertain inputs were varied directly in the sensitivity workflow, while the zoning variants described in Section 3.2.2 were used to test whether zoning-related effects remain visible under these uncertain conditions.

The sampled inputs and fixed boundary-condition choices are summarised in Table 3.5. The upper part of the table describes the general simulation design, while the lower part lists the uncertain physical and modelling inputs used in the sensitivity analysis.

The TABULA construction-age class and retrofit state represent typological enrichment uncertainty. In the TEASER workflow, these inputs determine the construction-related thermal properties assigned from the TABULA building typology (Loga et al. 2016). Since LoD2 geometry does not provide reliable construction-period information, and European building-stock data can be incomplete or inconsistent (European Commission, Joint Research Centre

Table 3.5: Boundary conditions and uncertain inputs used in the sensitivity analysis.

Category	**Setting or range**
General simulation design	
Zoning variants	V1–V8
Sampling structure	Building variant × uncertainty sample × weather condition × stochastic occupancy seed.
Uncertainty samples	60 samples per zoning variant.
Occupancy seeds	Four stochastic occupancy seeds per uncertainty sample.
Simulation period	Full annual simulation.
Uncertain physical and modelling inputs	
Construction state	Standard or retrofit state sampled with $p_{\text{standard}} = 0.75$.
TABULA age class	TABULA age class retained or shifted to a neighbouring class. The shift is represented as -1, 0, or $+1$ class.
Heating setpoint	Mean residential heating setpoint: 20.5–21.5 °C.
Zonal setpoint spread	0.5–2.0 K spread between residential zones in multi-zone variants.
Residential setpoint	Zone-level heating setpoints clipped to 18–24 °C.
Window-to-wall ratio	WWR scaling factor: 0.8–1.2.
Internal gains	Internal-gain scaling factor: 0.8–1.2.
Weather	Two annual weather cases: Napoli 2011 and Munich 2015.

2016), the TABULA class was either retained or shifted to a neighbouring class. The probability of retaining the original class was based on the residential building-age classification accuracy reported by Rosser et al. (2019). The retrofit-state probability $p_{\text{standard}} = 0.75$ was used as a proxy for the share of non-retrofitted or energy-inefficient buildings in the European building stock (European Commission 2020).

The heating setpoint and zonal setpoint spread represent operational uncertainty. A building-level mean heating setpoint was sampled first, and for multi-zone variants an additional zonal spread was applied. The resulting zone-level heating setpoints were limited to 18–24 °C, following literature-supported residential heating bounds and indoor-temperature guidance (IEA SHC Task 53 2018; Public Health England 2014). The zonal spread represents heterogeneous dwelling temperature preferences and is relevant because temperature differences between dwellings and the stairwell can drive interzonal heat exchange. For the single-zone variant, this heterogeneity cannot be represented explicitly, so a uniform building-level setpoint was used.

Stochastic household profiles were generated with the LoadProfileGenerator, which provides time-dependent profiles for occupancy, appliance use, and lighting (Pflugradt et al. 2022). The profiles were assigned to residential zones using stochastic seeds, with four occupancy seeds simulated for each uncertainty sample. The assignment followed a distribution of one- to five-person households, with ten household profiles used for variants with two residential zones per storey and twenty profiles for variants with four residential zones per storey. In aggregated variants, the same household profiles were combined according to the lower zoning resolution. The stairwell was not treated as a residential household zone.

Stochastic household profiles were generated with the LoadProfileGenerator, which is an agent-based tool for creating time-dependent residential load profiles for occupancy, appliance use, and lighting (Pflugradt et al. 2022). The tool represents household behaviour using occupant and household characteristics, such as household size and activity patterns, and generates hourly profiles based on stochastic activity modelling (Kühn et al. 2024; Pflugradt et al. 2022). In this thesis, the profiles were assigned to residential zones using stochastic seeds, with four occupancy seeds simulated for each uncertainty sample. The assignment followed a distribution of one- to five-person households, with ten household profiles used for variants with two residential zones per storey and twenty profiles for variants with four residential zones per storey. In aggregated variants, the same household profiles were combined according to the lower zoning resolution. The stairwell was not treated as a residential household zone.

The WWR factor was treated as a facade and window-area modifier. It represents uncertainty in opening geometry and facade assumptions where detailed window information is not available. The implementation follows the concept of Energy Design Measures, where facade and window-related properties are varied parametrically (Hale et al. 2008). The selected WWR range of 0.8–1.2 corresponds to a ±20 % variation around the baseline and follows previous work in which WWR was varied by the same order of magnitude (Rumnici and Abualdenien 2019).

Internal gains were additionally scaled by a factor of 0.8–1.2, corresponding to a ±20 % variation around the baseline. This range is consistent with the internal-gain scaling factor used by Brès et al. (2022). Weather was included as an external boundary-condition uncertainty. Two annual weather files, Napoli 2011 and Munich 2015, were used to represent warm and cold climatic conditions and to test whether zoning-related effects remain consistent across weather cases.

3.2.4. Input parameter sensitivity analysis

After the zoning-variant comparison, an additional input parameter sensitivity analysis was carried out to identify which uncertain enrichment and modelling assumptions had the strongest influence on the selected simulation outputs. While the zoning-variant comparison evaluates whether different zoning structures lead to different simulation results, the input parameter sensitivity analysis evaluates the relative importance of uncertain input assumptions within the simulation workflow. For this purpose, a two-step approach was used: Morris screening was first applied to rank a wider set of candidate inputs, and Sobol variance decomposition was then applied to a reduced set of influential inputs.

Table 3.6 summarises the design and input parameter settings used for the Morris screening and Sobol decomposition. Morris was used to screen a wider set of uncertain parameters, while Sobol was applied to the reduced set of influential parameters identified from the Morris results.

Table 3.6: Design and uncertain inputs used for the Morris screening and Sobol decomposition.

Category	Morris screening	Sobol decomposition
General simulation design		
Zoning variants	V3	V3
Sampling design	Morris elementary-effects design with $r = 15$ trajectories and 6 levels.	Saltelli sampling with $N_{\text{sob}} = 128$ and $K = 6$ selected parameters.
Weather	Fixed weather case: Berlin 2021.	Evaluated separately for Napoli 2011 and Munich 2015.
Stochastic occupancy	Controlled representative LPG realisation; one seed per trajectory.	Controlled representative LPG realisation with one fixed seed.
Construction state	Standard state fixed.	Standard state fixed.
Uncertain physical and modelling inputs		
TABULA age class shift	$-1, 0, +1$.	
Heating setpoint	Mean residential heating setpoint: 20.5–21.5 °C.	
Zonal setpoint spread	0.5–2.0 K.	
Window-to-wall ratio	WWR factor: 0.8–1.2.	
Internal gains	Internal-gain scaling factor: 0.8–1.2.	
Base air-change rate	0.05–0.8 h^{-1}.	
Shading factor	0.3–1.0.	
Window g-value	0.45–0.75.	
Window U-value	0.9–2.0 $\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$.	
Window convective coefficient	2.0–4.0 $\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$.	

Non-varied assumptions were fixed to representative values in the Morris and Sobol analyses to reduce stochastic noise. In the Morris screening, construction state, occupancy seed, and weather were fixed so that the elementary effects reflected the candidate input parameters. In the Sobol analysis, construction state and occupancy seed were fixed, while the analysis was repeated separately for Napoli 2011 and Munich 2015 to evaluate climate-dependent parameter importance.

The Morris method was used as a screening method because it requires fewer simulations than a full variance-based analysis and is therefore suitable for identifying influential parameters in a computationally expensive simulation workflow. The method evaluates elementary effects by changing one input parameter at a time along sampled trajectories in the input space. For an input parameter x_i, the elementary effect can be written as

$$EE_i = \frac{Y(x_1, \ldots, x_i + \Delta, \ldots, x_k) - Y(x_1, \ldots, x_i, \ldots, x_k)}{\Delta}, \tag{3.5}$$

where Y is the simulation output, k is the number of uncertain input parameters, and Δ is the step size in the discretised input space. For each input, the distribution of elementary effects was summarised using the absolute mean μ_i^* and the standard deviation σ_i. The value of μ_i^* indicates the overall importance of an input, while σ_i indicates whether the input effect is nonlinear or involved in interactions with other inputs.

Sobol analysis was then used to quantify the contribution of the selected input parameters to output variability. In contrast to Morris screening, Sobol analysis decomposes the variance of the model output into contributions from individual inputs and their interactions. The first-order Sobol index S_i measures the direct contribution of input x_i to the output variance:

$$S_i = \frac{\mathrm{Var}_{x_i}\left[\mathbb{E}_{\sim x_i}(Y \mid x_i)\right]}{\mathrm{Var}(Y)}, \tag{3.6}$$

where $\mathrm{Var}(Y)$ is the total output variance and $\mathbb{E}_{\sim x_i}(Y \mid x_i)$ is the expected output when all other inputs are averaged out. The total-order Sobol index S_{T_i} measures the total contribution of input x_i, including interaction effects with other parameters:

$$S_{T_i} = \frac{\mathbb{E}_{\sim x_i}\left[\mathrm{Var}_{x_i}(Y \mid \sim x_i)\right]}{\mathrm{Var}(Y)}. \tag{3.7}$$

The Sobol analysis was applied to the reduced parameter set identified from the Morris screening. This allowed the analysis to distinguish between parameters that mainly act independently and parameters whose influence appears through interactions with other inputs. The comparison between first-order and total-order indices was used to interpret whether an input has a direct effect on the output or whether its effect depends on the values of other uncertain inputs.

4. Results

This chapter presents the outcomes of the validation and sensitivity analyses described in the methodology. The results are reported descriptively, focusing on the calculated metrics, diagnostic classifications, and observed distributions before their implications are interpreted in the discussion chapter.

4.1. Validation results

The validation results evaluate the geometric and topological agreement between the MZA-generated dwelling and stairwell zones and the corresponding reference ground-truth zones. The results are presented separately for the MSD and real estate validation datasets because the two datasets differ in their source formats and ground-truth extraction procedures. For each dataset, the ranked distribution of mean overall IoU values provides an overview of geometric agreement across all validation cases. This is followed by the diagnostic classification of the cases and representative reference–prediction overlays, which illustrate the spatial meaning of the numerical categories.

4.1.1. Validation results for MSD

Figure 4.1 shows the validation cases ranked by their mean overall IoU value. Each point represents one validation case, with the horizontal position indicating its percentile rank and the vertical position showing the corresponding mean overall IoU. The marker symbols indicate the diagnostic category assigned to each case. The dashed horizontal lines indicate the two IoU thresholds used for diagnostic classification.

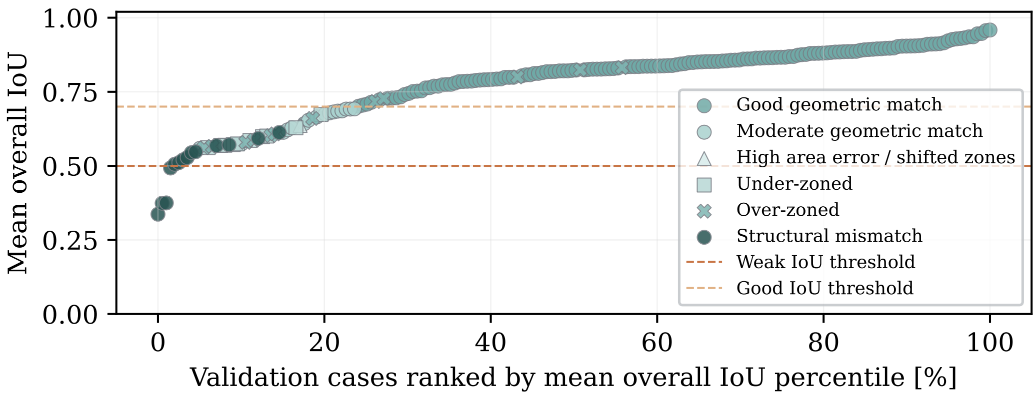


Figure 4.1: Ranked distribution of mean overall IoU values for the MSD validation cases. Marker symbols indicate the assigned diagnostic categories. The dashed horizontal lines indicate the weak IoU threshold of 0.50 and the good IoU threshold of 0.70 used for classification.

The mean overall IoU values range from approximately 0.33 to 0.96. Only a small number of cases fall below the weak IoU threshold of 0.50, and these cases are located at the lower end of the ranked distribution. Cases between the weak and good IoU thresholds are mainly classified as moderate geometric matches or as cases with diagnostic deviations, such as under-zoning, over-zoning, or structural mismatch. Above the good IoU threshold of 0.70, most cases are classified as good geometric matches, with the highest-ranked cases reaching mean overall IoU values above 0.90.

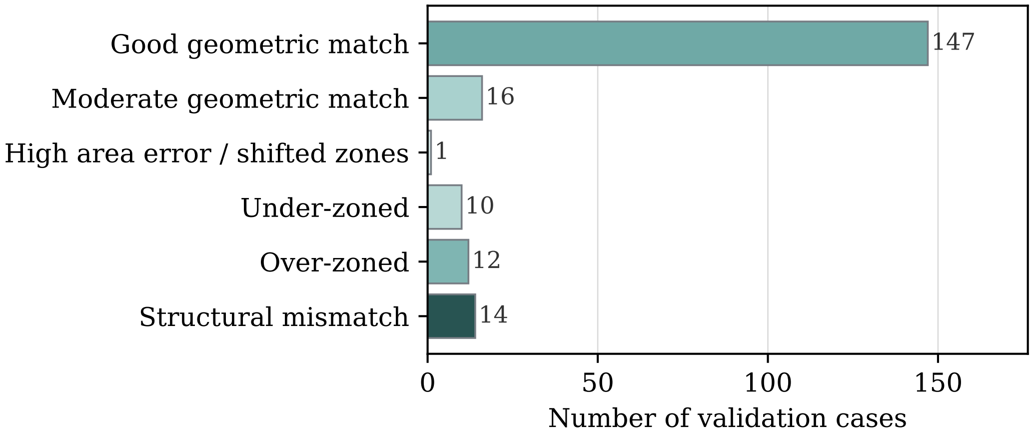


Figure 4.2: Diagnostic classification of the MSD validation cases, showing the number of cases assigned to each diagnostic category.

Figure 4.2 summarises the diagnostic classification of the MSD validation cases. The largest group is the good geometric match category, with 147 out of 200 cases. A further 16 cases are classified as moderate geometric matches. The remaining cases are distributed across deviation-related categories, including 10 under-zoned cases, 12 over-zoned cases, 14 structural mismatches, and one case with high area error or shifted zones. Overall, the classification shows that most MSD cases are successfully reproduced at dwelling–stairwell level. In contrast, the non-matching cases are mainly related to zone-count differences or structural mismatch rather than isolated area-error problems.

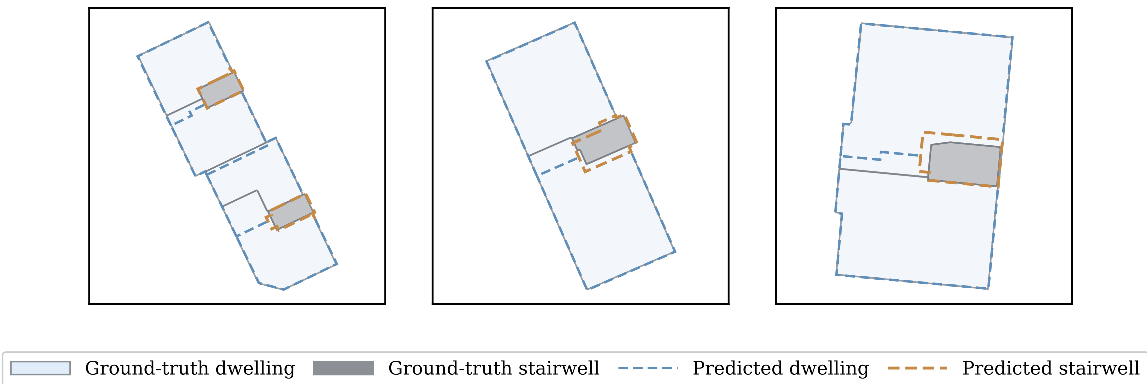


Figure 4.3: Representative MSD validation cases with good geometric agreement between reference ground-truth zones and MZA-predicted zones. Solid filled polygons show the ground-truth dwelling and stairwell zones, while dashed outlines show the predicted dwelling and stairwell zones.

Figure 4.3 shows representative MSD cases classified as good geometric matches. In these examples, the predicted dwelling outlines closely follow the reference dwelling geometries, and the predicted stairwell is located close to the reference stairwell. Small deviations remain along some internal boundaries and stairwell edges, but the overall dwelling–stairwell arrangement is reproduced. These examples visually support the high share of good geometric matches reported in the diagnostic classification.

Figure 4.4 shows representative MSD validation cases assigned to deviation-related diagnostic categories. The structural mismatch examples illustrate cases where the predicted dwelling–stairwell arrangement differs from the reference ground-truth layout. The over-zoned example shows a case where the MZA predicts more subdivisions than the reference structure. In contrast, the under-zoned example shows a more aggregated predicted zoning than the ground truth. The high-area-error or shifted-zone example shows that the general dwelling–stairwell structure is partly reproduced. Still, the predicted zone position or area differs visibly from the reference geometry. These overlays illustrate the spatial meaning of the non-matching categories reported in the diagnostic classification.

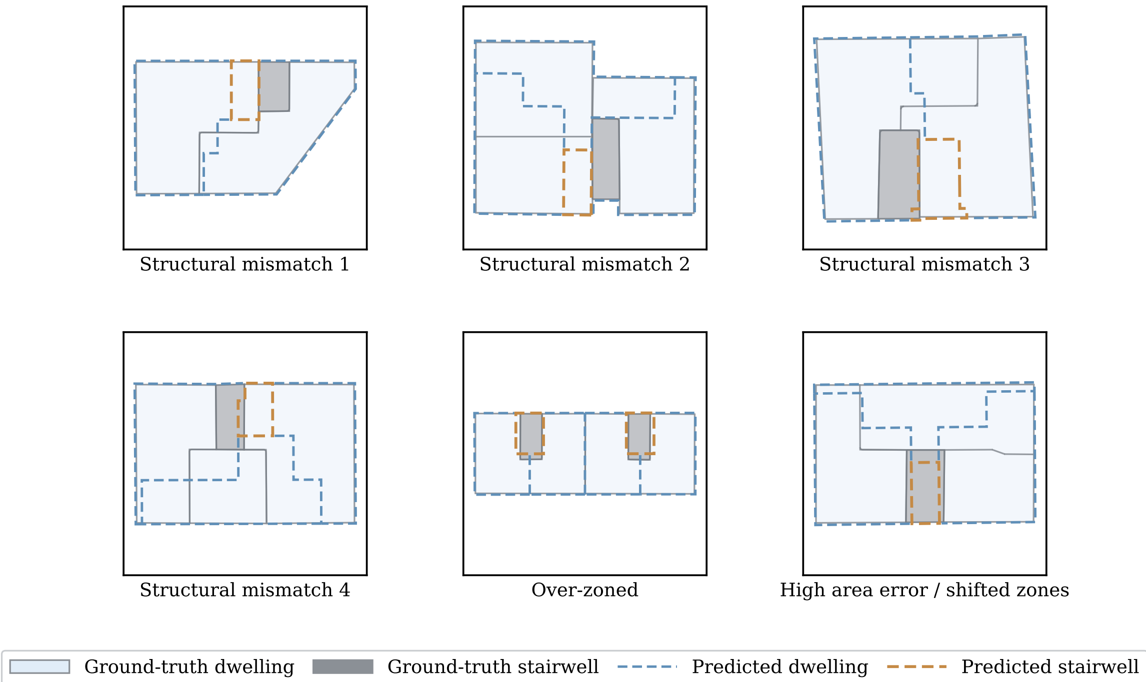


Figure 4.4: Representative MSD validation cases with diagnostic deviations between reference ground-truth zones and MZA-predicted zones. Solid filled polygons show the ground-truth dwelling and stairwell zones, while dashed outlines show the predicted dwelling and stairwell zones.

4.1.2. Validation results for real estate data

The real estate validation cases are presented in the same format as the MSD results to compare the distributions of geometric agreement and diagnostic outcomes across the second validation dataset.

Figure 4.5 shows the ranked distribution of mean overall IoU values for the real estate validation cases. The validation cases are ordered from the lowest to the highest mean overall IoU, so the figure provides an overview of the complete geometric-agreement distribution for the real estate dataset.

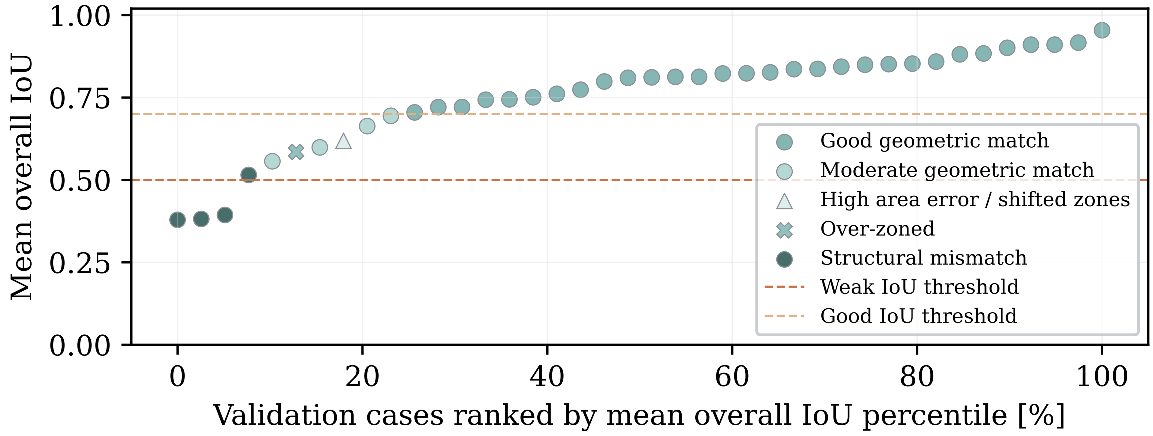


Figure 4.5: Ranked distribution of mean overall IoU values for the real estate validation cases. Marker symbols indicate the assigned diagnostic categories. The dashed horizontal lines indicate the weak IoU threshold of 0.50 and the good IoU threshold of 0.70 used for classification.

The mean overall IoU values range from approximately 0.38 to 0.96. Only a small number of cases fall below the weak IoU threshold of 0.50, and these cases are located at the lower end of the ranked distribution. Cases between the weak and good IoU thresholds are mainly classified as moderate geometric matches or as cases with diagnostic deviations, such as over-zoning, high area error, shifted zones, or structural mismatch. Above the good IoU threshold of 0.70, most cases are classified as good geometric matches, with the highest-ranked cases reaching mean overall IoU values above 0.90.

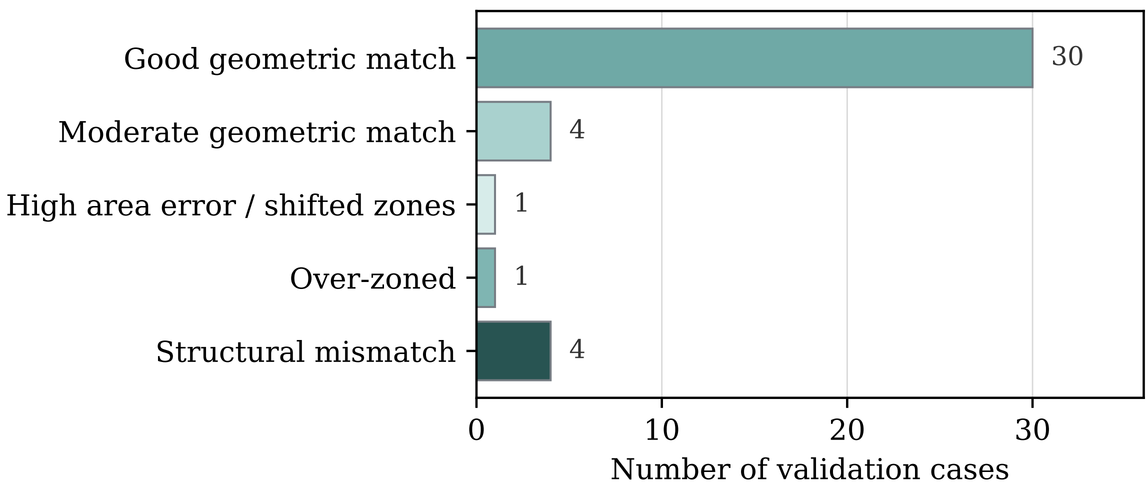


Figure 4.6: Diagnostic classification of the real estate validation cases, showing the number of cases assigned to each diagnostic category.

Figure 4.6 summarises the diagnostic classification of the real estate validation cases. The largest group is the good geometric match category, with 30 out of 40 cases. A further four cases are classified as moderate geometric matches. The remaining cases are distributed across deviation-related categories, including one over-zoned case, one case with high area error or shifted zones, and four structural mismatches. Overall, the classification shows that most real estate cases are successfully reproduced at dwelling–stairwell level. In contrast, the non-matching cases are mainly associated with structural mismatch rather than isolated geometric area errors.

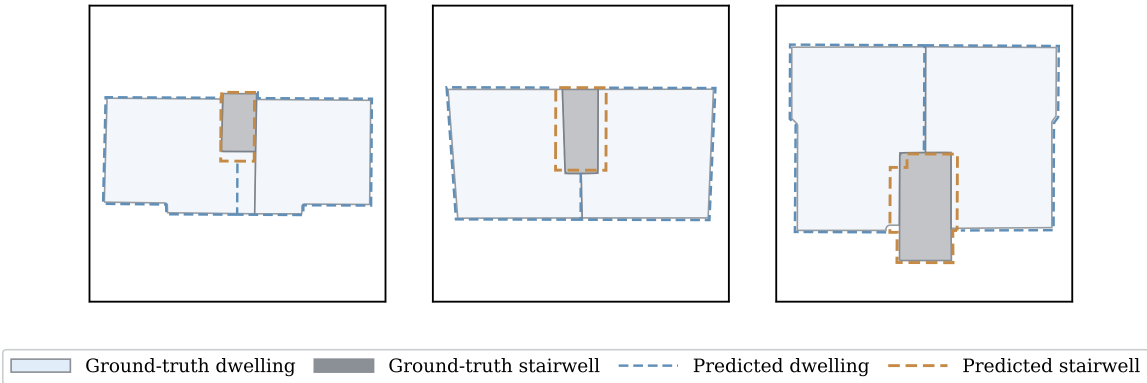


Figure 4.7: Representative real estate validation cases with good geometric agreement between reference ground-truth zones and MZA-predicted zones. Solid filled polygons show the ground-truth dwelling and stairwell zones, while dashed outlines show the predicted dwelling and stairwell zones.

Figure 4.7 shows representative real estate validation cases classified as good geometric matches. In these examples, the predicted dwelling outlines closely follow the reference dwelling geometries, and the predicted stairwell is located close to the reference stairwell. Small deviations remain in the exact position of some internal boundaries and stairwell edges, but the overall dwelling–stairwell arrangement is reproduced well. These examples visually support the high number of good geometric matches reported in the real estate diagnostic classification.

Figure 4.8 shows representative real estate validation cases assigned to deviation-related diagnostic categories. The structural mismatch examples illustrate cases where the predicted dwelling–stairwell arrangement differs from the reference ground-truth layout. The over-zoned example shows a case where the MZA generates more predicted subdivisions than the reference structure. The high-area-error or shifted-zone example shows that the general dwelling–stairwell structure is partly reproduced. Still, the predicted zone position or area differs visibly from the reference geometry. These overlays illustrate the spatial meaning of the non-matching categories reported in the real estate diagnostic classification.

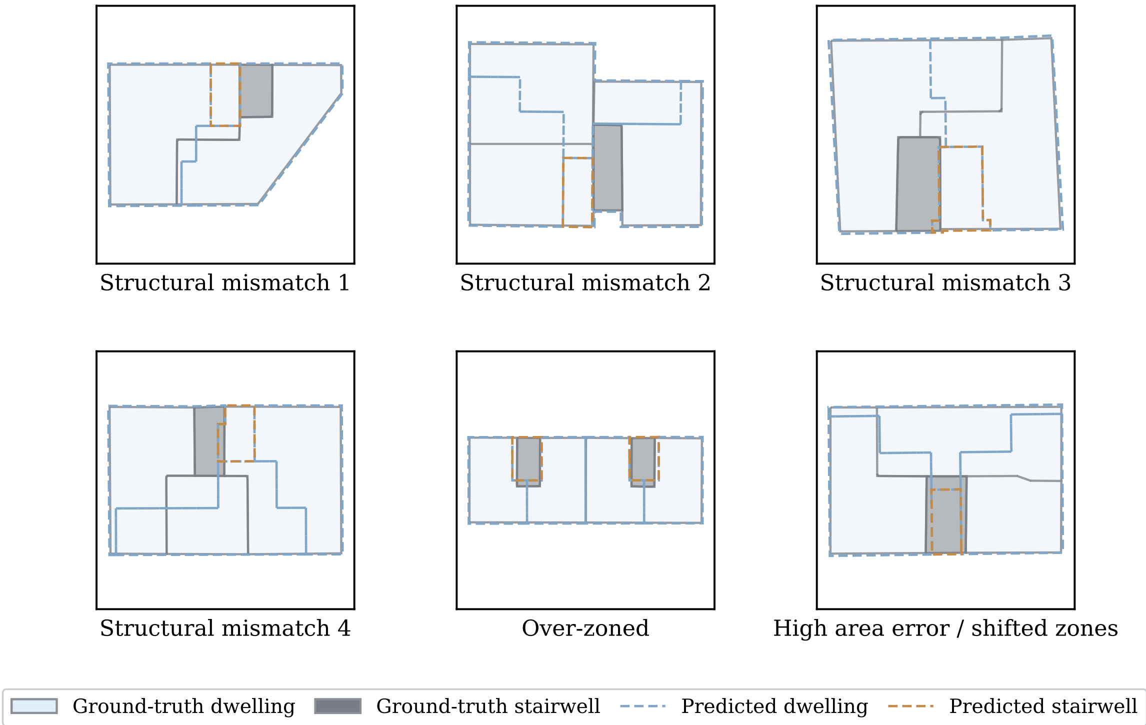


Figure 4.8: Representative real estate validation cases with diagnostic deviations between reference ground-truth zones and MZA-predicted zones. Solid filled polygons show the ground-truth dwelling and stairwell zones, while dashed outlines show the predicted dwelling and stairwell zones.

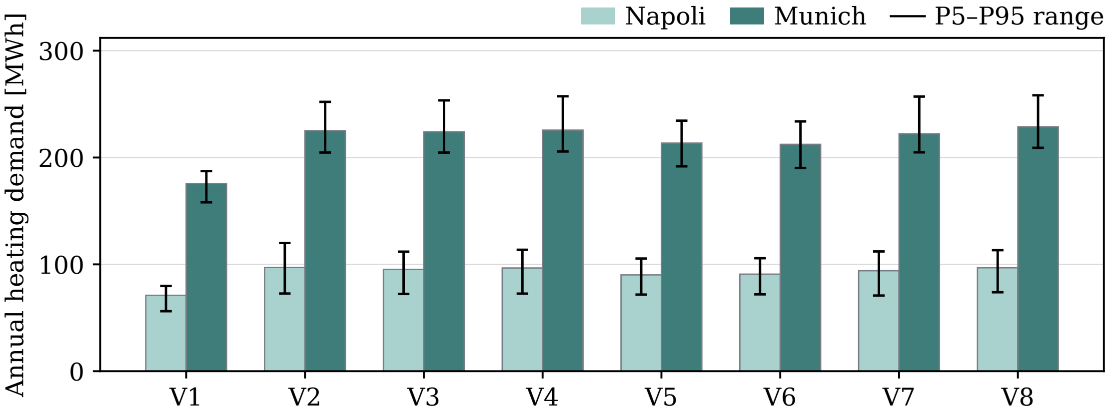
4.2. Zoning-variant sensitivity analysis

The zoning-variant results are presented using the same figure structure for each performance indicator. Each figure compares the zoning variants V1–V8 for the standard and retrofit construction states and for the Napoli and Munich weather cases. Bars represent the median value (P50), while error bars show the P5–P95 range of the propagated uncertainty samples.

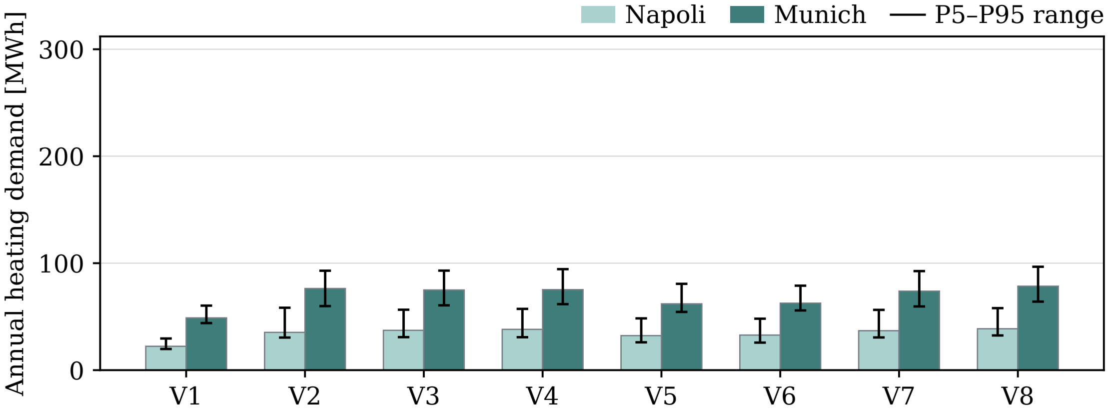
4.2.1. Annual heating demand

Figure 4.9 shows the effect of the zoning variants on annual heating demand for the standard and retrofit construction states. In both construction states, the Munich weather case produces substantially higher annual heating demand than the Napoli weather case. For the standard construction state, V1 has a median annual heating demand of 71.0 MWh under Napoli and 175.5 MWh under Munich. The multi-zone variants range from 90.1–97.1 MWh under Napoli and from 212.3–228.8 MWh under Munich.

The variant comparison shows that annual heating demand is mainly affected by the level of zoning aggregation and the treatment of the stairwell. The fully aggregated variant V1 produces the lowest annual heating demand in both weather cases. The storey-level variant V2 produces higher values than V1 and is close to the two-zone-per-floor variants V3 and V4. The alternative-layout comparison at fixed zone count shows only small differences: V3 and V4 differ by about 1.3 MWh under Napoli and 1.6 MWh under Munich in the standard construction state.



(a) Standard construction state.



(b) Retrofit construction state.

Figure 4.9: Annual heating demand by zoning variant and weather case for the standard and retrofit construction states. Bars show the median value (P50), while error bars indicate the P5–P95 range of the remaining propagated uncertainties.

The finer subdivision variants V5 and V6 produce lower annual heating demand than V3 and V4, with median values of about 90.1–90.8 MWh under Napoli and 212.3–213.5 MWh under Munich. The stairwell-related variants also show a visible difference: V7, without a separate stairwell zone, is slightly lower than V3, while V8, with a heated stairwell, gives the highest or near-highest annual heating demand among the multi-zone variants.

In the retrofit construction state, the same relative pattern remains visible, but the absolute annual heating demand is substantially lower. V1 has a median annual heating demand of 22.3 MWh under Napoli and 48.9 MWh under Munich, while the multi-zone variants range from 32.3–38.7 MWh under Napoli and from 62.0–78.5 MWh under Munich.

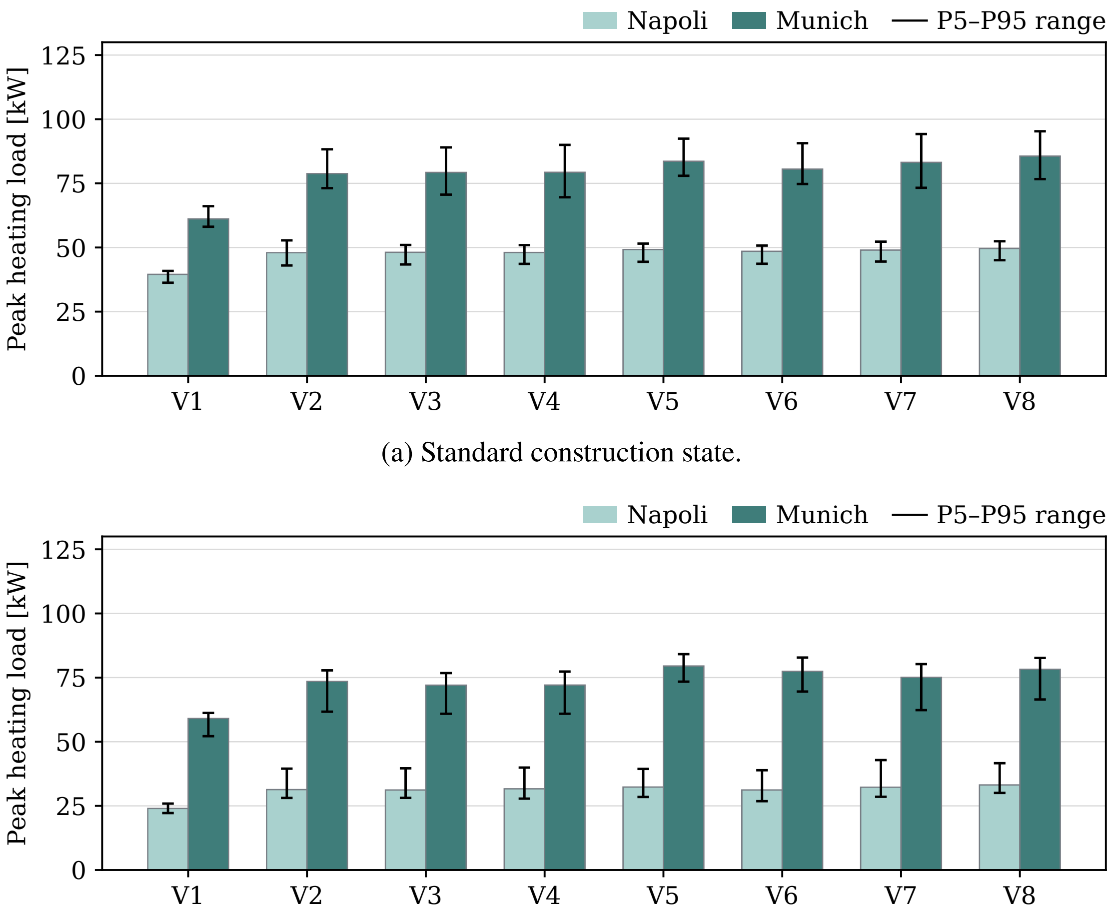
The P5–P95 uncertainty ranges show overlap among several multi-zone variants. For example, V2 and V3 overlap under both standard Napoli conditions, with ranges of 72.7–120.0 MWh and 72.3–111.9 MWh, and retrofit Napoli conditions, with ranges of 30.5–58.4 MWh and

30.9–56.5 MWh. The position of the median within the uncertainty range also varies: V5 under standard Munich conditions is nearly balanced around the median of 213.5 MWh within a range of 191.7–234.5 MWh, whereas retrofit cases such as V2 under Napoli and V5 under Munich show a stronger upper-side spread above the median.

Overall, the annual heating demand results show that the largest zoning-related difference occurs between the fully aggregated single-zone model and the multi-zone representations. At the same time, finer subdivision and stairwell treatment introduce additional differences.

4.2.2. Peak heating demand

Figure 4.10 shows the effect of the zoning variants on peak heating load for the standard and retrofit construction states. In both construction states, the Munich weather case produces higher peak heating loads than the Napoli weather case. For the standard construction state, V1 has a median peak heating load of 39.5 kW under Napoli and 61.1 kW under Munich. The multi-zone variants range from 48.0–49.6 kW under Napoli and from 78.8–85.6 kW under Munich.



(a) Standard construction state.

(b) Retrofit construction state.

Figure 4.10: Peak heating load by zoning variant and weather case for the standard and retrofit construction states. Bars show the median value (P50), while error bars indicate the P5–P95 range of the remaining propagated uncertainties.

The variant comparison shows that peak heating demand is strongly affected by the transition from the fully aggregated model to the multi-zone representations. V1 produces the lowest peak heating load in both weather cases. The storey-level variant V2 is close to the two-zone-per-floor variants V3 and V4, while the alternative-layout comparison at fixed zone count shows only very small differences. In the standard construction state, V3 and V4 differ by less than 0.1 kW in both weather cases.

The finer subdivision variants V5 and V6 generally produce higher peak loads than V3 and V4, especially under the Munich weather case. The stairwell-related variants also show visible differences: V7, without a separate stairwell zone, is higher than V3, while V8, with a heated stairwell, gives the highest or near-highest peak heating load among the multi-zone variants.

In the retrofit construction state, the same relative pattern remains visible, although the absolute peak heating loads are generally lower than in the standard construction state. V1 has a median peak heating load of 24.0 kW under Napoli and 59.1 kW under Munich, while the multi-zone variants range from 31.2–33.2 kW under Napoli and from 72.0–79.5 kW under Munich. The peak heating demand results show a clearer zoning effect than the annual heating demand results, with the multi-zone variants producing higher peak loads than the fully aggregated model.

The P5–P95 uncertainty ranges show overlap among several multi-zone variants. In the standard Napoli case, V2 spans 43.0–52.8 kW and V3 spans 43.4–51.0 kW, while in the retrofit Napoli case V2 spans 28.1–39.5 kW and V3 spans 28.1–39.6 kW. The median position within the uncertainty range is not always centred: for example, under retrofit Napoli conditions, V3 has a median of 31.2 kW within a range of 28.1–39.6 kW, showing a wider upper-side spread, while V5 under standard Munich conditions has a median of 83.6 kW within a range of 78.0–92.4 kW.

Figure 4.11 shows the heating load profile and the corresponding weighted mean air temperature during a selected three-day peak heating event window. The vertical dashed line marks the time of the peak heating load on 22.11 at 02:00. At this time, the fully aggregated variant V1 has a heating load of 38.39 kW, while the multi-zone variants range from 47.79 kW in V3 to 49.76 kW in V5. The highest value is therefore observed for the finer subdivision variant V5, closely followed by V8 with 49.73 kW.

The weighted mean air temperatures at the same time range from 19.41 °C in V2 to 21.11 °C in V7. Compared with the heating load values, the weighted mean air temperatures vary within a narrower absolute range. The event-window result shows that the zoning variants produce similar temporal heating load patterns, but the magnitude at the peak time differs clearly between the fully aggregated model and the multi-zone representations.

4.2.3. Overheating hours

Figure 4.12 shows the effect of the zoning variants on overheating hours for the standard and retrofit construction states. In contrast to the heating-related outputs, the Napoli weather case produces substantially higher overheating hours than the Munich weather case. For the standard construction state, V1 has a median of 2035.5 h under Napoli and 16.5 h under

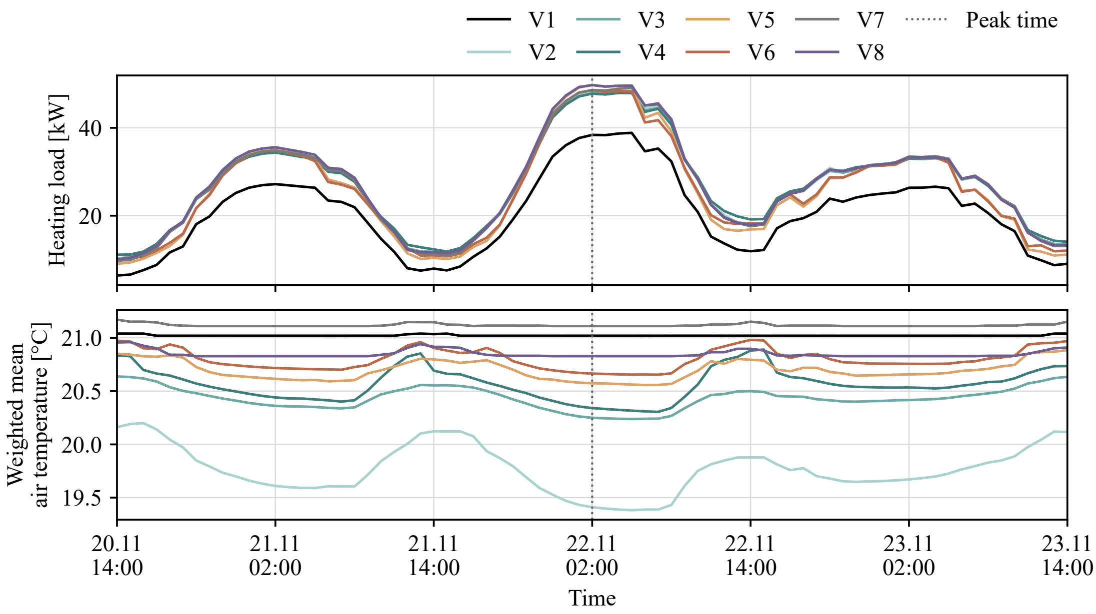


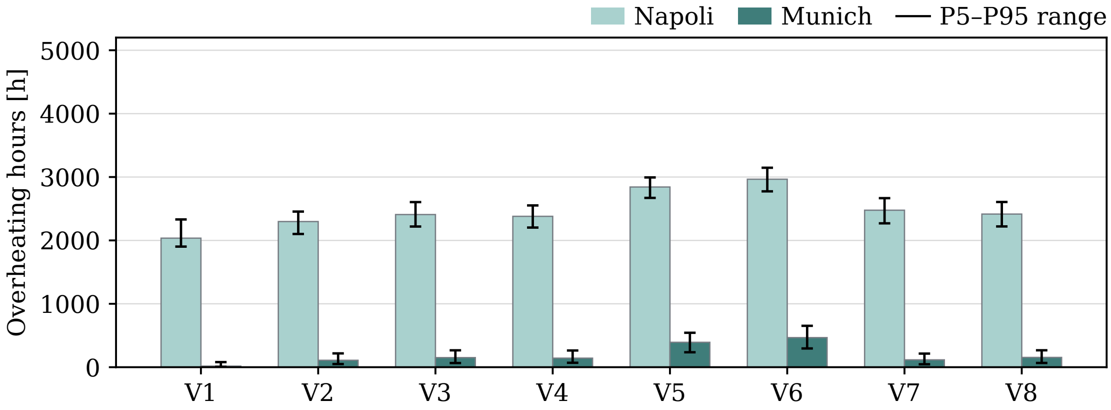
Figure 4.11: Heating load and weighted mean air temperature during a selected peak heating event window for the zoning variants. The upper panel shows the heating load profiles, while the lower panel shows the corresponding weighted mean air temperatures. The vertical dashed line indicates the time of peak heating load.

Munich. The multi-zone variants range from 1866.5 to 1995.0 h under Napoli and from 10.0 to 22.5 h under Munich.

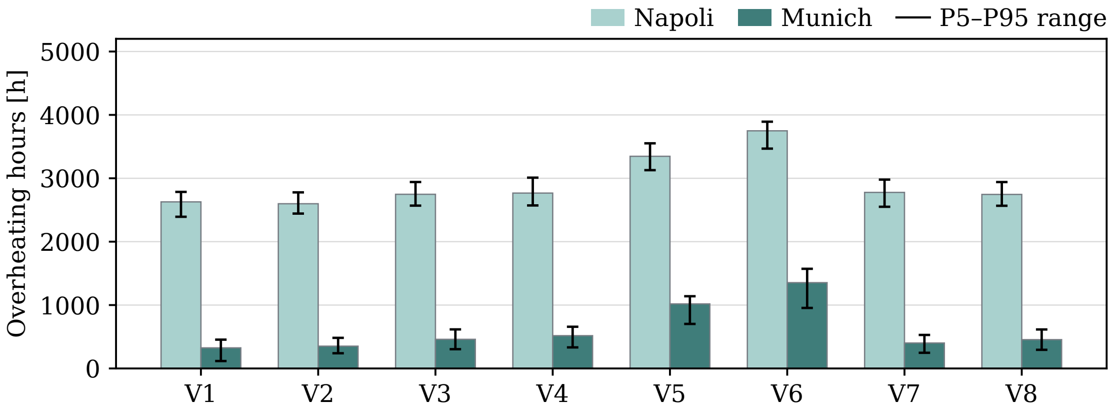
The variant comparison shows that overheating hours respond differently from annual and peak heating demand. The fully aggregated variant V1 produces the highest or near-highest overheating hours, especially under the Napoli weather case. The storey-level and two-zone-per-floor variants show lower overheating hours than V1 in most cases. At fixed zone count, the alternative layout variants show visible but moderate differences. In the standard construction state, V3 and V4 differ by about 15 h under Napoli and 6 h under Munich.

The finer subdivision variants V5 and V6 produce more overheating hours than the two-zone-per-floor variants V3 and V4, indicating that the change from coarser to finer residential subdivision affects the overheating response. The stairwell-related variants show smaller differences: V7 is close to or lower than V3, while V8 remains close to V3 for this KPI.

In the retrofit construction state, overheating hours increase for all variants, especially under the Napoli weather case. V1 has a median of 2628.0 h under Napoli and 325.0 h under Munich, while the multi-zone variants range from 2279.5–2427.0 h under Napoli and from 162.5–238.0 h under Munich. The results show that zoning representation affects the duration of threshold exceedance, especially under the warmer Napoli weather case and in the retrofit construction state.



(a) Standard construction state.



(b) Retrofit construction state.

Figure 4.12: Overheating hours by zoning variant and weather case for the standard and retrofit construction states. Bars show the median value (P50), while error bars indicate the P5–P95 range of the remaining propagated uncertainties.

The P5–P95 uncertainty ranges also overlap among several multi-zone variants for overheating hours. In the standard Napoli case, V2 spans 1732.3–2103.9 h and V3 spans 1755.8–2127.3 h, while in the retrofit Napoli case V2 spans 2158.9–2452.9 h and V3 spans 2168.6–2485.1 h. The position of the median within the uncertainty band varies more strongly under the Munich weather case: for example, V3 under standard Munich conditions has a median of 16.0 h within a range of 5.8–52.2 h, and V5 has a median of 22.0 h within a range of 6.8–64.2 h, showing a wider upper-side spread.

Figure 4.13 shows the weighted mean air temperature during a selected three-day overheating event window. The dashed horizontal line marks the 26 °C overheating threshold. All zoning variants follow a similar diurnal pattern, with temperatures increasing during the daytime and decreasing at night and in the early morning. The weighted mean air temperature exceeds the threshold during the warm periods of all three days.

Over the full event window, the fully aggregated variant V1 remains above the 26 °C threshold for 66 hours. The multi-zone variants remain above the threshold for shorter durations, ranging from 41 hours in V2 to 54 hours in V5 and V6. The maximum weighted mean air temperature

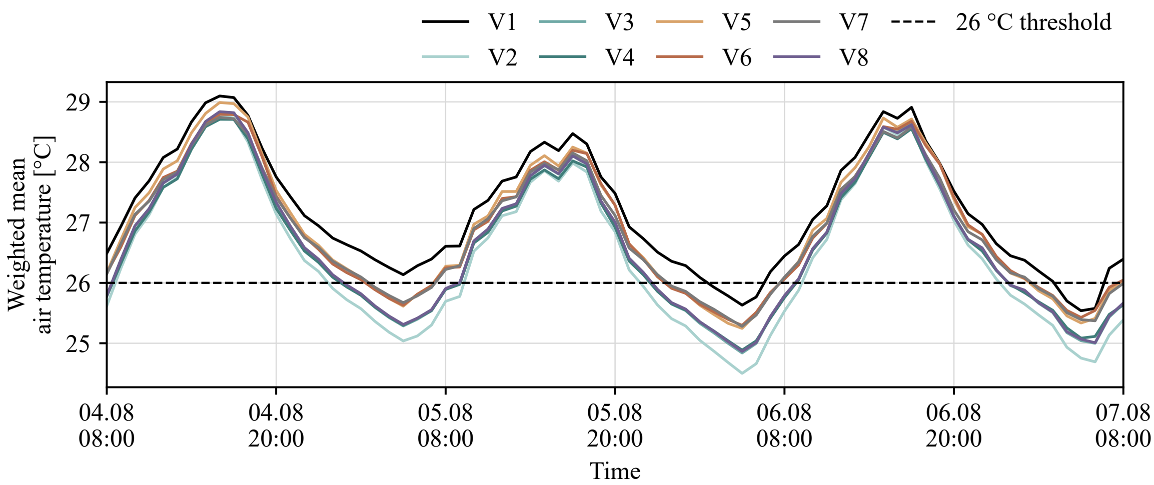


Figure 4.13: Weighted mean air temperature during a selected overheating event window for the zoning variants. The dashed horizontal line indicates the 26 °C overheating threshold.

occurs on 04.08 at 16:00, with values ranging from 28.71 °C in V4 to 29.10 °C in V1. Although the maximum temperatures are close across the variants, the duration of threshold exceedance differs more clearly.

The cooling periods show the largest differences between variants. The lowest weighted mean air temperature occurs on 06.08 at 05:00, with values ranging from 24.50 °C in V2 to 25.63 °C in V1. The event-window result shows that all variants reproduce the same overheating event pattern, but the zoning representation affects the duration of overheating and the night-time cooling behaviour.

4.3. Input parameter sensitivity analysis

4.3.1. Morris screening

The input parameter sensitivity analysis first uses the Morris screening method to identify the most influential uncertain inputs, then applies the Sobol variance-decomposition analysis.

Figure 4.14 shows the Morris screening results for annual heating demand in variant V3. The base air-change rate has the largest Morris μ^* value, at 53.7 MWh, indicating the strongest overall influence among the tested input parameters. The next most influential parameters are the year of construction shift and the mean setpoint, with μ^* values of 27.5 MWh and 25.5 MWh, respectively.

The shading factor and window g-value also show a noticeable influence, with μ^* values of 16.2 MWh and 11.1 MWh, respectively. The remaining parameters have smaller effects: gains scale contributes 6.4 MWh, the window convective heat-transfer coefficient contributes 3.8 MWh, and the window-to-wall ratio factor contributes 2.2 MWh. Setpoint spread and window U-value show the lowest influence, with μ^* values below 1 MWh.

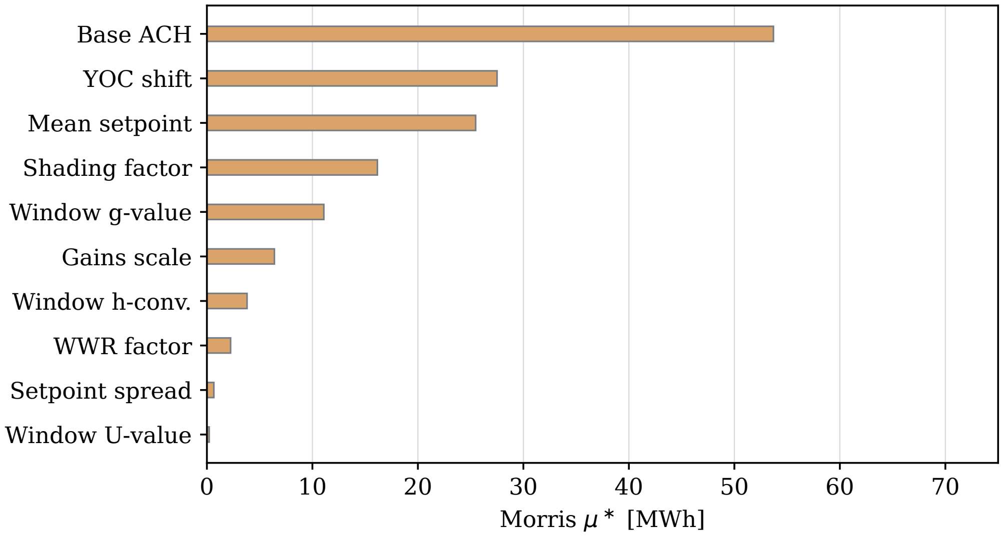


Figure 4.14: Morris screening results for annual heating demand in variant V3. Bars show the absolute elementary-effect mean μ^* for each uncertain input parameter.

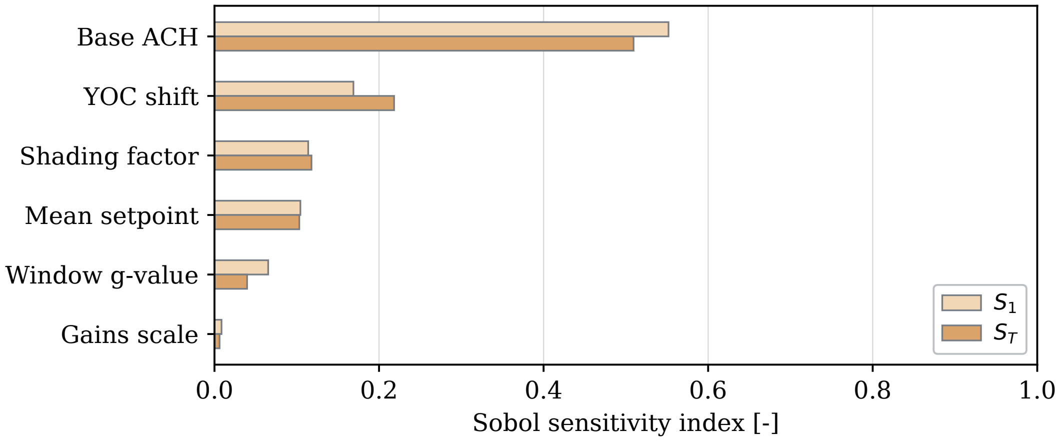
4.3.2. Sobol analysis

The Sobol analysis was used to quantify the contribution of the selected uncertain input parameters to the output variability in variant V3. For each output, the first-order index S_1 indicates the direct contribution of an input parameter, while the total-order index S_T includes both direct and interaction effects. Larger differences between S_1 and S_T therefore indicate stronger interaction effects with other uncertain inputs.

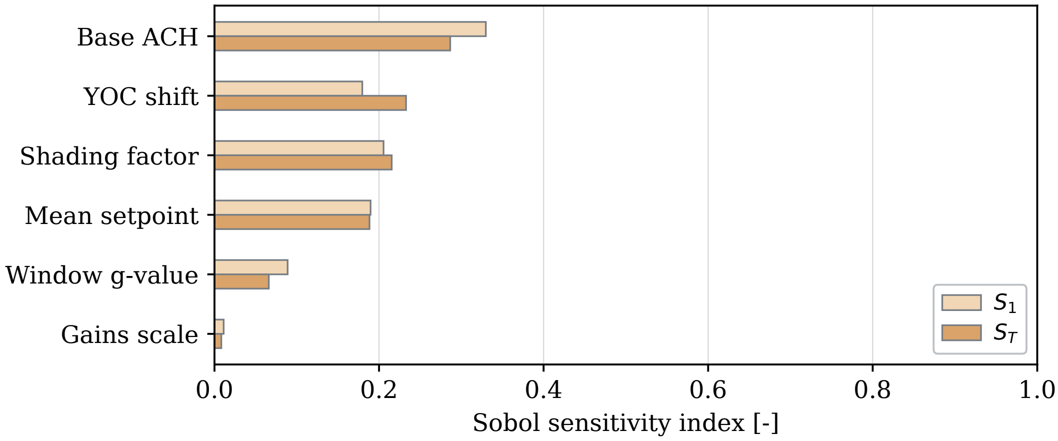
Figure 4.15 shows the Sobol sensitivity indices for annual heating demand in variant V3 under the Munich and Napoli weather cases. In both weather cases, the base air-change rate is the dominant input parameter. For Munich, the base air-change rate has the highest sensitivity, with $S_1 = 0.552$ and $S_T = 0.509$. For Napoli, it is also the most influential parameter, although with lower values of $S_1 = 0.330$ and $S_T = 0.287$.

The remaining influential parameters differ slightly between the two weather cases. Under Munich, the year of construction shift is the second most influential parameter, with $S_T = 0.218$, followed by the shading factor ($S_T = 0.118$) and mean setpoint ($S_T = 0.103$). Under Napoli, the year of construction shift also has a high total-order index ($S_T = 0.233$), but the shading factor ($S_T = 0.215$) and mean setpoint ($S_T = 0.188$) are closer in magnitude. The window g-value and gains scale show lower sensitivity in both weather cases.

Figure 4.16 shows the Sobol sensitivity indices for peak heating load in variant V3 under the Munich and Napoli weather cases. Under the Munich weather case, peak heating load is mainly influenced by the base air-change rate and the mean setpoint. The base air-change rate has the highest sensitivity, with $S_1 = 0.547$ and $S_T = 0.550$, while the mean setpoint has a similar but slightly lower influence, with $S_1 = 0.445$ and $S_T = 0.446$. The remaining parameters show very low sensitivity, with the year of construction shift reaching only $S_T = 0.008$.



(a) Munich weather case.

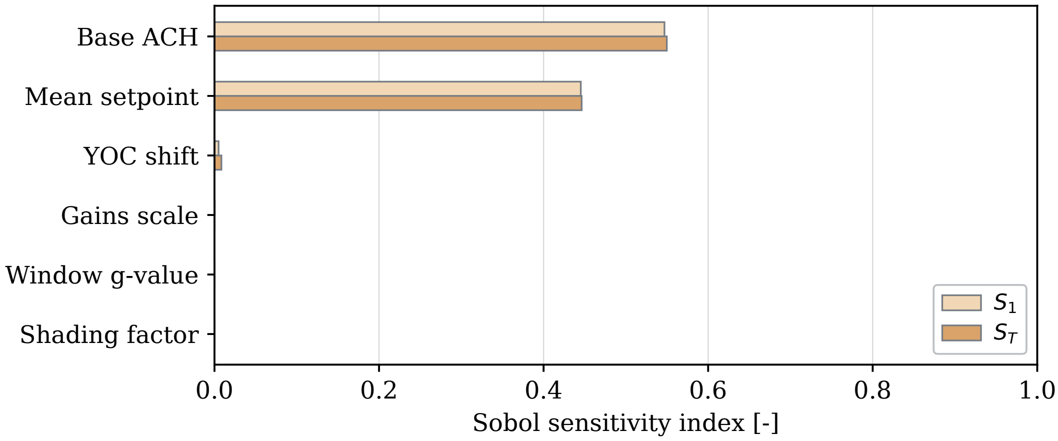


(b) Napoli weather case.

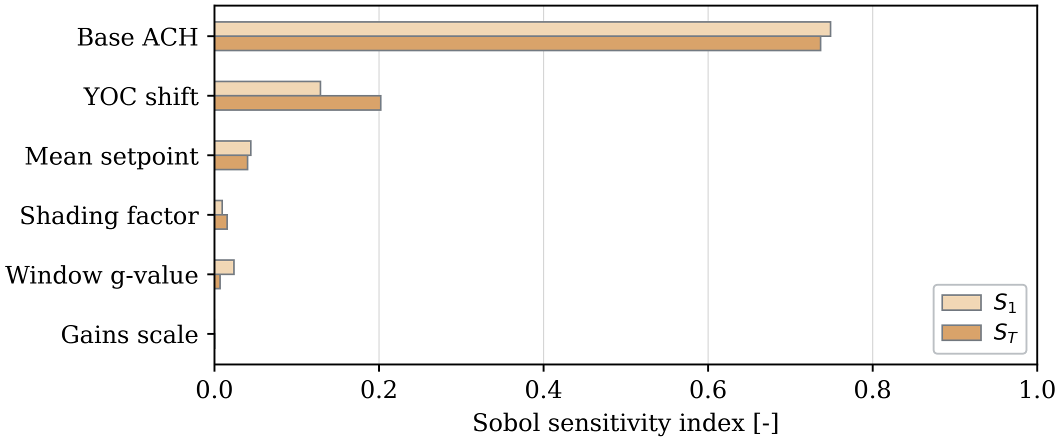
Figure 4.15: Sobol sensitivity indices for annual heating demand in variant V3 under the Munich and Napoli weather cases. Bars show the first-order index S_1 and total-order index S_T for each uncertain input parameter.

Under the Napoli weather case, the sensitivity pattern is different. The base air-change rate is clearly dominant, with $S_1 = 0.749$ and $S_T = 0.737$. The year of construction shift is the second most influential parameter, with $S_1 = 0.129$ and $S_T = 0.202$, while the mean setpoint has a smaller influence, with $S_1 = 0.044$ and $S_T = 0.040$. The remaining parameters, including the shading factor, window g-value, and gains scale, contribute only weakly to the peak heating load variation.

Figure 4.17 shows the Sobol sensitivity indices for overheating hours in variant V3 under the Munich and Napoli weather cases. In both weather cases, the shading factor is the dominant input parameter. Under the Munich weather case, the shading factor has the highest sensitivity, with both S_1 and S_T slightly above 0.5. The window g-value is the second most influential parameter, with a total-order index of approximately 0.35. The remaining parameters, including gains scale, year of construction shift, mean setpoint, and base air-change rate, show much smaller effects.



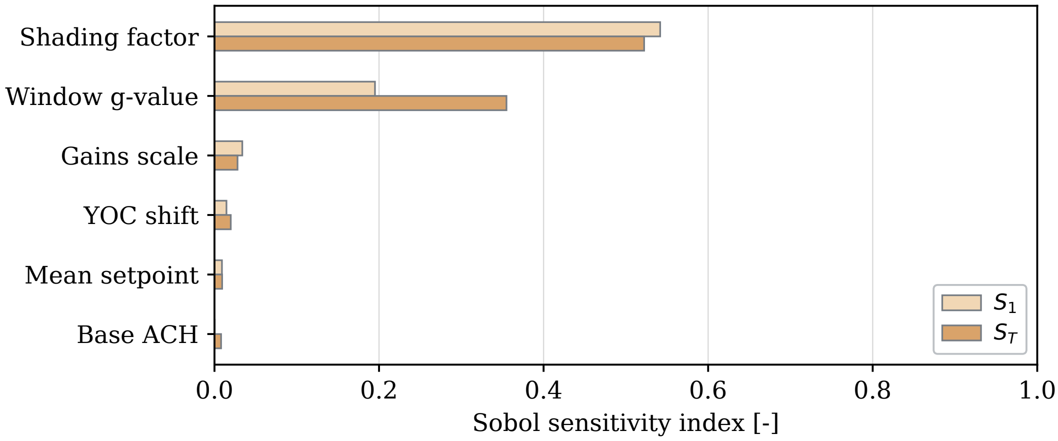
(a) Munich weather case.



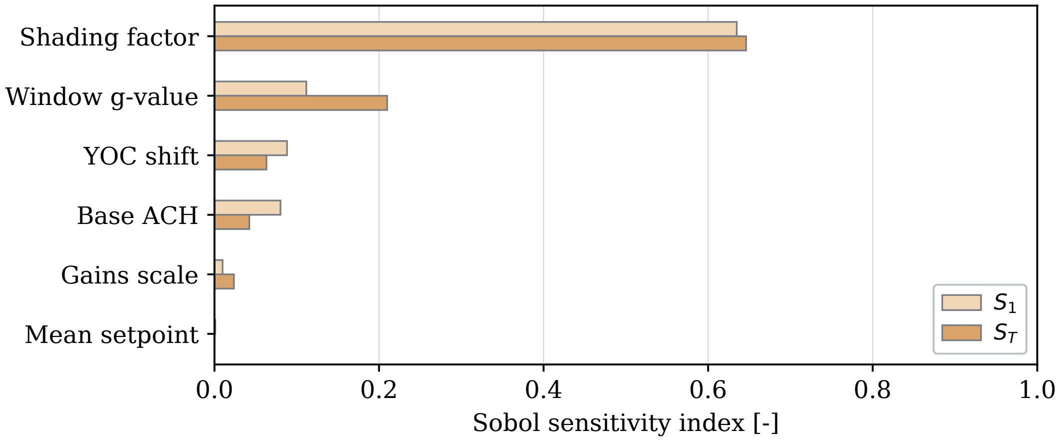
(b) Napoli weather case.

Figure 4.16: Sobol sensitivity indices for peak heating load in variant V3 under the Munich and Napoli weather cases. Bars show the first-order index S_1 and total-order index S_T for each uncertain input parameter.

For the Napoli weather case, the same overall pattern remains visible. The shading factor is again the dominant parameter, with S_1 and S_T around 0.65. The window g-value is the second most influential input, with a total-order index of approximately 0.21. The year of construction shift and the base air-change rate show smaller secondary effects, while gains scale and the mean setpoint have only minor influence.



(a) Munich weather case.



(b) Napoli weather case.

Figure 4.17: Sobol sensitivity indices for overheating hours in variant V3 under the Munich and Napoli weather cases. Bars show the first-order index S_1 and total-order index S_T for each uncertain input parameter.

5. Discussion

This chapter interprets the results of the geometric validation, zoning-deviation sensitivity analysis, and input parameter sensitivity analysis. The discussion first evaluates the reliability of the MZA at the dwelling–stairwell abstraction level. It then examines how MZA-related zoning deviations influence annual heating demand, peak heating demand, and overheating hours. Finally, it discusses the role of uncertain input parameters in the MZA-based simulation workflow before synthesising the thesis contribution, impact, limitations, and future research directions.

5.1. Geometric and topological reliability of the MZA

5.1.1. Validation performance of the MZA zoning output

The validation results indicate that the MZA can reproduce the dominant dwelling–stairwell zoning structure for a large share of the investigated multi-family residential floor plans. This is visible in the ranked IoU distributions and diagnostic classifications presented in Figures 4.1–4.6, and in the aggregated outcome shares shown in Figure 5.1. For the MSD dataset, 73.5% of the cases were classified as good geometric matches, and a further 8.0% were classified as moderate matches. For the real estate dataset, 75.0% of the cases were classified as good matches and an additional 10.0% as moderate matches. Therefore, more than four-fifths of the cases in both validation datasets achieved at least a moderate level of agreement.

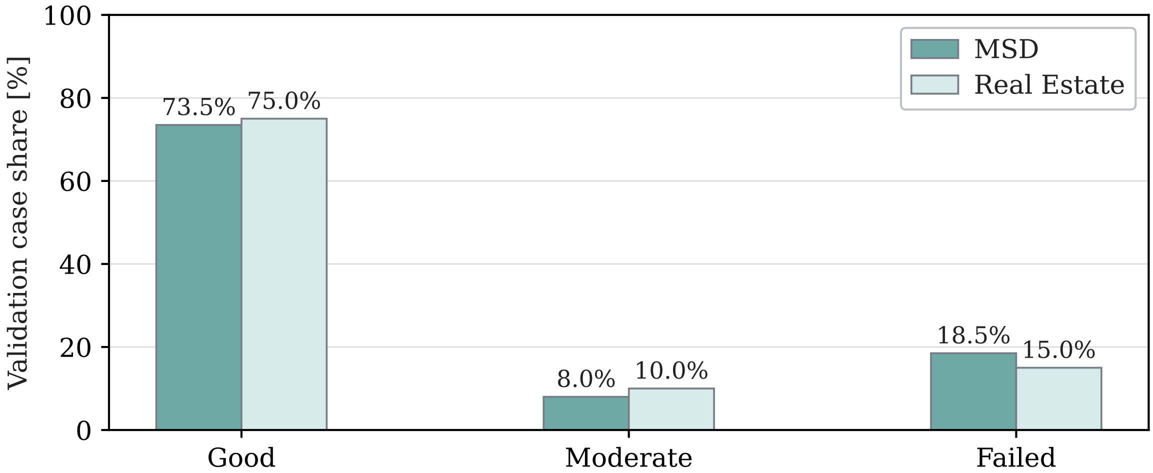


Figure 5.1: Comparison of validation outcome shares between the MSD and real estate validation datasets. The diagnostic categories are aggregated into good, moderate, and failed outcomes.

The comparable outcome pattern across the MSD and real estate datasets strengthens the interpretation that the validation results are not only a consequence of one specific dataset format or extraction procedure. This is important because the two datasets differ substantially in their origins, representations, and ground-truth extraction workflows. The MSD cases provide a larger structured Swiss reference source, whereas the real estate cases provide a smaller but more heterogeneous set of German floor plan documents. Since both datasets show a similar dominance of good and moderate outcomes, the observed MZA performance appears robust across these two forms of floor plan evidence.

This result suggests that the MZA is generally able to approximate dwelling and stairwell zoning at the intended abstraction level of the validation. It also provides a stronger basis for interpreting the validation results within a Swiss–German residential building context than either dataset would provide alone. At the same time, the remaining failed and deviation-related cases show that overall performance cannot be evaluated solely by the success rate. The diagnostic categories and representative overlays are therefore needed to understand where and why the generated zoning differs from the reference layouts. These deviations are discussed in the following subsections.

5.1.2. Interpretation of successful and moderate validation cases

In the representative good-match cases shown in Figures 4.3 and 4.7, the predicted dwelling zones generally follow the reference dwelling extents, and the predicted stairwell is located close to the reference stairwell. These cases indicate that the MZA can preserve the main spatial structure required for dwelling-level thermal zoning, including the approximate number and locations of dwelling zones and their relationship to the stairwell.

This outcome is consistent with the design of the MZA. The algorithm groups discretised footprint elements into dwelling zones using the inferred stairwell, access relations, façade contact, and area-balance constraints. Good matches are therefore expected when the reference layout has a clear dwelling–stairwell organisation and when the estimated subdivision of the floor area corresponds closely to the actual dwelling structure (Waiz et al. 2025).

The good-match cases should therefore be interpreted as successful reconstruction at the intended dwelling–stairwell abstraction level. Small local deviations along internal boundaries or stairwell edges can remain acceptable when the dominant dwelling–stairwell organisation is preserved, because the validation focuses on dwelling- and stairwell-level zoning rather than exact room-level geometry.

The moderate validation cases represent an intermediate level of agreement. In these cases, the MZA usually captures the general dwelling–stairwell structure, but with weaker geometric overlap, local boundary deviations, or less accurate zone-area distribution. Therefore, moderate cases should not be interpreted as complete failures. Instead, they show that the MZA can preserve the main thermal zone organisation even when the geometric precision of individual zone boundaries is reduced.

5.1.3. Qualitative analysis of failed cases

The failed validation cases provide additional insight into the conditions under which the MZA output becomes less reliable. Although the validation performance is positive, the representative failed cases shown in Figures 4.4 and 4.8 indicate that the deviations are not random. They are mainly related to structural mismatch, under-zoning, over-zoning, and cases where the predicted zones are shifted or have high area error.

The failed cases can be interpreted through two main mechanisms. The first mechanism concerns the structural assumptions made before and during zone generation, particularly the inferred location of the stairwell. Since the MZA uses the stairwell location as a starting condition for generating dwelling zones, a mismatch between the assumed and actual stairwell organisation can change the resulting dwelling–stairwell topology. These cases are more critical than local boundary deviations because they affect the spatial relationship between dwellings and the stairwell.

The second mechanism concerns subdivision and boundary allocation. In many cases, the deviation can be explained by a combination of mean-area estimation and area-allocation effects. Since the MZA derives the expected dwelling subdivision from estimated auxiliary information, a mismatch between this estimate and the actual reference layout can produce too few or too many dwelling zones, resulting in under-zoned or over-zoned cases. Such estimation uncertainty is common in urban-scale residential modelling, where household areas and counts are often estimated rather than directly observed (Köhler et al. 2021).

Even when the number of generated zones is broadly correct, the MZA may still deviate from the reference layout because it favours an area-balanced subdivision based on the specified average dwelling area and size constraints. In real multi-dwelling floor plans, however, dwelling units can vary substantially in area within the same building (Massafra et al. 2025). Therefore, the generated subdivision may preserve the correct number of dwellings but still differ from the actual dwelling-size distribution, leading to shifted internal boundaries or high area error.

The failed cases show that the reliability of the MZA is mainly conditional on three factors: the external footprint geometry, the inferred stairwell representation, and the estimated dwelling subdivision. Therefore, the failed cases do not only represent unsuccessful outputs; they also identify the main boundary conditions under which MZA-based dwelling–stairwell zoning should be interpreted with caution.

5.1.4. Reliability of the MZA for dwelling–stairwell zoning

Taken together, the validation results indicate that the MZA is generally reliable for reproducing dwelling- and stairwell-based zoning at the intended dwelling–stairwell abstraction level. In both validation datasets, the majority of cases were classified as good geometric matches.

However, the reliability of the MZA should be interpreted as conditional rather than universal. The successful and moderate cases show that the method performs well when the external footprint, stairwell representation, and estimated dwelling subdivision correspond closely to the actual floor plan structure. In such cases, remaining deviations are mostly local boundary differences and do not fundamentally change the dwelling–stairwell organisation. The failed cases, however, show that the output becomes less reliable when these assumptions do not adequately represent the reference layout, particularly when the predefined stairwell location or the estimated dwelling subdivision differs from the actual floor plan.

Therefore, the MZA can be considered reliable as an automated dwelling–stairwell zoning approximation for many multi-family residential buildings, but not as a full reconstruction of the actual architectural floor plan. This interpretation is consistent with the role of thermal zoning as a modelling abstraction, in which zones represent heat balance and control units rather than exact architectural room or wall layouts (Shin and Haberl 2019; Johari et al. 2022). It also reflects the intended purpose of the MZA, which is to generate dwelling- and stairwell-based thermal zone layouts from limited geometric input for simulation use, rather than to reconstruct complete internal floor plan geometry.

5.2. Sensitivity analysis of zoning deviations

The validation results identify the main geometric and topological conditions under which MZA-related zoning deviations can occur. The following sensitivity analysis examines the influence of these deviation mechanisms on the simulation using controlled zoning variants.

5.2.1. Annual heating demand

As shown in Figure 4.9, the dominant zoning-related effect for annual heating demand is the transition from the fully aggregated V1 model to explicit multi-zone representations. V1 consistently produces the lowest median annual heating demand, but this should not be interpreted as higher accuracy or better model performance. The difference should also be read in relation to the TEASER–AixLib model formulation, where the multi-zone variants include explicit internal separating surfaces and interzonal heat-transfer connections, whereas these effects are absent or aggregated in the single-zone model (**groesdonk2023**).

At the same time, aggregation smooths local effects such as dwelling-level temperature differences, internal gains, heat losses, setpoint differences, and interzonal heat exchange within one heat-balance unit. This is also reflected in the uncertainty bands: the separation between V1 and the multi-zone variants is clearer than most differences within the multi-zone group, where the P5–P95 ranges often overlap. Therefore, annual heating demand is more sensitive to the change from aggregated to multi-zone modelling than to most layout differences within the multi-zone variants (Johari et al. 2022; Kühn et al. 2024).

Within the multi-zone group, the effect of zoning granularity is not linear. The difference between the coarser V2 variant and V3 remains small across the weather and construction states, ranging from about -5.1% to +1.8% relative to V3. In contrast, the finer V5 variant produces lower median annual heating demand than V3 in all cases, with reductions of about 5% in the standard state and up to approximately 13–17% in the retrofit state. This shows that increased zoning granularity does not automatically increase annual heating demand. The effect depends on how the additional zones redistribute floor area, façade exposure, internal gains, adjacency to the stairwell, and interzonal heat exchange. Therefore, the V3–V5 comparison should be interpreted as a configuration-dependent over-zoning effect rather than as a direct rule that more zones always increase or decrease demand.

The layout comparisons indicate that shifted-zone, high-area-error, and moderate stairwell-position effects have a smaller influence on annual heating demand. The difference between

V3 and V4 remains below about 2.5%, while the difference between V5 and V6 remains around 1.6% or less. Since these differences are small and fall within partially overlapping uncertainty ranges, they suggest that layout and boundary-placement deviations are partially averaged out in annual heating demand when the overall zoning structure remains similar.

In addition, the stairwell-related variants indicate that the stairwell's thermal representation becomes relevant primarily when its thermal status changes. Omitting the separate unheated stairwell representation in V7 has only a small effect compared with V3, mostly around 1–2%. In contrast, treating the stairwell as heated in V8 increases annual heating demand, with the relative difference reaching about 4–5% in the retrofit state. This indicates that the stairwell is less influential when it remains unheated or omitted as a separate zone, but becomes more relevant when it is treated as part of the heated volume.

This behaviour is expected because annual heating demand is an aggregated energy-balance indicator over the full simulation year. When the same external geometry, construction state, weather case, and overall residential use are retained, many local zoning differences are partly averaged over time and across the building. Therefore, larger deviations are mainly expected when the zoning change alters the effective thermal structure of the model, such as the distinction between single-zone and multi-zone representation or the treatment of the stairwell.

This behaviour is expected because annual heating demand is an aggregated energy-balance indicator over the full simulation year. When the same external geometry, construction state, weather case, and overall residential use are retained, many local zoning differences are partly averaged over time and across the building. Therefore, the annual heating demand results show that MZA-related zoning deviations are most relevant when they alter the effective thermal structure of the model, such as the distinction between single-zone and multi-zone representation or the treatment of the stairwell.

Within the multi-zone group, reduced granularity has only a small effect, increased subdivision produces a clearer but configuration-dependent change, and layout-related differences remain limited when the uncertainty ranges overlap. Retrofit lowers the absolute demand level, but it makes the remaining variability more important in relative terms because the demand is reduced more strongly than the uncertainty range.

5.2.2. Peak heating demand

In Figure 4.10, the dominant zoning-related effect for peak heating demand is again the transition from the fully aggregated V1 model to explicit multi-zone representations. V1 yields the lowest median peak load across all weather and construction states, whereas the multi-zone variants produce higher values. This indicates that aggregation and strong under-zoning can smooth short-term and zone-specific load peaks. Therefore, the lower peak load of V1 should not be interpreted as evidence of better model performance, especially since peak heating demand is relevant to heating-system sizing.

The event-window analysis in Figure 4.11 supports this interpretation from a temporal perspective. The peak occurs at approximately the same time across variants, indicating that the event is primarily weather-driven. However, the magnitude of the heating-load response differs between the zoning variants. The aggregated V1 model produces the smoothest and lowest load profile, while the multi-zone variants show higher short-term peaks. This means that MZA-related zoning deviations mainly affect the peak load magnitude, rather than creating a different peak heating event.

Among variants with similar zoning resolution, the differences are smaller and the P5–P95 uncertainty ranges overlap strongly. The coarser V2 variant remains close to V3, suggesting that moderate reduction in zoning granularity has only a limited effect on peak heating demand when the main dwelling–stairwell structure remains comparable. In contrast, the finer V5 and V6 variants generally produce higher peak loads than V3 and V4, especially under the cold-weather case. This suggests that finer subdivision can make local peak-load contributions more visible, because zones can respond more independently to setpoints, heat losses, interzonal heat exchange, and stochastic internal gains.

The layout comparisons indicate that shifted-zone, high-area-error, and moderate stairwell-position effects are less important for peak heating demand than the broader change in zoning resolution. The differences between V3 and V4, and between V5 and V6, remain limited compared with the difference between V1 and the multi-zone variants. Since these layout-related differences are small and partly covered by overlapping uncertainty bands, the exact placement of internal boundaries appears less influential for peak heating demand than the representation of multiple thermal zones itself.

The stairwell-related variants show that the stairwell's thermal representation can still affect peak-load estimation. V7, where the stairwell is not represented as a separate unheated zone, and V8, where the stairwell is treated as heated, change the peak-load response relative to V3. This indicates that the stairwell is relevant to peak heating demand because it can either serve as an unheated buffer zone or be part of the heated volume. The peak-load results therefore indicate that peak heating demand is more sensitive to zoning aggregation, finer subdivision, and stairwell representation than to small layout shifts. However, the overlapping P5–P95 ranges indicate that some zoning-deviation effects are comparable in magnitude to the propagated input uncertainty.

5.2.3. Overheating hours

Figure 4.12 shows that the overheating hours respond differently to zoning deviations than the heating-related indicators. While annual and peak heating demand are mainly driven by the transition from V1 to the multi-zone variants, overheating hours depend more strongly on the weather case, construction state, and the representation of local thermal zones. The hot-weather case produces substantially higher overheating hours than the cold-weather case, and the retrofit state increases overheating hours even though it reduces heating demand and peak heating load. This indicates a target conflict: a retrofit improves heating performance but can increase the risk of overheating when heat losses are reduced.

The aggregated V1 variant produces the highest or near-highest overheating hours, especially under the hot-weather case. This indicates that aggregation can distort the assessment of overheating, but in a different way than it does for heating demand. In V1, the building is represented as one thermal zone, so cooler and warmer internal areas cannot be distinguished. Local differences among dwellings, the stairwell, and differently exposed zones are averaged into a single temperature condition. Therefore, V1 should not be interpreted as a reliable representation of local overheating risk, even if it provides a simple building-level indicator.

At the multi-zone level, overheating hours are more sensitive to spatial zoning effects than annual heating demand. Reduced granularity, finer subdivision, and alternative layouts influence the distribution of internal gains, façade exposure, heat storage, and interzonal heat exchange among zones. Therefore, layout-related differences such as V3–V4 and V5–V6 are more relevant to overheating than to annual heating demand, because overheating is a threshold-based indicator that depends on local temperature exceedance rather than solely on the total building heat balance.

The zone-level event window in Figure 5.2 supports this interpretation. The selected variants follow the same external weather-driven daily pattern, but the individual zone temperatures differ in both peak magnitude and duration above the 26 °C threshold. This is especially relevant for finer zoning variants, where individual zones can show stronger local overheating than the building-level average would suggest (see Figure 4.13). Therefore, over-zoning or finer subdivision does not only change the number of thermal zones; it can also reveal local overheating behaviour that may be hidden when zones are aggregated.

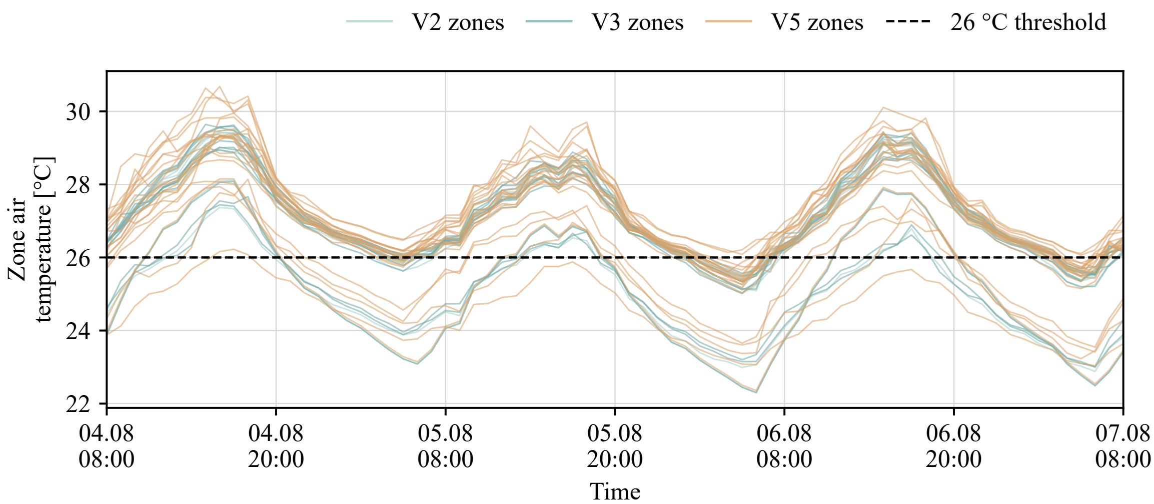


Figure 5.2: Zone-level air temperatures during a selected overheating event window for selected multi-zone variants. The dashed horizontal line indicates the 26 °C overheating threshold.

The P5–P95 uncertainty ranges are important for interpreting these differences. Under the cold-weather case, the absolute overheating hours are low, and some variant differences are small relative to the uncertainty range. In the hot-weather case, especially in the retrofit state, overheating levels are much higher, so zoning-related differences become more relevant to interpreting thermal risk. This means that MZA-related zoning deviations are most important for overheating when the building is already close to, or frequently above, the overheating threshold.

The overheating response of the finer V5 and V6 variants should also be read in relation to the occupancy-profile assignment. These variants use more household profiles than the coarser multi-zone variants, which can increase and redistribute internal gains across the building, thereby affecting the duration of threshold exceedance.

The overheating-hour results show that MZA-related zoning deviations are more relevant for comfort-related threshold indicators than for annual heating demand. Aggregation can hide local thermal differences, while finer subdivision and alternative layouts can change the duration and magnitude of threshold exceedance. This supports the interpretation that overheating assessment requires more careful attention to zoning structure than aggregated heating-energy indicators.

The sensitivity analysis shows that MZA-related zoning deviations affect simulation outputs, but their importance depends on the type of deviation and the performance indicator evaluated. Aggregation and strong reductions in zoning detail primarily influence the results by smoothing zone-specific temperature differences and heating-load peaks, which can lead to lower annual and peak heating demand compared with explicit multi-zone representations. Changes in stairwell representation are relevant because the stairwell can either act as an unheated buffer zone or become part of the heated volume, thereby affecting both heat losses and heating demand. In contrast, smaller layout shifts and boundary-placement differences have a more limited effect on annual heating demand, especially when the uncertainty ranges overlap. For overheating assessment, however, local zoning differences become more important because threshold exceedance depends on the temperature behaviour of individual zones rather than only on the building-level average. Therefore, zoning deviations are most relevant to simulation when they change heat-loss pathways, heated volume, interzonal heat exchange, short-term peak-load behaviour, or local temperature exceedance.

5.3. Influence of uncertain input parameters

The Morris screening was used as a first step to reduce the wider set of uncertain inputs to the most relevant parameters for further analysis. Figure 4.14 shows that the screening identified six dominant inputs for the subsequent Sobol analysis: base air-change rate, year of construction shift, mean heating setpoint, shading factor, window g-value, and internal-gain scaling factor. This result shows that the most relevant uncertainties are not limited to one input category. Instead, they cover ventilation, typological enrichment, operational assumptions, solar gains, and internal loads. For an MZA-based simulation workflow, this means that reliable output prediction depends not only on geometric model preparation, but also on the quality of the thermal and operational enrichment data assigned to the generated model.

The Sobol results for annual heating demand confirm that ventilation-related uncertainty is the dominant input category. As shown in Figure 4.15, the base air-change rate has the largest influence in both weather cases. In the cold-weather case, its first-order and total-order indices are both around 0.5, indicating that approximately half of the output variance is attributable to this parameter. In the hot-weather case, the influence is lower but still dominant, with indices around 0.3. This shows that uncertainty in infiltration or ventilation assumptions can strongly affect predicted annual heating demand.

However, the numerical sensitivity values are also conditioned by the selected parameter ranges, because a wider sampled range can increase the variance attributed to a parameter, while a narrower range can reduce its apparent influence. Therefore, the dominance of the base air-change rate reflects both its physical relevance for heat loss and the uncertainty range assigned to it in the analysis. Construction-period assumptions and the mean heating setpoint form a second group of influential inputs, which is plausible because annual heating demand is governed not only by ventilation losses, but also by envelope properties and the indoor temperature level maintained by the heating system.

A similar heat-loss-related pattern appears for peak heating demand, but with a stronger focus on short-term heating response. In Figure 4.16, the base air-change rate remains the most important input. In the hot-weather case, it accounts for roughly three-quarters of the output variance when first-order and total-order effects are considered. This indicates that peak-load estimation is highly sensitive to ventilation-related assumptions. The year of construction shift exerts a secondary influence, with total-order effects around 0.2 in the hot-weather case, while the mean heating setpoint contributes less strongly. In the cold-weather case, the base air-change rate and heating setpoint are also important, as they directly affect the heating power required during cold periods. Therefore, for heating-system sizing or peak-load analysis, ventilation, construction-period enrichment, and setpoint assumptions should be treated with particular care.

The sensitivity pattern clearly changes during overheating hours. Figure 4.17 indicates that overheating is controlled mainly by solar-gain-related parameters rather than by the inputs that dominate the heating indicators. The shading factor is the most influential parameter in both weather cases. In the cold-weather case, its first-order and total-order indices are slightly above 0.5. At the same time, the window g-value is the second most important input, with a total-order contribution of approximately 0.35. In the hot-weather case, the shading factor becomes even more dominant, with indices around 0.65. This shows that overheating hours are mainly controlled by solar gains and window-related assumptions, whereas the base air-change rate and mean heating setpoint are much less relevant for this output.

The sensitivity pattern clearly changes for overheating hours. Figure 4.17 indicates that overheating is controlled mainly by solar-gain-related parameters rather than by the inputs that dominate the heating indicators. In the cold-weather case, the shading factor has first-order and total-order indices slightly above 0.5, while the window g-value has a total-order contribution of approximately 0.35. In the hot-weather case, the shading factor becomes even more dominant, with indices around 0.65, while the window g-value remains the second most influential parameter with a total-order index of approximately 0.21. In contrast, the base air-change rate and mean heating setpoint have much smaller sensitivity indices in both weather cases, showing that overheating-hour variability is mainly associated with solar gains and window-related assumptions.

The comparison between first-order and total-order Sobol indices provides additional information about interaction effects. When the total-order index is clearly larger than the first-order index, part of the parameter influence occurs through interactions with other inputs. This is visible, for example, for the year of construction shift in the peak heating-load analysis, where the total-order effect is larger than the direct first-order effect. During overheating hours, the dominance of the shading factor and the window g-value suggests that solar-control assumptions interact with the building's thermal response. Therefore, overheating risk should not be interpreted as the consequence of a single parameter, but as the combined effect of solar gains, window properties, internal gains, weather conditions, and thermal storage.

Taken together, the input-sensitivity results show that the priority of missing or uncertain input information depends strongly on the simulation output under evaluation. Ventilation, construction-period assumptions, and heating setpoint mainly influence heating-related indicators. In contrast, overheating is mainly influenced by shading and the solar-gain properties of windows.

5.4. Contribution to automated thermal zoning evaluation

This thesis establishes a combined geometric and simulation-oriented evaluation framework for MZA-generated dwelling–stairwell zoning in multi-family residential buildings. The contribution is not limited to applying the MZA; it also evaluates whether the generated dwelling and stairwell zones are geometrically reliable, how deviations in these zones affect simulation outputs, and which uncertain input parameters are most important in the resulting simulation workflow.

A central part of this contribution is the construction of dwelling–stairwell ground truth for validating automated zoning outputs. The thesis develops a methodology for deriving reference dwelling and stairwell zones from both structured floor plan information and real estate floor plan documents. This is important because automated zoning methods require reference layouts for evaluation, but such ground-truth data are rarely directly available in urban building energy modelling workflows. The resulting validation basis can also support the future evaluation of other automated thermal zoning approaches.

Building on this reference basis, the thesis provides a systematic geometric and topological validation of the MZA. The validation shows that the MZA can reproduce the dominant dwelling–stairwell structure for more than four-fifths of the investigated cases across both validation datasets. At the same time, the diagnostic interpretation identifies the main conditions under which deviations occur, such as stairwell-location assumptions, estimated dwelling subdivision, under-zoning, over-zoning, shifted zones, and high area error. This provides a more detailed understanding of MZA reliability than a success rate alone.

The thesis also links geometric zoning deviations to simulation relevance. Controlled zoning variants represent MZA-related deviation mechanisms such as aggregation, changes in zoning granularity, alternative layout arrangements, and stairwell representation. The results show that these deviations do not influence all simulation outputs in the same way: aggregation and reduced zoning detail mainly affect heating-related indicators, while local zoning differences become especially relevant for overheating because threshold exceedance depends on individual zone temperatures.

Finally, the input-sensitivity analysis shows that reliable MZA-based simulation also depends on the quality of uncertain enrichment inputs. Heating-related indicators are mainly influenced by ventilation, construction-period assumptions, and heating setpoint, whereas overheating is mainly influenced by shading and window solar-gain properties. Together, these contributions provide a framework for assessing when MZA-generated zoning is geometrically reliable and when its deviations become relevant for simulation-based assessment.

5.5. Contribution to automated thermal zoning evaluation

The contribution of this thesis lies in framing automated thermal zoning as both a geometric and a simulation-oriented modelling problem. MZA-generated dwelling–stairwell zones are not treated only as spatial outputs, but also as assumptions that influence the thermal representation of multi-family residential buildings. This perspective extends the evaluation of automated zoning from the question of whether zones can be generated to the question of whether the generated zones are sufficiently reliable for simulation-based assessment.

A central element of this contribution is the development of a validation basis for dwelling–stairwell zoning. The thesis demonstrates how reference zoning structures can be derived from different types of floor plan information, including structured floor plan data and real estate floor plan documents. This addresses a practical limitation in urban building energy modelling, where detailed internal layouts and ready-made thermal zoning references are usually unavailable. The resulting ground-truth construction workflow provides a transferable basis for evaluating automated zoning methods beyond the specific cases analysed in this thesis.

The validation approach also contributes a diagnostic interpretation of automated zoning quality. Instead of reducing the evaluation to a single agreement score, the thesis combines geometric, topological, and structural indicators to characterise the type and severity of zoning deviations. This makes it possible to distinguish between acceptable geometric approximation, local deviations, and more fundamental structural mismatches. The evaluation therefore provides a more informative basis for interpreting MZA performance than a purely numerical success criterion.

The simulation-oriented part of the thesis shows that geometric deviations should not be interpreted independently from their thermal consequences. The relevance of a zoning deviation depends on whether it changes the effective thermal structure of the model, such as the level of aggregation, the representation of the stairwell, interzonal relationships, or the zone-level behaviour that determines threshold-based comfort indicators. This provides a more nuanced basis for deciding when automated zoning deviations are likely to matter for building-performance simulation.

The thesis further shows that automated zoning evaluation must include the uncertain enrichment inputs required to create simulation-ready models. Even when the zoning geometry is plausible, the resulting simulation outputs can still be strongly affected by assumptions that define the thermal and operational behaviour of the model. The contribution is therefore not only a validation of geometric zoning quality, but also an evaluation of the wider MZA-based simulation workflow.

Overall, the thesis contributes a framework for assessing automated dwelling–stairwell zoning as a simulation-relevant modelling decision. It connects reference-data construction, diagnostic geometric validation, zoning-deviation analysis, and input-parameter sensitivity into a single evaluation logic. This framework can support future work on automated thermal zoning by clarifying when generated zoning structures are geometrically reliable, when deviations are simulation-relevant, and which additional input uncertainties need to be considered.

5.6. Limitations

This thesis has several limitations that should be considered when interpreting the results. The first limitation concerns the scope and regional coverage of the validation data. The validation is intended to examine the behaviour of the MZA in a DACH-oriented residential building context. However, the available validation data cover only two national contexts within this region: Switzerland, through the Modified Swiss Dwellings dataset, and Germany, through real estate floor plan documents. Austrian floor plan data are not included. The validation results should therefore be interpreted as evidence for a Swiss–German subset of the DACH region rather than as a complete validation across the entire DACH building stock.

A second limitation concerns the construction of the real estate reference data. The real estate floor plans were obtained from heterogeneous PDF or image-based sources and required manual digitisation, scaling, classification, and correction. Although this procedure made it possible to create reference dwelling–stairwell geometries from otherwise unavailable data, it also introduces potential interpretation and vectorisation uncertainties.

The use of controlled zoning variants limits the scope of the sensitivity analysis. These variants represent MZA-related deviation mechanisms such as aggregation, reduced and increased zoning granularity, alternative layout arrangements, and stairwell representation. However, they do not reproduce each failed validation case directly. Therefore, the sensitivity results should be interpreted as an analysis of representative zoning-deviation mechanisms rather than as a direct simulation of every observed validation error.

Another limitation concerns the scope of the simulation outputs evaluated. The analysis focuses on annual heating demand, peak heating demand, and overheating hours. These indicators are relevant for energy performance and thermal comfort, but they do not cover all possible consequences of zoning deviations. Cooling demand, emissions, detailed HVAC operation, room-level comfort metrics, and system-control effects were not evaluated.

The input-parameter sensitivity analysis is also limited by the selection of uncertain parameters and by the ranges assigned to them. The Morris and Sobol analyses identify influential inputs only within the parameter set and uncertainty ranges defined for this study. Parameters that were not included cannot be assessed, and parameters with wider or more sensitive sampled ranges may appear more influential than parameters with narrower ranges. The resulting sensitivity ranking should therefore be interpreted as conditional on the assumptions used in the analysis.

A further limitation is that the simulation results are not validated against measured building-performance data. Without measured reference data, the analysis quantifies differences between zoning representations and identifies sensitivity patterns, rather than estimating absolute prediction errors. The results therefore show how MZA-related zoning assumptions influence simulation outputs within the tested modelling framework, but they do not indicate how far each variant would deviate from a building's actual performance. The transferability of these findings to other residential building types, construction systems, climates, and modelling contexts remains an additional limitation.

5.7. Future work

Future work can extend the validation to a broader and more diverse residential building dataset. A larger reference sample would make it possible to test whether the observed MZA performance and deviation mechanisms remain consistent across different building typologies, access configurations, footprint shapes, and stairwell arrangements.

The validation framework could also be extended toward room-level reference data. Such an extension would allow the evaluation of more detailed thermal zoning methods, layout reconstruction workflows, and other zoning algorithms. This would require the extraction of room boundaries, internal doors, corridors, and adjacency relations from floor plans, but would provide a richer basis for comparing different levels of zoning detail.

More automated extraction of zoning reference data from floor plan sources is another important direction. Floor plan parsing, semantic segmentation, and vectorisation methods could be used to detect dwelling areas, stairwells, circulation spaces, and relevant boundaries with less manual correction (Kalervo et al. 2019; Pizarro et al. 2023; Wu et al. 2019). This would support larger and more consistent ground-truth datasets for validating automated thermal zoning methods.

The failed validation cases could be used to refine the assumptions that guide MZA zone generation and thermal enrichment. By analysing stairwell positions across a larger set of floor plans, recurring relationships between footprint shape, entrance location, façade orientation, plot access, and stairwell placement could be identified. These patterns, together with richer contextual and thermal input data, could support more reliable stairwell placement, dwelling subdivision, and enrichment assumptions without changing the basic MZA zoning logic.

The simulation-based analysis could be extended to additional outputs and validation settings. Future studies should test whether the observed zoning effects remain consistent across other residential buildings, HVAC assumptions, occupancy profiles, weather files, and future climate scenarios. Where measured data are available, simulated heating demand, peak loads, and indoor temperatures should be compared with monitored building performance. The scope of performance indicators could also be expanded to include cooling demand, emissions, system operation, and more detailed thermal-comfort metrics.

5.8. Impact

The impact of this thesis lies in improving the reliability and interpretability of automated building-performance simulation workflows. By validating MZA-generated dwelling–stairwell zoning and analysing how zoning deviations and uncertain input parameters influence simulation outputs, the thesis provides evidence that can support better modelling decisions in urban building energy modelling.

Multi-family residential buildings are directly connected to thermal comfort, heating costs, and overheating risk. If automated zoning methods produce misleading thermal zone representations, simulations may underestimate or overestimate heating demand, peak heating loads, or overheating hours. By showing when MZA-generated zoning is reliable and when deviations are relevant, this thesis supports better interpretation of simulation results used for renovation planning, comfort assessment, and overheating risk evaluation. This is especially important for occupants because poor modelling assumptions can lead to retrofit decisions that reduce heating demand but unintentionally increase the risk of overheating.

The thesis supports more robust analysis of residential building stocks by showing that zoning representation and uncertain enrichment assumptions can affect outputs used in urban building energy models. Since these models can improve the credibility of simulation-based retrofit planning and decarbonisation strategies. The findings also show that overheating should be considered alongside heating reduction, because energy-efficient retrofits can change the thermal behaviour of residential buildings under warm conditions.

The thesis supports more robust analysis of residential building stocks by showing that zoning representation and uncertain enrichment assumptions can affect outputs used in urban building energy models. This improves the interpretation of simulation-based retrofit planning and decarbonisation strategies, especially where limited building data are used. The findings also show that overheating should be considered alongside heating reduction, because energy-efficient retrofits can change the thermal behaviour of residential buildings under warm conditions.

Municipalities, planners, housing companies, and energy consultants increasingly rely on automated modelling workflows because detailed building data are often unavailable at urban scale. The validation and sensitivity methods developed in this thesis help identify which geometric assumptions, zoning deviations, and input parameters require the most attention. This can reduce uncertainty in large-scale energy assessments and help decision-makers prioritise where better data are needed, for example in dwelling-size information, stairwell representation, ventilation assumptions, window properties, or shading. Economically, this can support a more efficient allocation of modelling effort and renovation investment by focusing resources on assumptions that most strongly influence simulation outputs.

Taken together, the thesis not only improves understanding of the MZA as an automated zoning method. It also contributes to the broader goal of making urban building energy modelling more transparent, reliable, and useful for residential retrofit planning, climate adaptation, and evidence-based policy support.

6. Conclusion

This thesis evaluated the suitability of the Multizone Assignment Algorithm (MZA) for generating simulation-relevant dwelling- and stairwell-based thermal zoning structures for multi-family residential buildings from limited geometric input. The point of departure was the challenge that urban building energy models often rely on LoD2-based external geometry and typological enrichment. At the same time, detailed internal layouts are usually unavailable at district or city scale. Since thermal zoning affects heat balances, interzonal heat transfer, zone temperatures, internal gains, and control assumptions, the absence of this internal structure can compromise the reliability of building energy simulation results.

The literature review showed that automated zoning methods are needed to balance modelling detail and scalability. The MZA provides a promising approach because it aims to generate dwelling- and stairwell-based zoning for residential buildings rather than only generic single-zone or perimeter–core representations. However, three gaps were identified: the generated zoning geometry had not yet been systematically validated against independent reference dwelling–stairwell structures; the simulation relevance of MZA-related zoning deviations was unclear; and the influence of uncertain enrichment inputs in an MZA-based workflow still needed to be assessed.

To address these gaps, this thesis combined geometric validation with simulation-based sensitivity analysis. Reference dwelling–stairwell ground truth was constructed from the Modified Swiss Dwellings dataset and German real estate floor plan documents. The extracted ground-truth geometries were converted into simplified external footprints, inserted into CityGML template buildings, and processed through the MZA workflow. The generated zones were then compared with the reference layouts using geometric, topological, and diagnostic metrics.

The validation results showed that the MZA can reproduce the dominant dwelling–stairwell structure for a large share of the investigated cases. In both validation datasets, more than four-fifths of the cases achieved at least moderate agreement. This indicates that the MZA can provide useful dwelling- and stairwell-level zoning approximations when the external footprint, stairwell representation, and estimated dwelling subdivision are compatible with the actual floor plan structure. At the same time, the failed cases showed that the method is not universally reliable. Deviations mainly resulted from structural mismatches, under-zoning, over-zoning, shifted zones, high area errors, and differences in stairwell locations.

The sensitivity analysis showed that these zoning deviations are relevant, but not equally for all simulation outputs. Annual heating demand was primarily driven by the difference between the fully aggregated single-zone model and the multi-zone representations. In contrast, smaller layout shifts within the multi-zone group had a more limited effect. Peak heating demand was more sensitive to zoning aggregation and finer subdivision because aggregated models can smooth zone-specific heating-load peaks. Overheating hours were especially sensitive to local zoning behaviour because threshold exceedance depends on individual zone temperatures rather than only on a building-level average.

The input parameter sensitivity analysis showed that reliable MZA-based simulation depends not only on the generated zoning geometry, but also on the quality of the enrichment inputs. Base air-change rate, construction-age assumptions, and heating setpoint mainly influenced heating-related outputs. In contrast, overheating hours were mainly controlled by shading factor and window g-value. This means that the most important missing data depend on the simulation question: heating demand studies require careful treatment of ventilation and construction-period assumptions, while overheating studies require more reliable information on solar gains, shading, and window properties.

The main contribution of this thesis is the development of a combined evaluation logic for automated thermal zoning. The results show that an automated zoning output should be assessed in two steps: first, whether the generated geometry represents the intended dwelling–stairwell structure; and second, whether the remaining deviations are relevant for the simulation outputs of interest. This distinction is important because not every geometric deviation has the same thermal consequence.

Overall, this thesis shows that automated dwelling–stairwell zoning can form a useful bridge between limited urban geometry and more realistic building energy simulation. Detailed internal layouts are not always necessary to improve urban-scale modelling; approximate yet validated zoning structures can already provide additional value for simulation. With better ground-truth datasets, improved stairwell-placement assumptions, and more reliable enrichment inputs, MZA-based workflows can support more transparent, scalable, and decision-relevant simulation of residential building stocks. This knowledge can help future urban energy models move beyond simple external geometry towards models that better reflect how buildings are divided, used, and experienced.

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A. Appendix

A.1. MSD ground-truth layouts

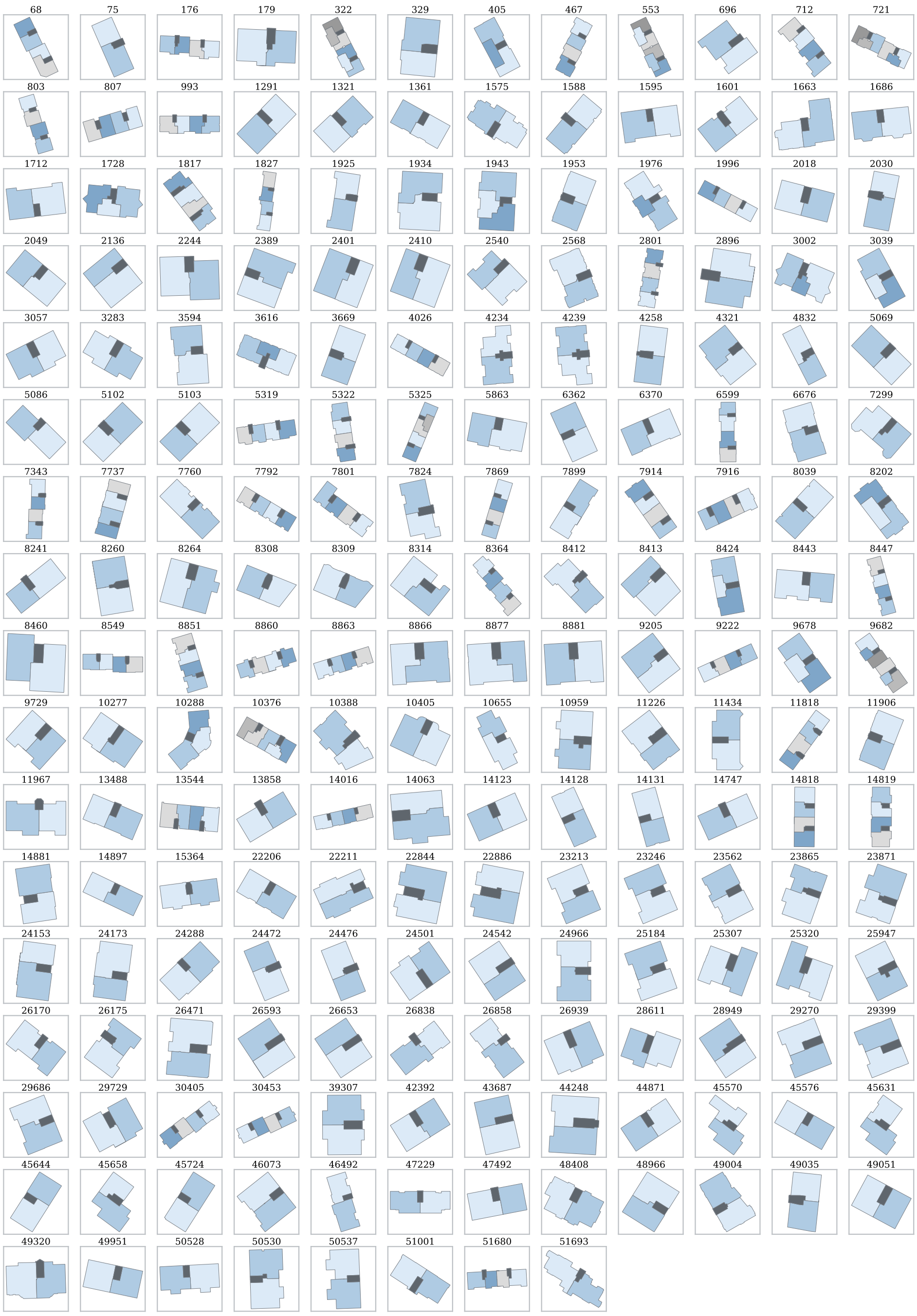

Figure A.1: Selected MSD ground-truth thermal zoning layouts with corresponding building IDs.

Figure A.1 provides an overview of the selected MSD reference layouts used for the geometric validation. The figure shows the extracted dwelling and stairwell zoning representation after the room-level MSD graph data were processed and aggregated to the validation abstraction level.

A.2. Real estate ground-truth layouts

Figure A.2: Selected real estate ground-truth thermal zoning layouts with corresponding case IDs.

Figure A.2 shows the selected real estate reference layouts used in the validation. These layouts were manually digitised from PDF or image-based floor plans, scaled to metric coordinates, and classified into dwelling and stairwell zones.

A.3. Auxiliary input estimation for MSD cases

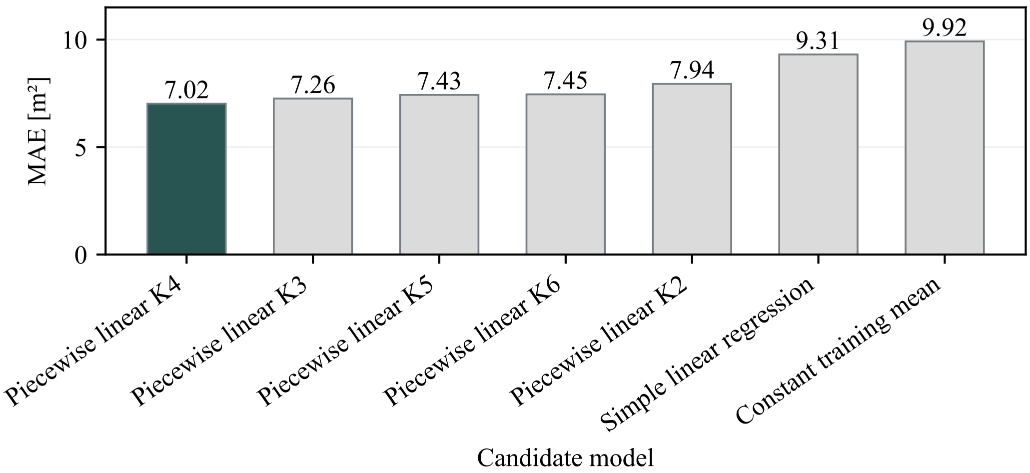

Figure A.3: Comparison of candidate models for estimating the missing mean dwelling-area input from external footprint area.

Figure A.3 compares the prediction error of the candidate estimation models using the mean absolute error (MAE). The comparison was used to select the piecewise-linear model for estimating the missing mean dwelling-area input in the MSD-based MZA preprocessing workflow.

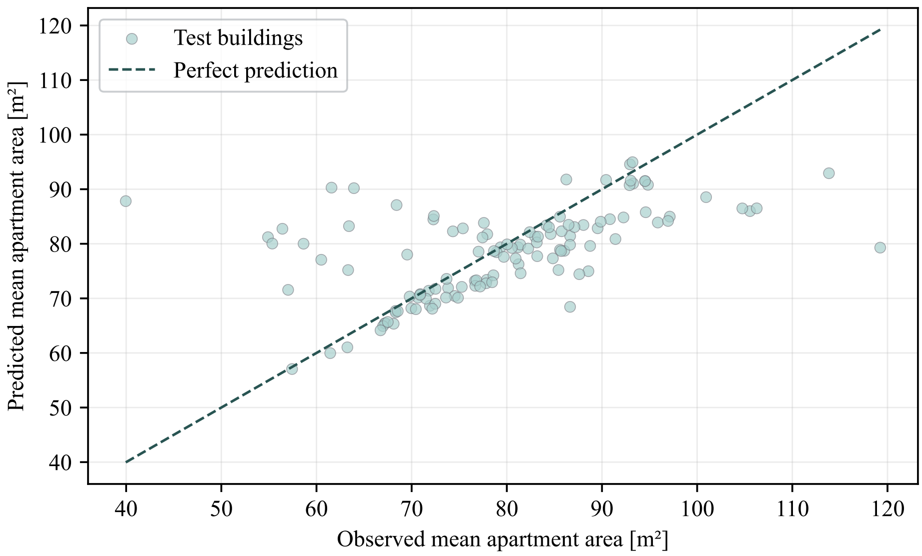

Figure A.4: Predicted versus observed mean dwelling area for the selected auxiliary input estimation model.

Figure A.4 shows the agreement between observed mean dwelling areas and the values predicted by the selected estimation model. The figure is used to assess whether the model provides a reasonable estimate of the auxiliary input for the MZA workflow.

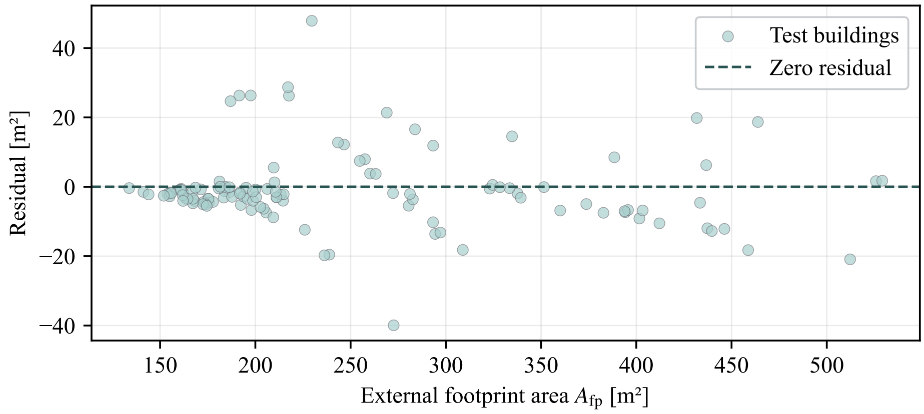

Figure A.5: Residuals of the selected mean dwelling-area estimation model plotted against external footprint area.

Figure A.5 shows the residuals of the selected estimation model across the external footprint-area range. The residual pattern indicates that the footprint area explains only part of the variation in mean dwelling area, which is why the model is used as a pragmatic auxiliary input estimate rather than as a complete explanatory model.

A.4. Additional sensitivity analysis results

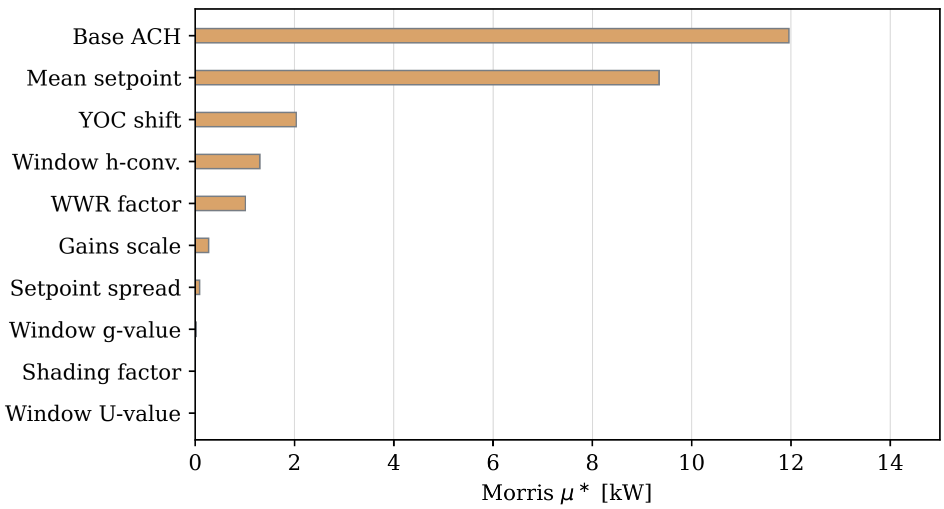

Figure A.6: Morris screening results for peak heating load in variant V3. Bars show the absolute elementary-effect mean μ^* for each uncertain input parameter.

Figure A.6 provides the additional Morris screening result for peak heating load. The plot supports the input parameter sensitivity analysis by showing which uncertain inputs have the strongest overall influence on the peak-load output before the reduced Sobol parameter set is selected.